

(12) **United States Patent**
R. et al.

(10) **Patent No.:** **US 12,414,261 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **THERMAL MANAGEMENT SYSTEMS FOR ELECTRONIC DEVICES AND RELATED METHODS**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 445 days.

(21) Appl. No.: **17/711,821**

(22) Filed: **Apr. 1, 2022**

(65) **Prior Publication Data**

US 2022/0225536 A1 Jul. 14, 2022

(30) **Foreign Application Priority Data**

Jun. 25, 2021 (IN) 202141028573

(51) **Int. Cl.**
H05K 7/20 (2006.01)
G06F 1/20 (2006.01)

(52) **U.S. Cl.**
CPC **H05K 7/20172** (2013.01); **G06F 1/203** (2013.01); **H05K 7/20145** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC H05K 7/20172; H05K 7/20145; H05K 7/20154; H05K 7/20509; H05K 7/20518; G06F 1/203

(Continued)

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Primary Examiner — Anatoly Vortman

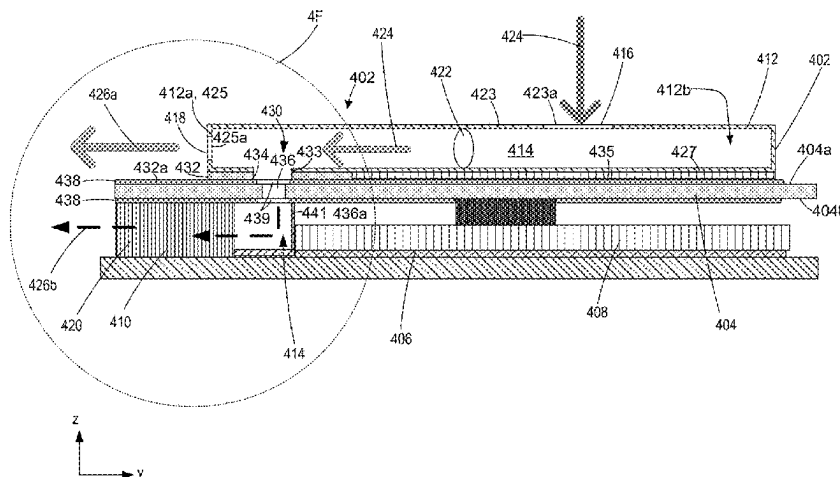
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(57) **ABSTRACT**

Synthetic jet systems and related methods are disclosed. An electronic device includes a heat spreader including a first surface and a second surface opposite the first surface. A synthetic jet is coupled to the first surface of the heat spreader. The synthetic jet and the heat spreader define a fluid flow passageway having an inlet, a first outlet and a second outlet. The synthetic jet and the heat spreader bifurcate airflow from the inlet such that the first outlet is to exhaust airflow from the passageway adjacent the first surface of the heat spreader and the second outlet is to exhaust airflow from the passageway adjacent the second surface of the heat spreader.

25 Claims, 25 Drawing Sheets



- (52) **U.S. Cl.**
CPC *H05K 7/20154* (2013.01); *H05K 7/20509*
(2013.01); *H05K 7/20518* (2013.01)

- (58) **Field of Classification Search**
USPC 361/696
See application file for complete search history.

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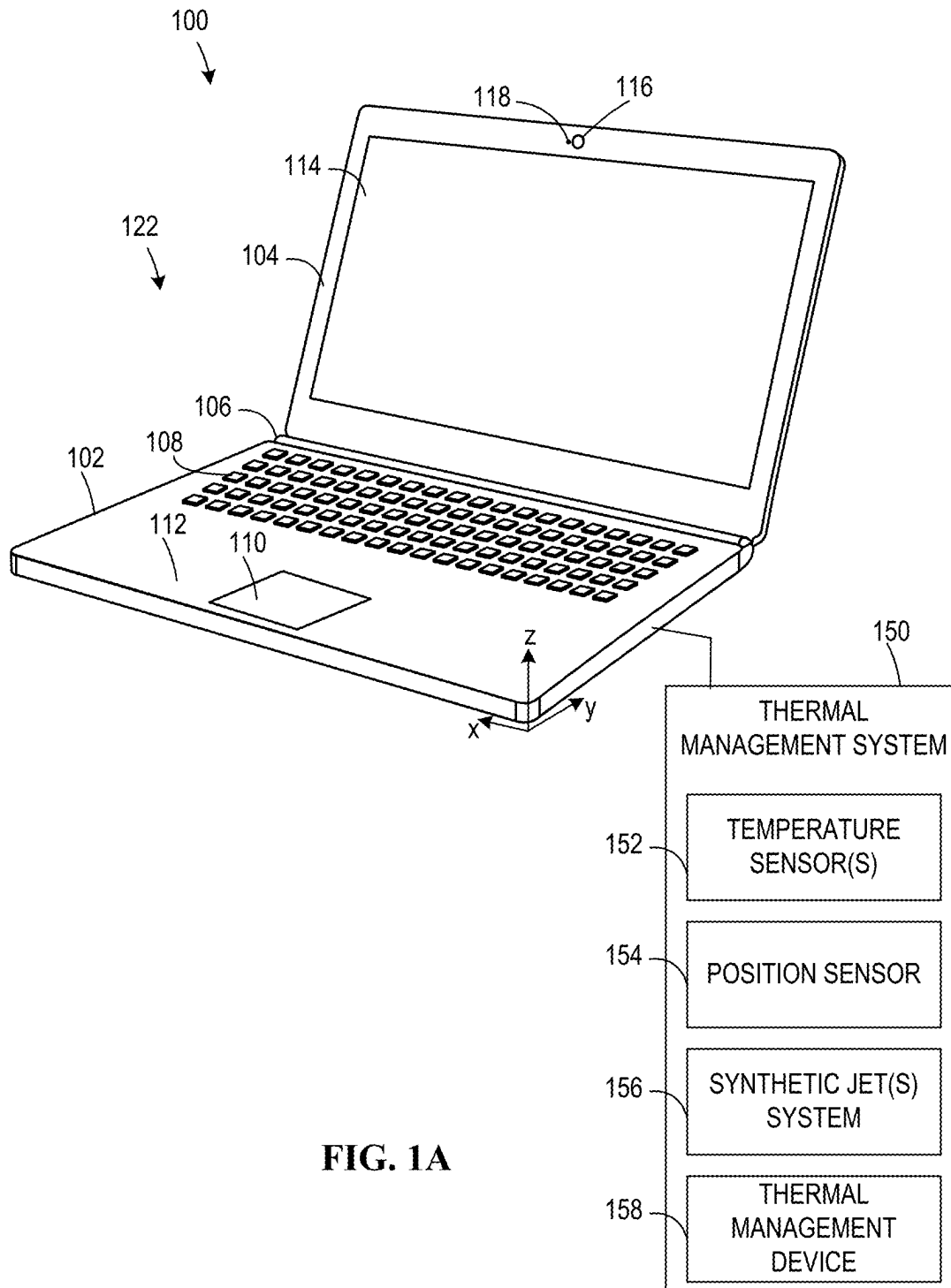


FIG. 1A

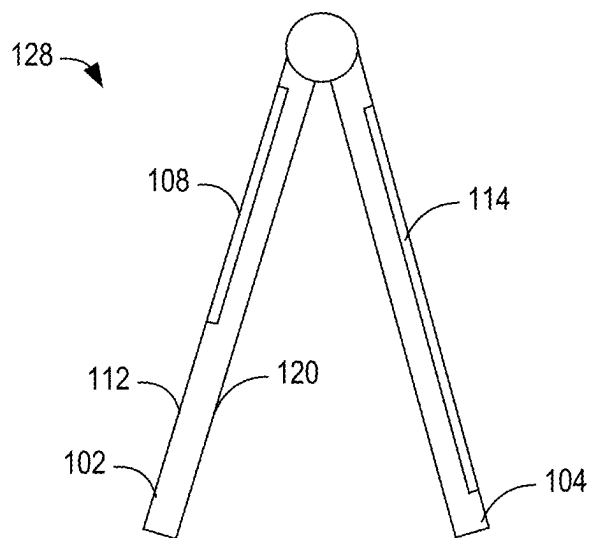


FIG. 1D

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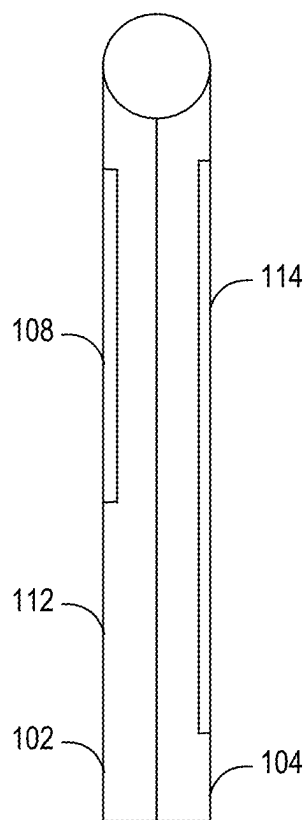


FIG. 1B

126

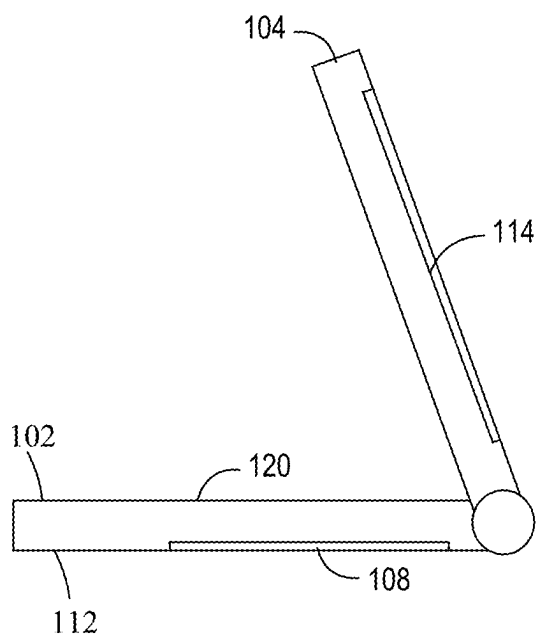
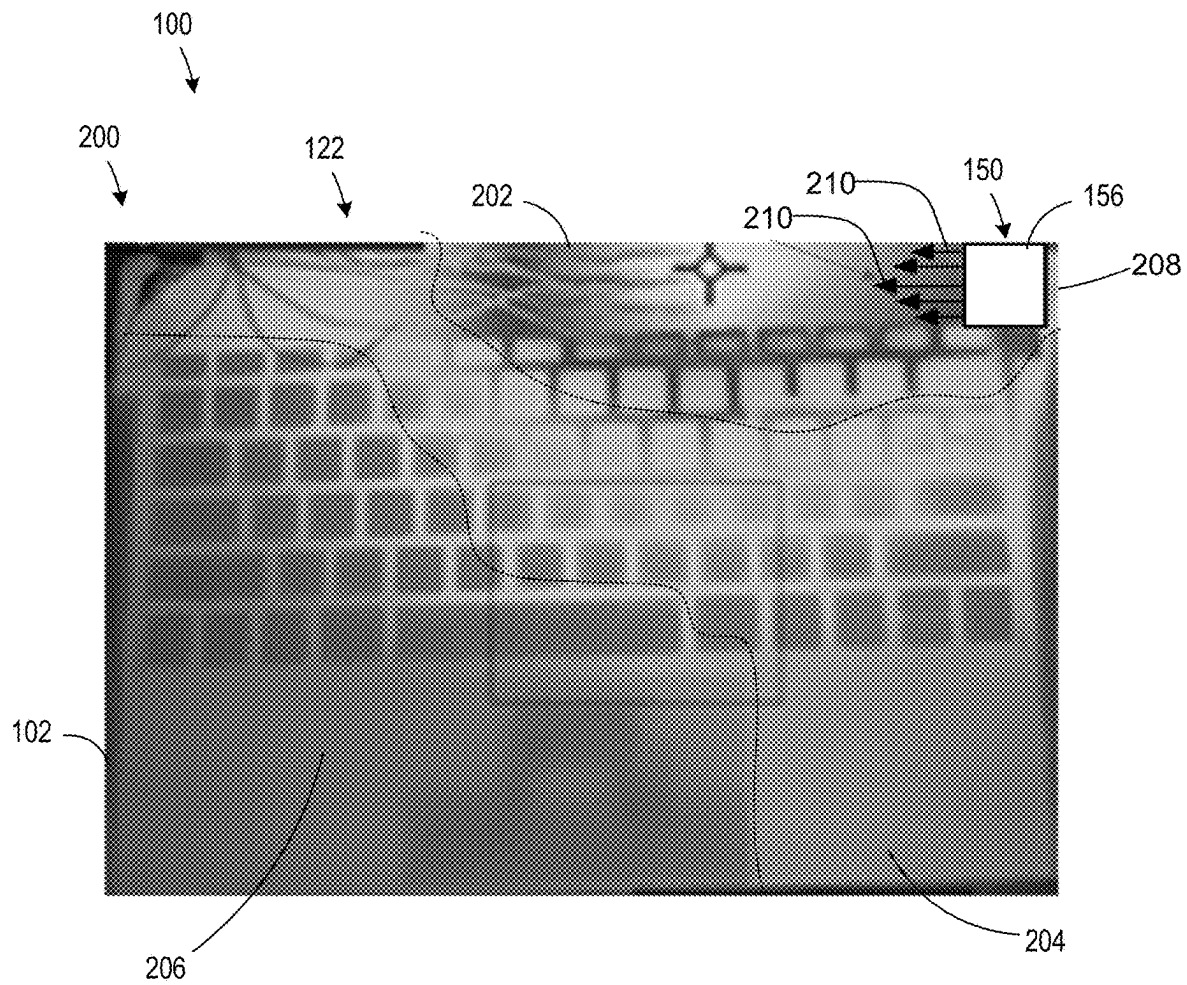


FIG. 1C

**FIG. 2**

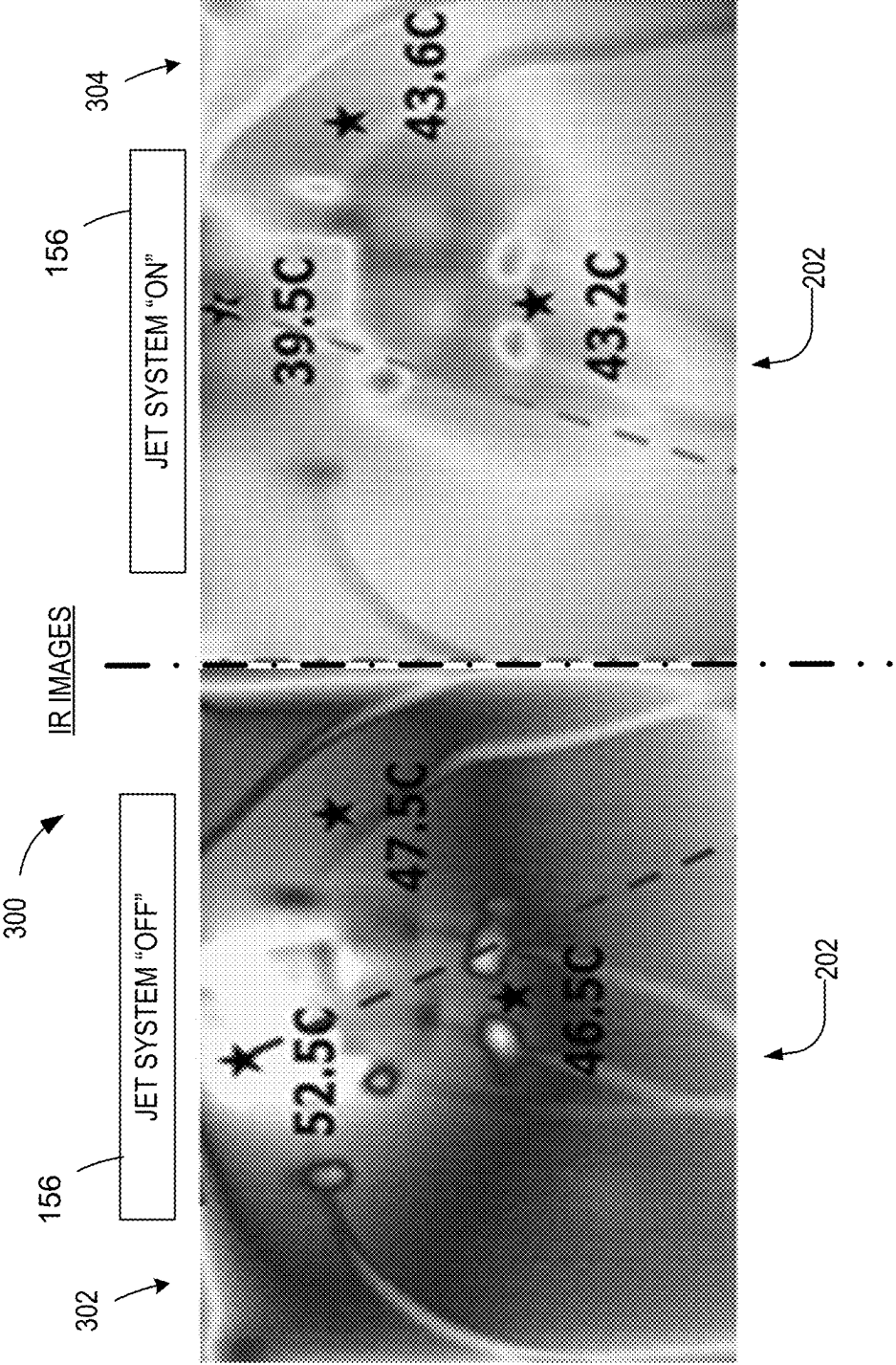


FIG. 3

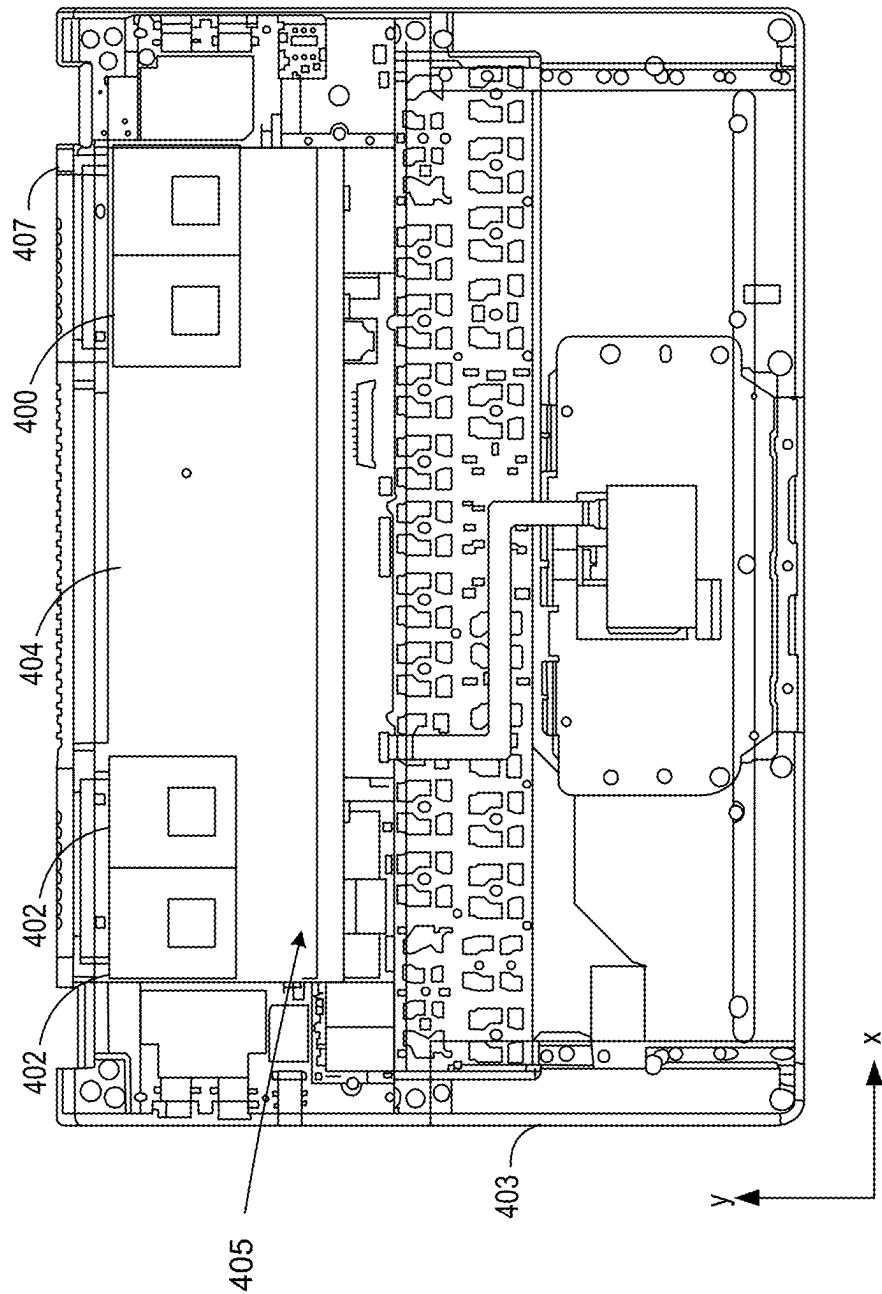
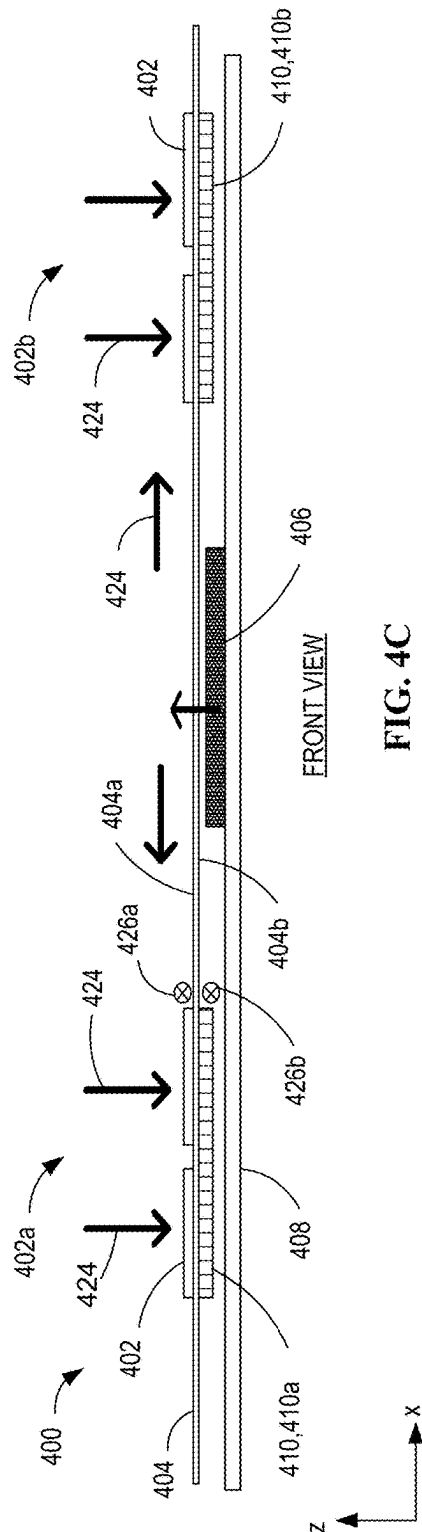
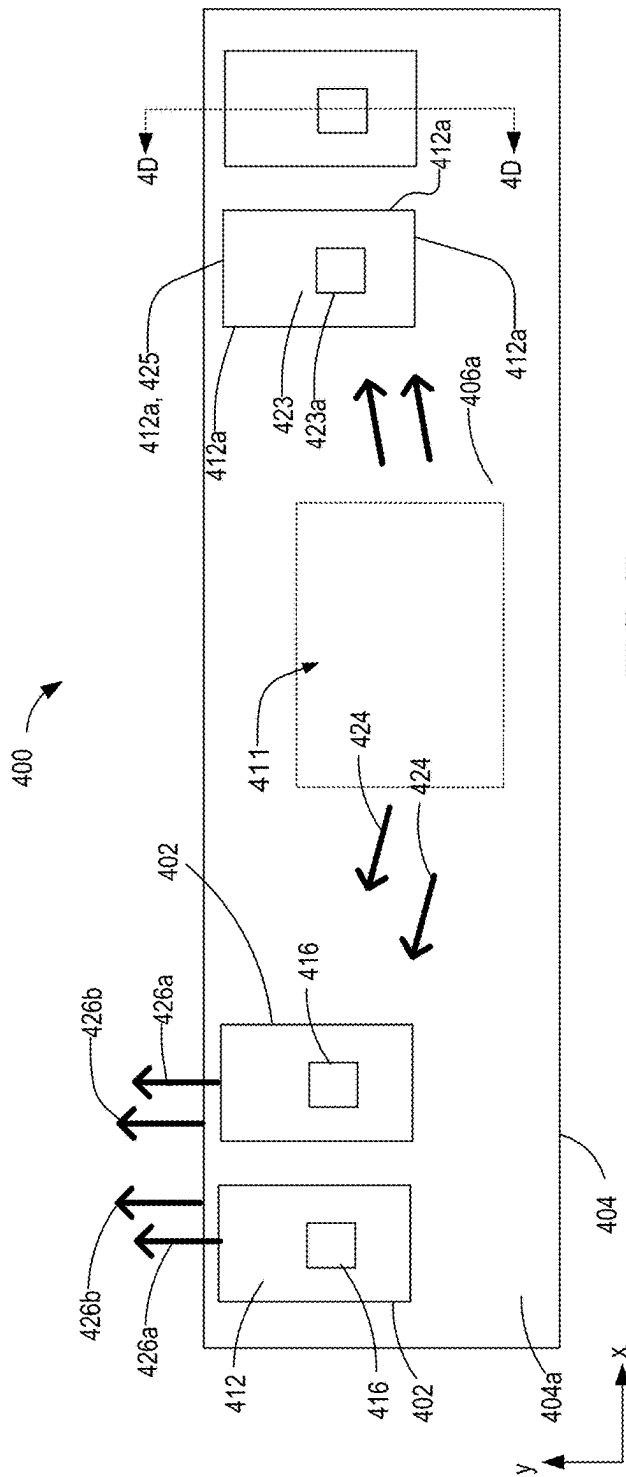


FIG. 4A



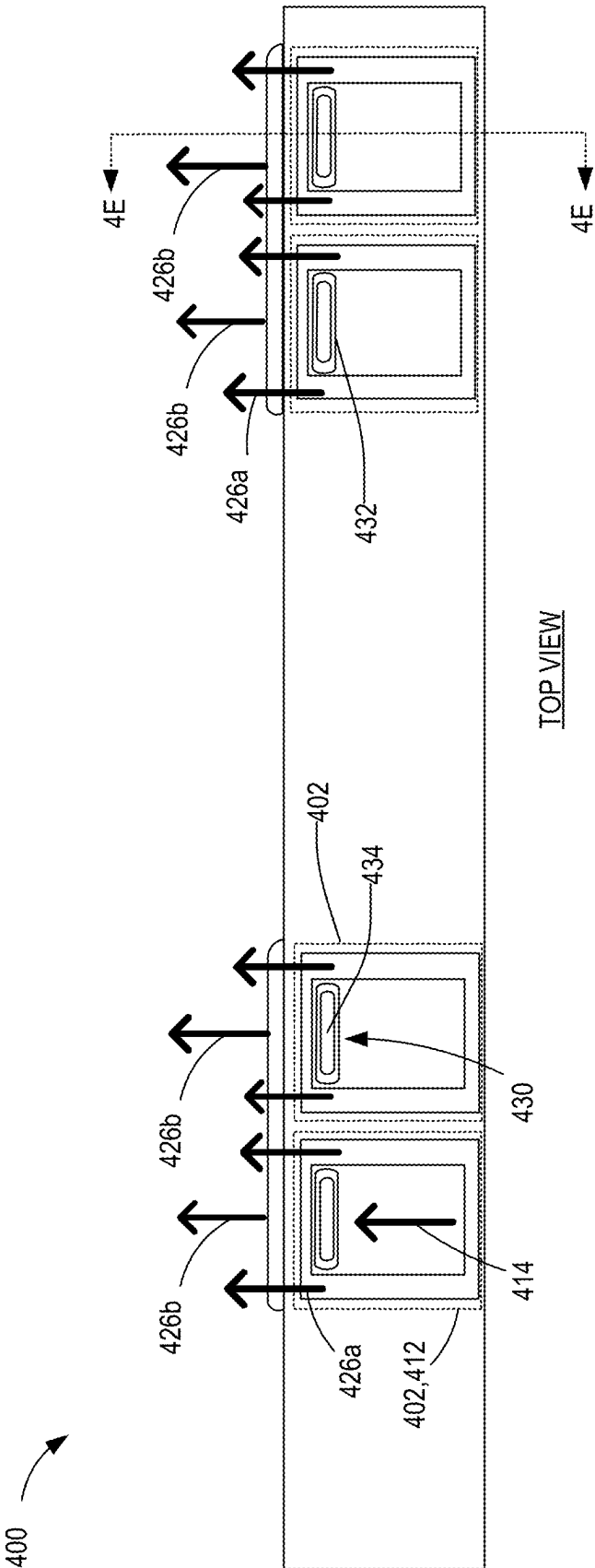


FIG. 4D

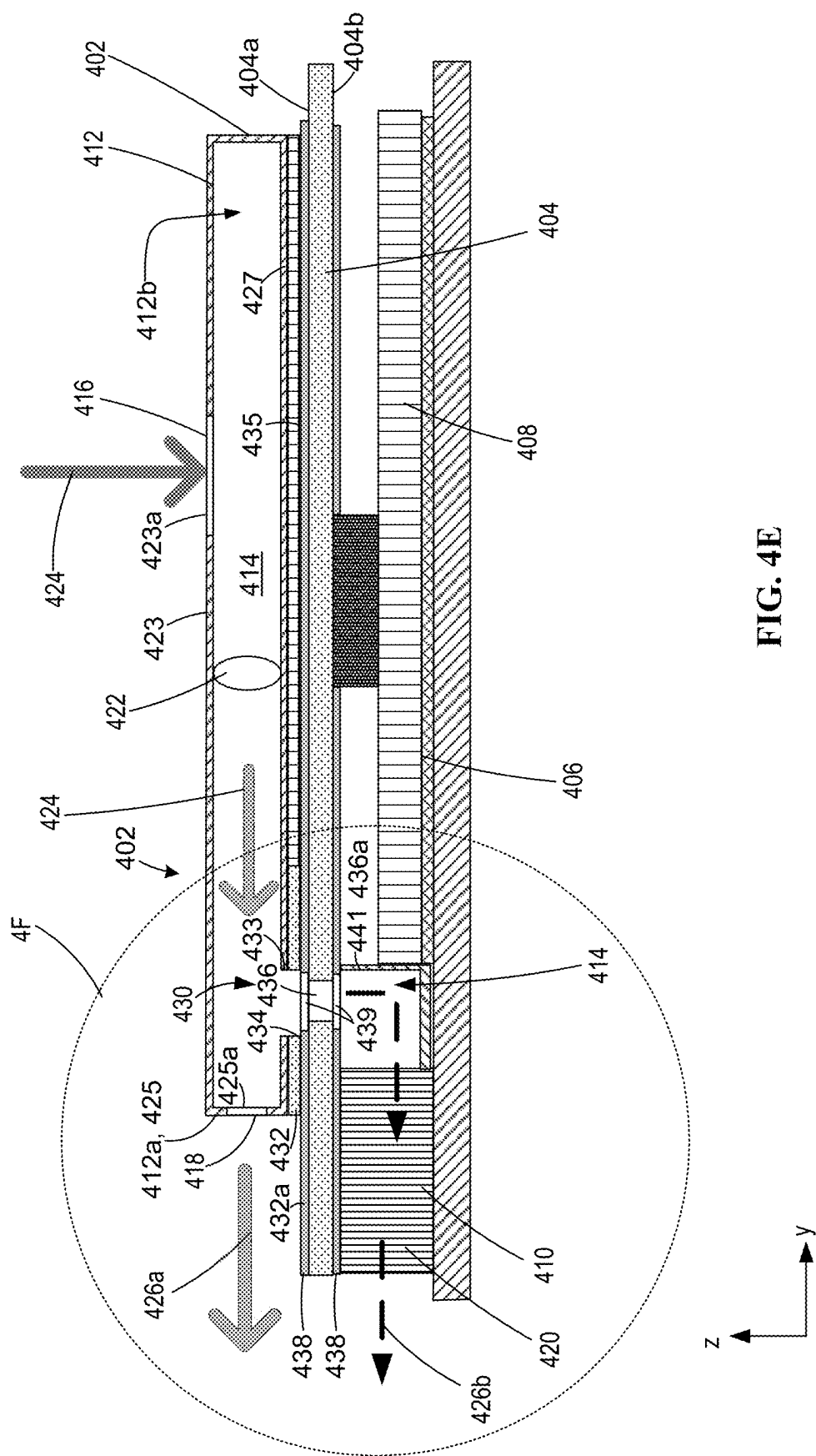


FIG. 4E

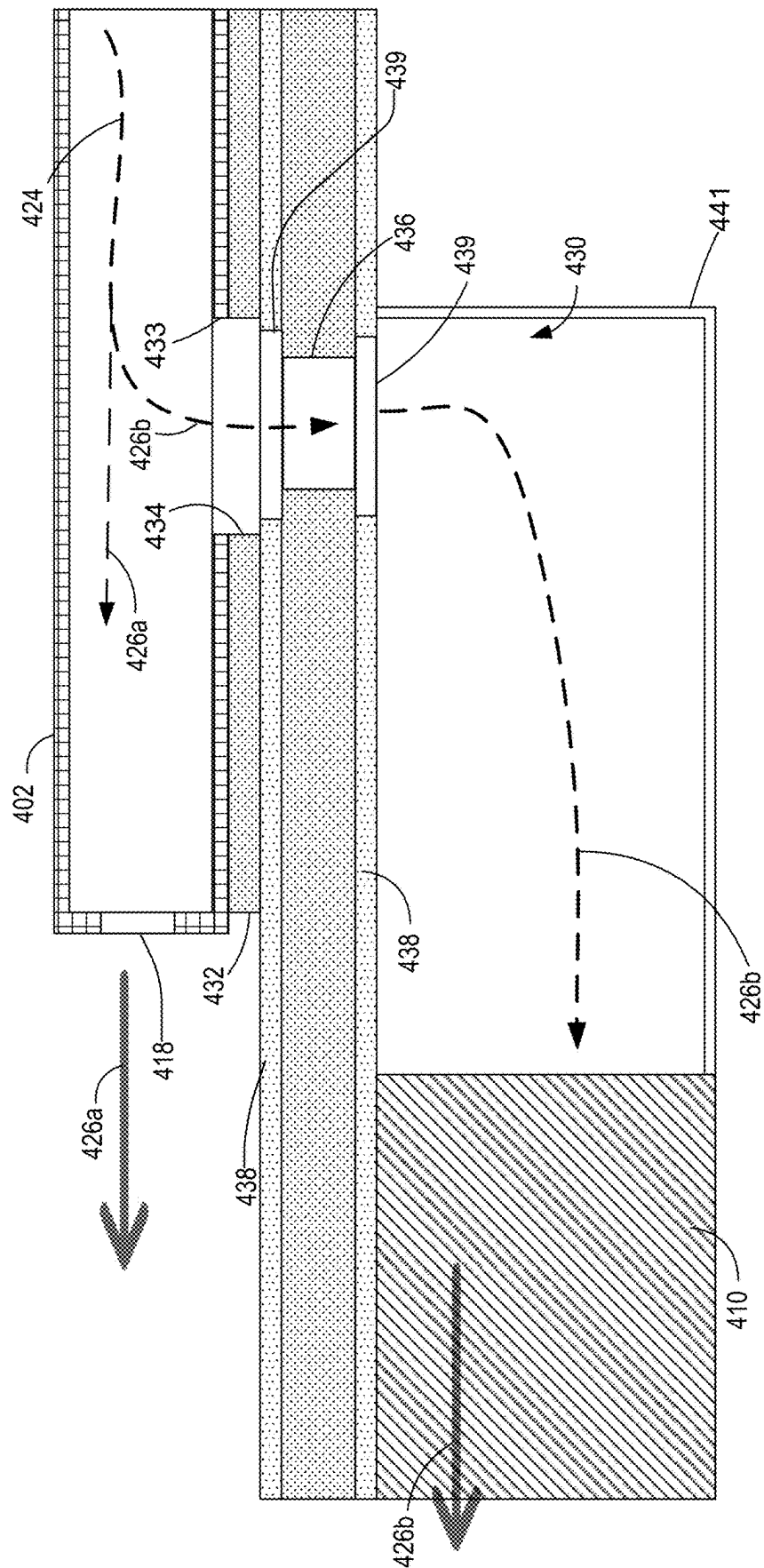
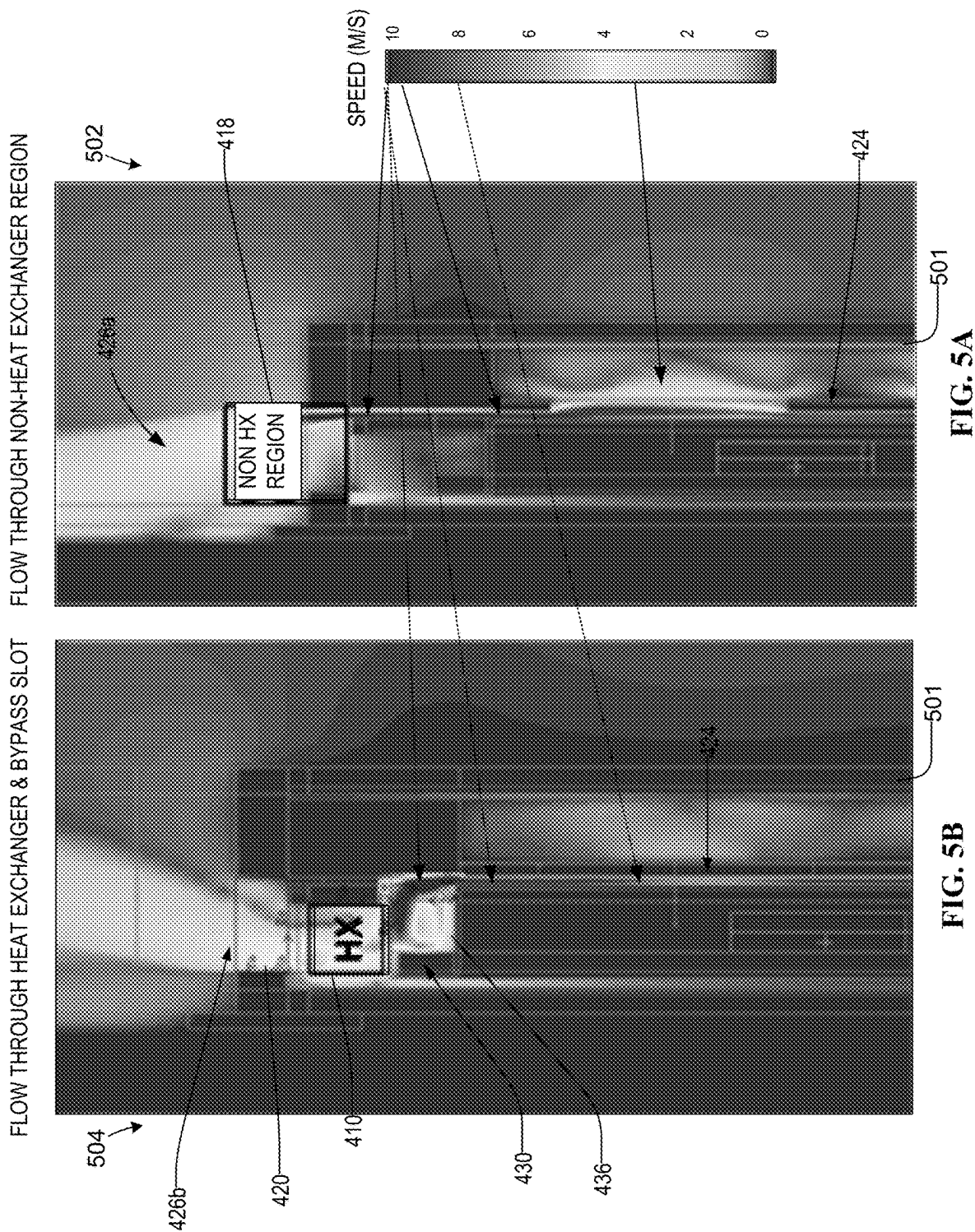
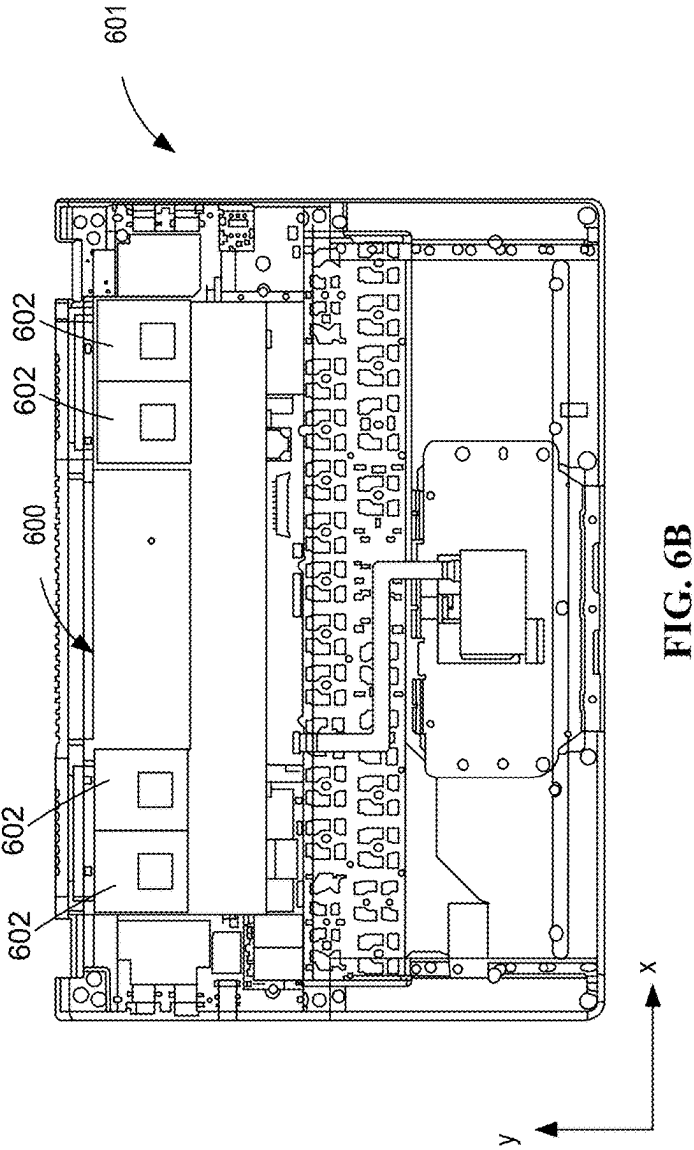
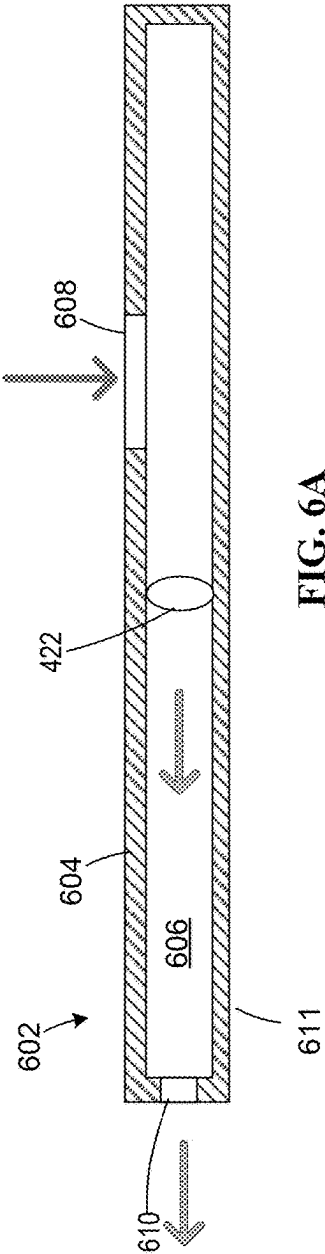


FIG. 4F





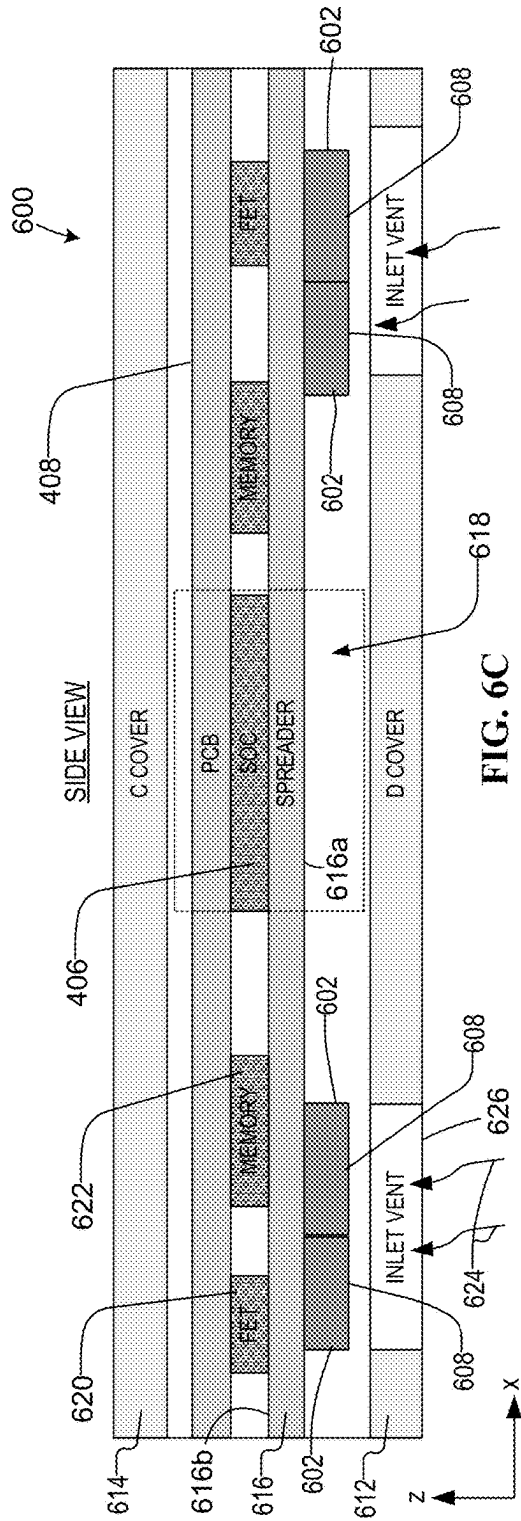


FIG. 6C

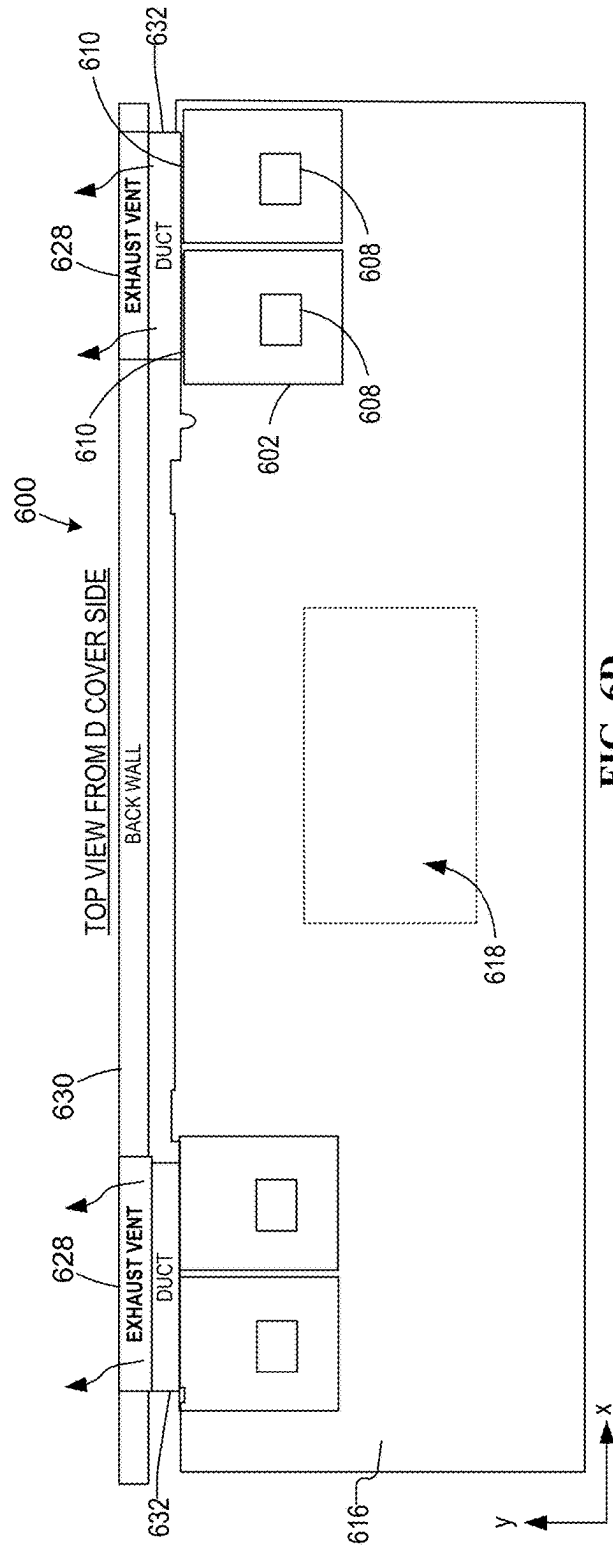


FIG. 6D

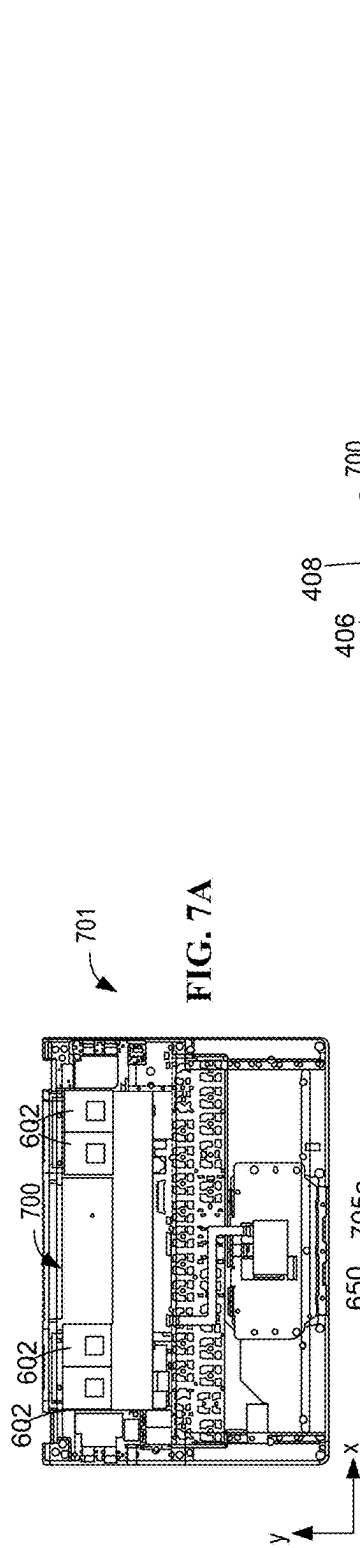


FIG. 7A

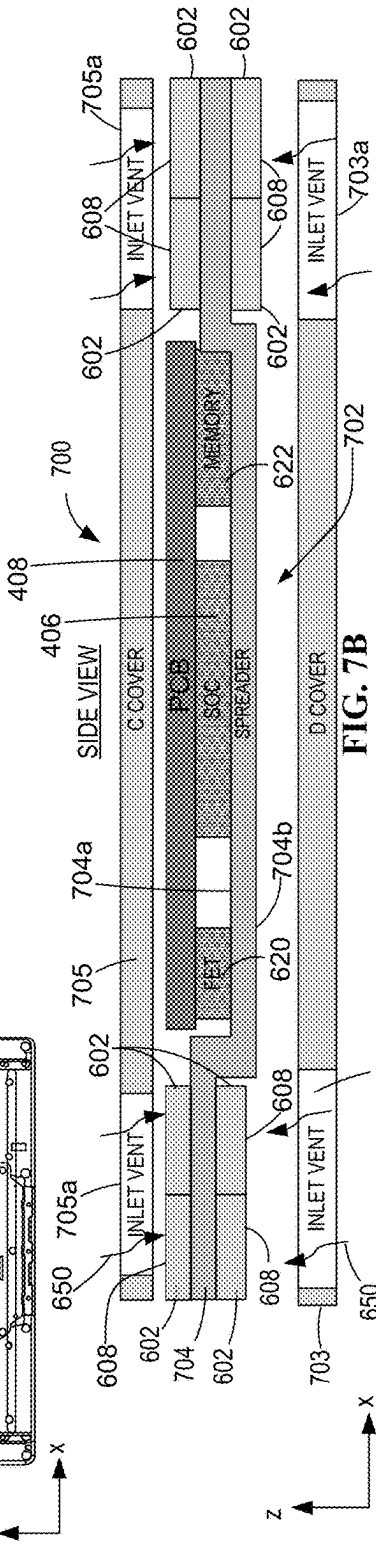


FIG. 7B

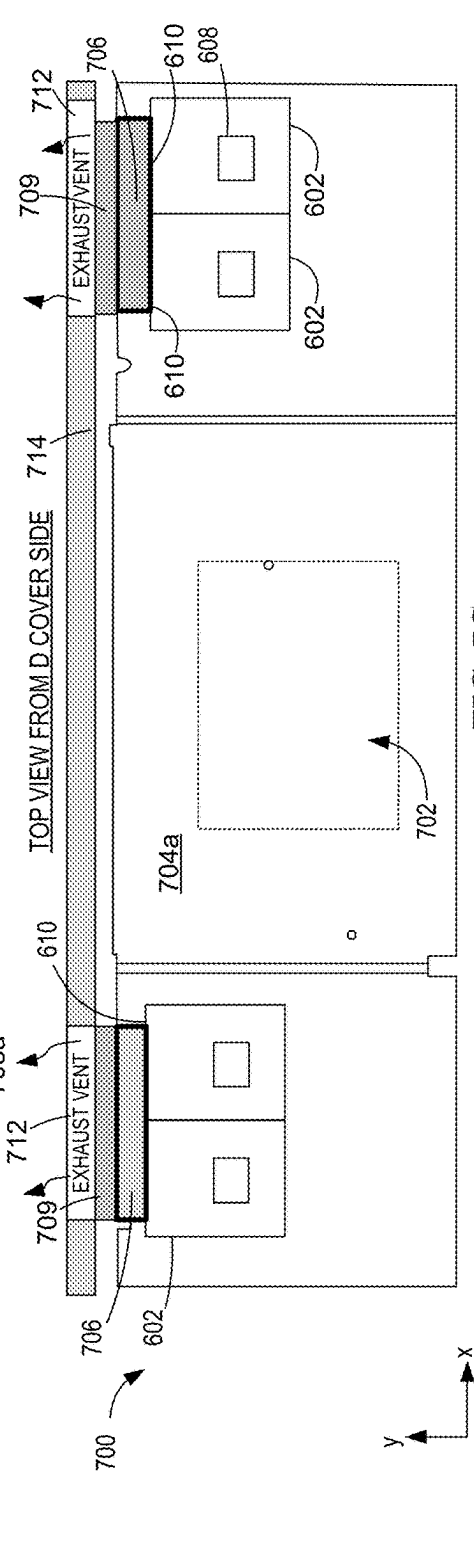
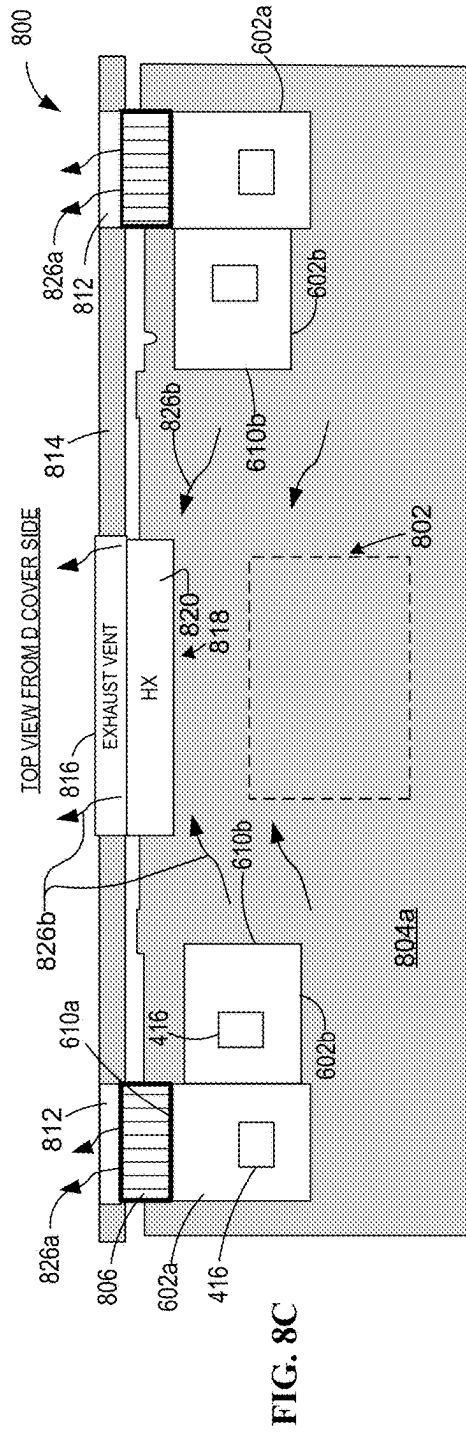
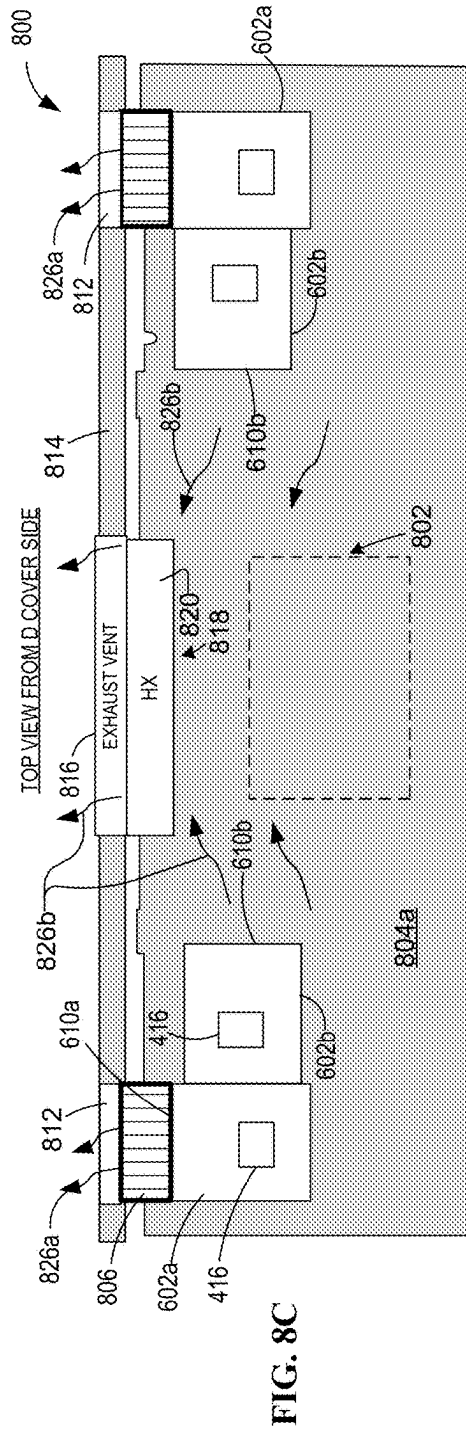
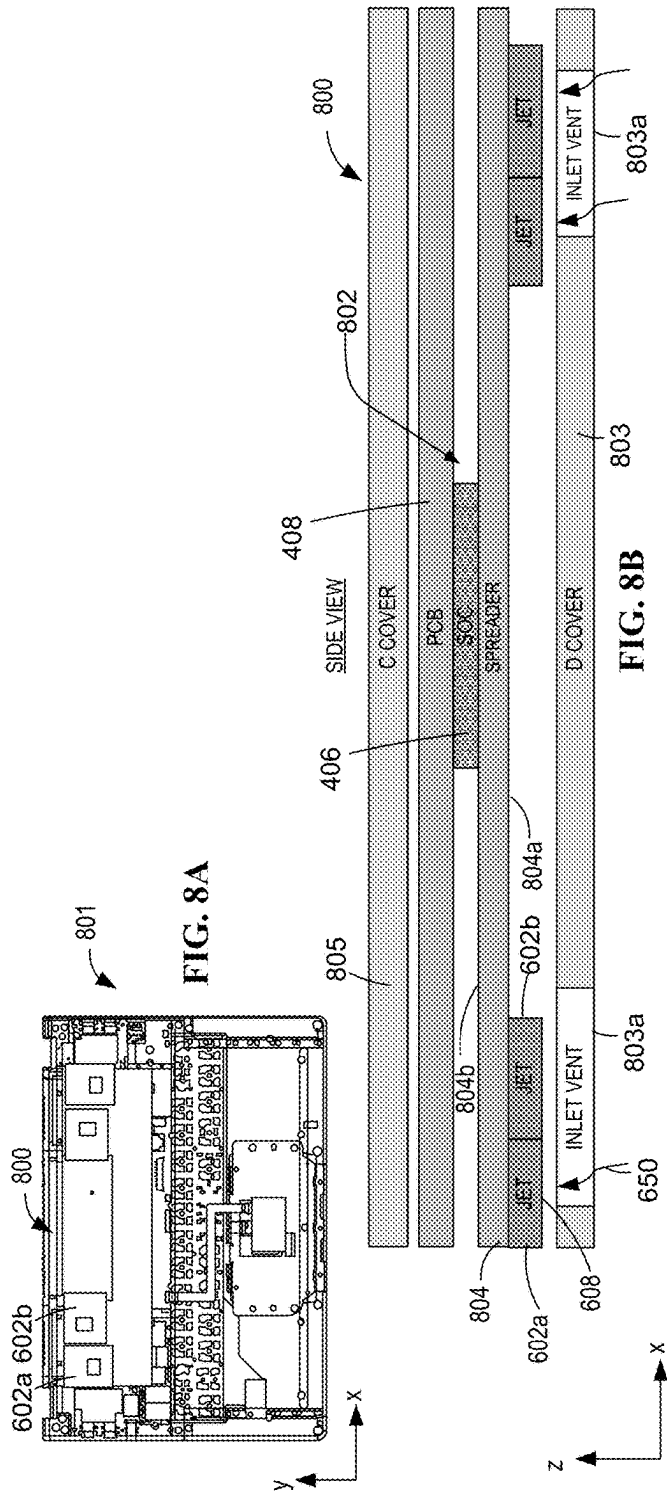
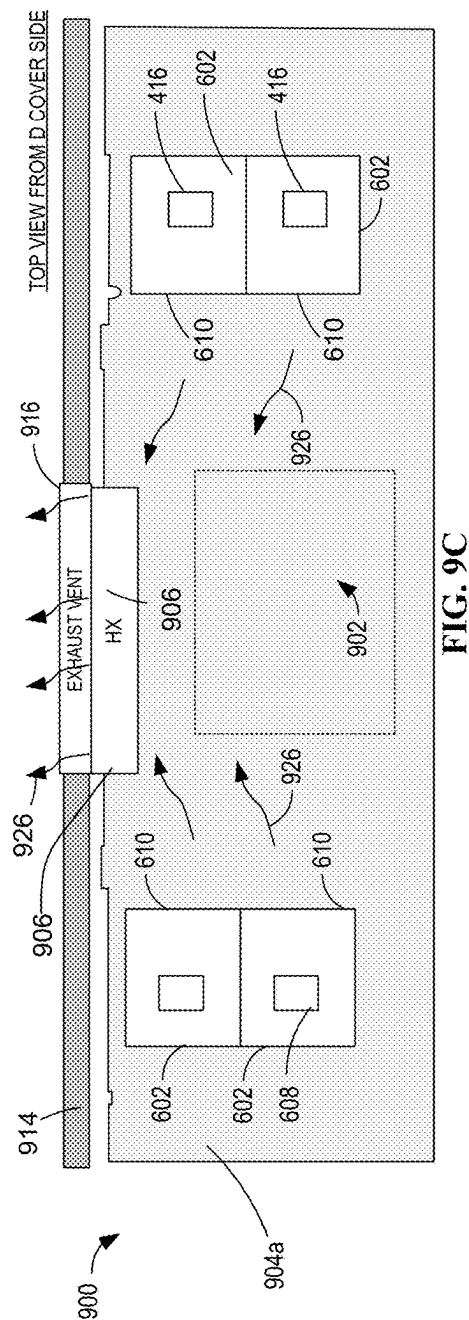
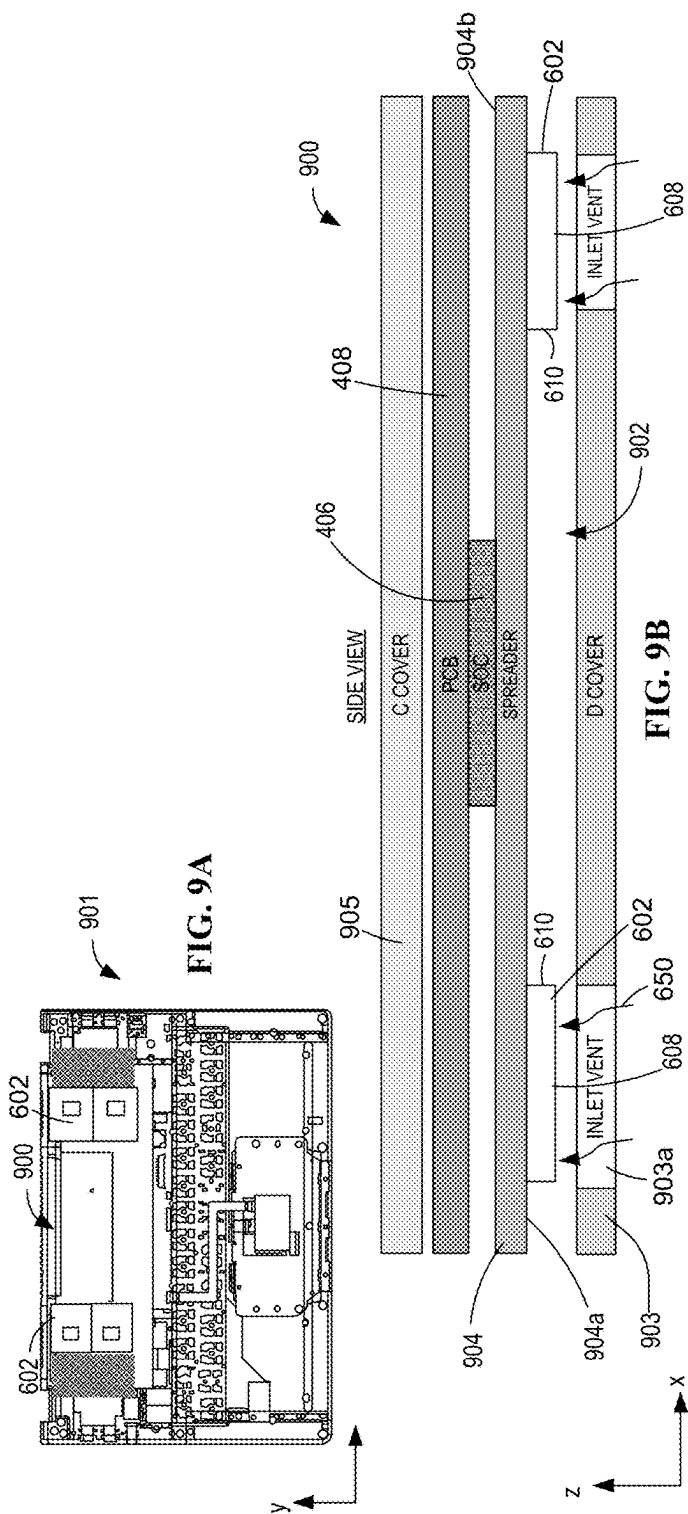


FIG. 7C





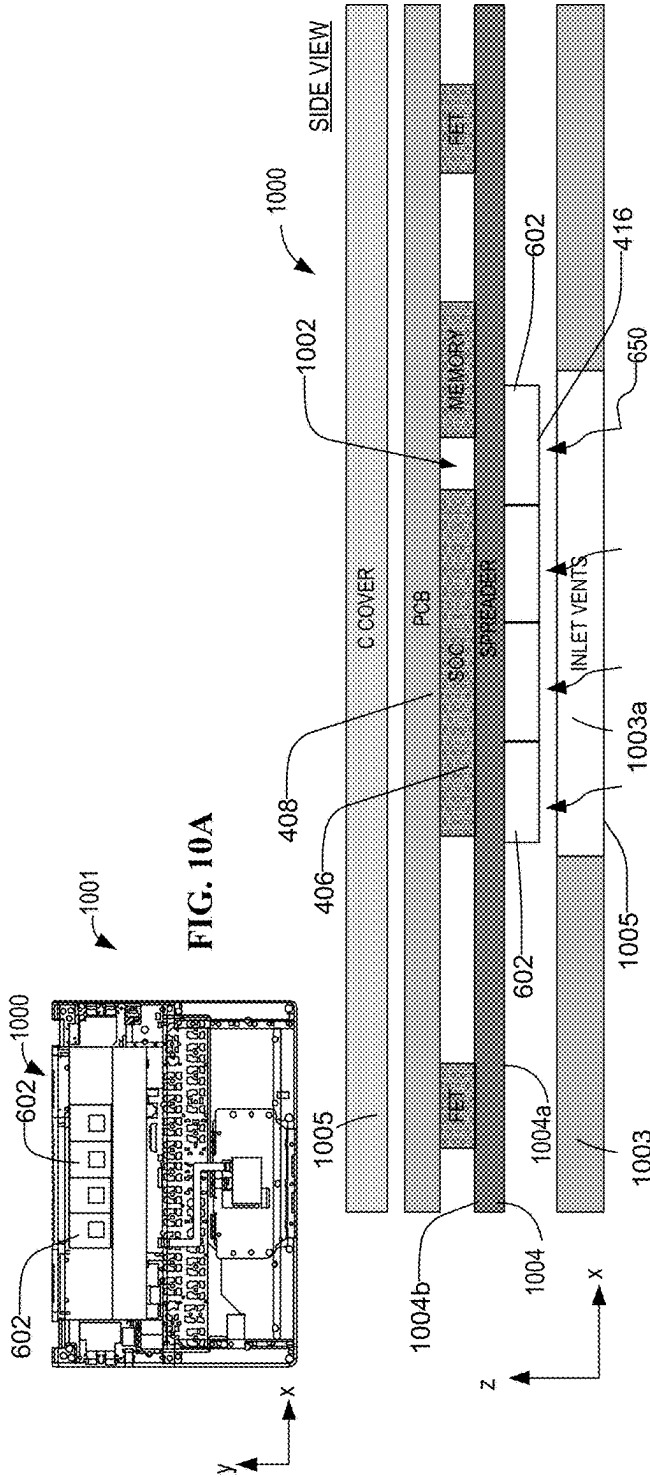


FIG. 10B

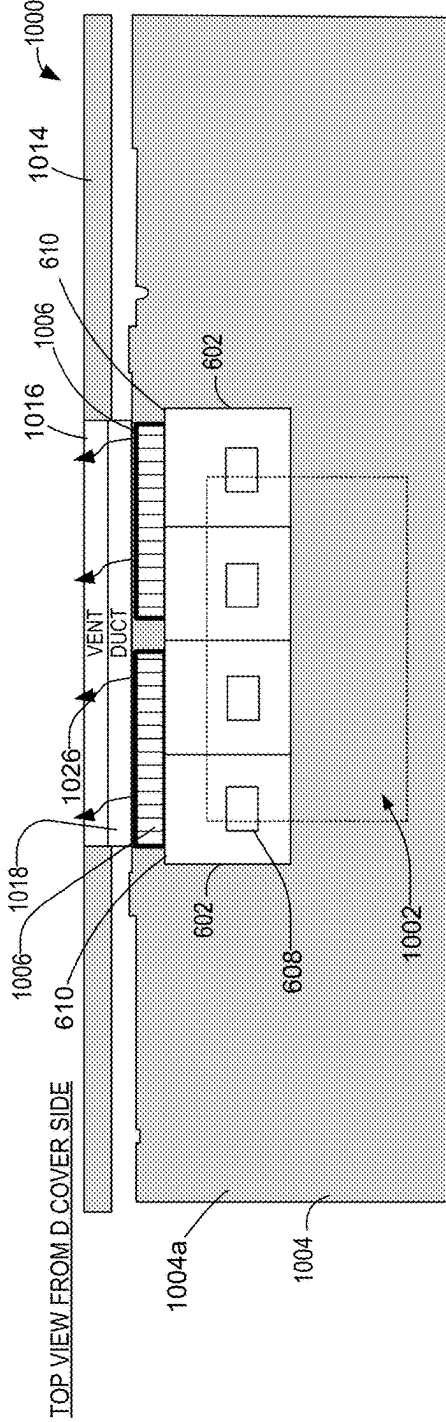
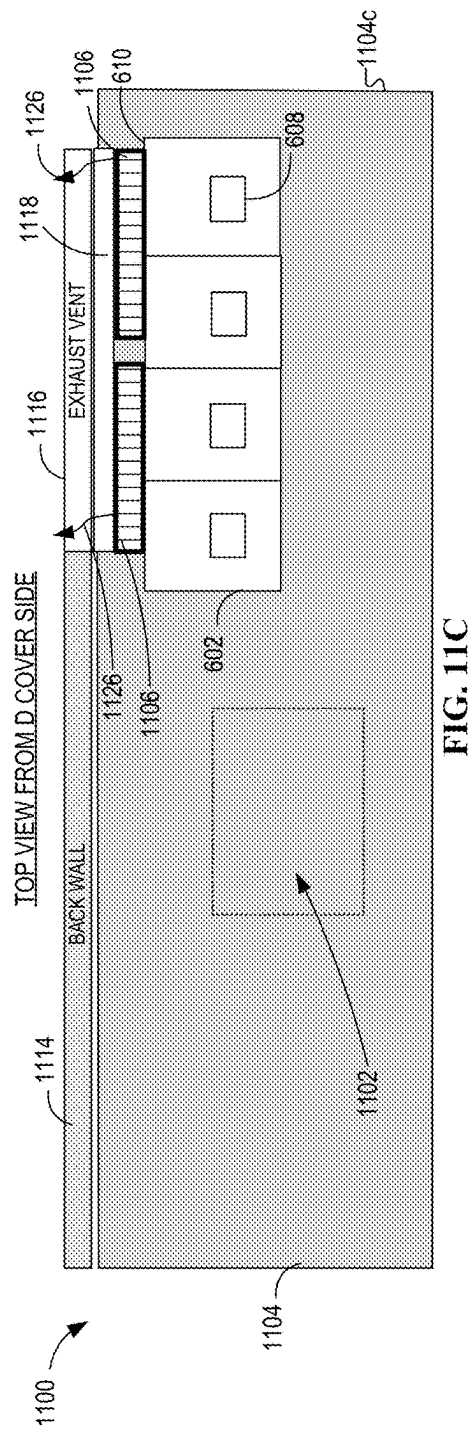
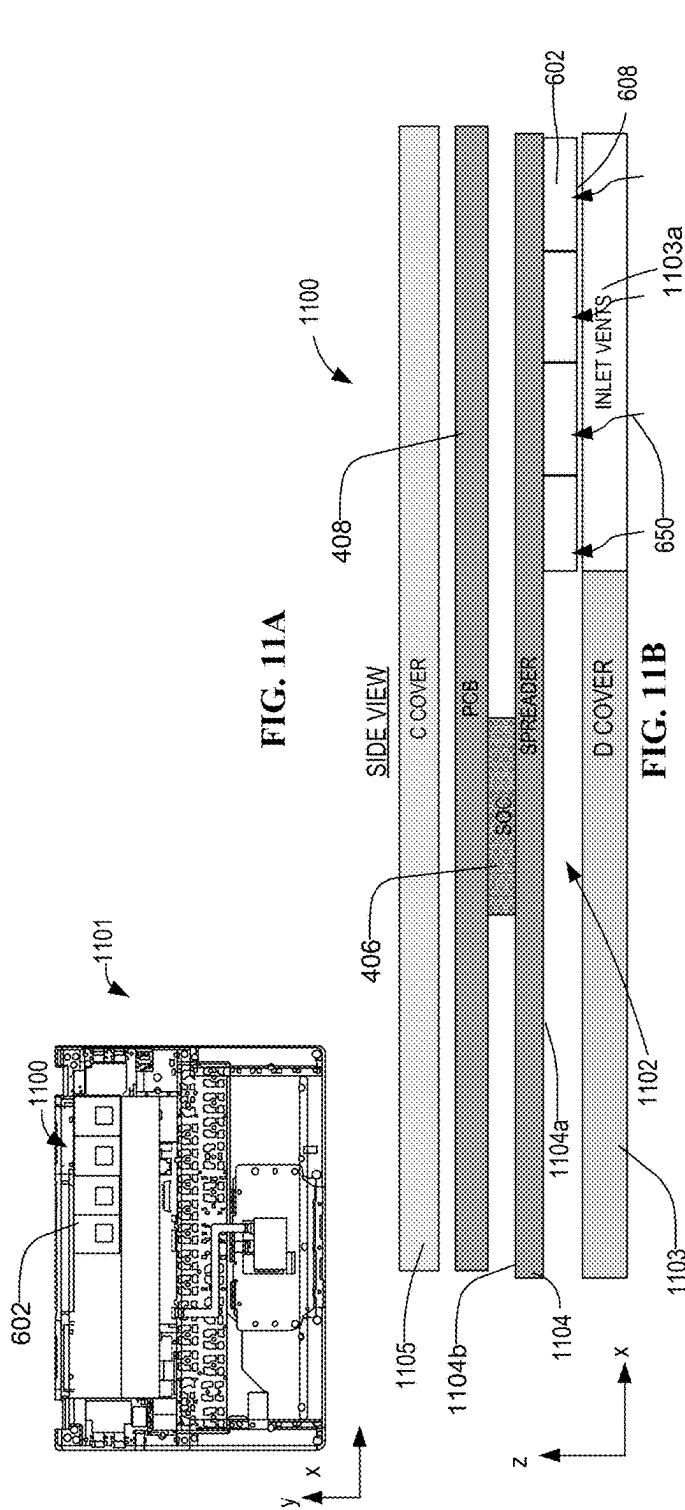


FIG. 10C



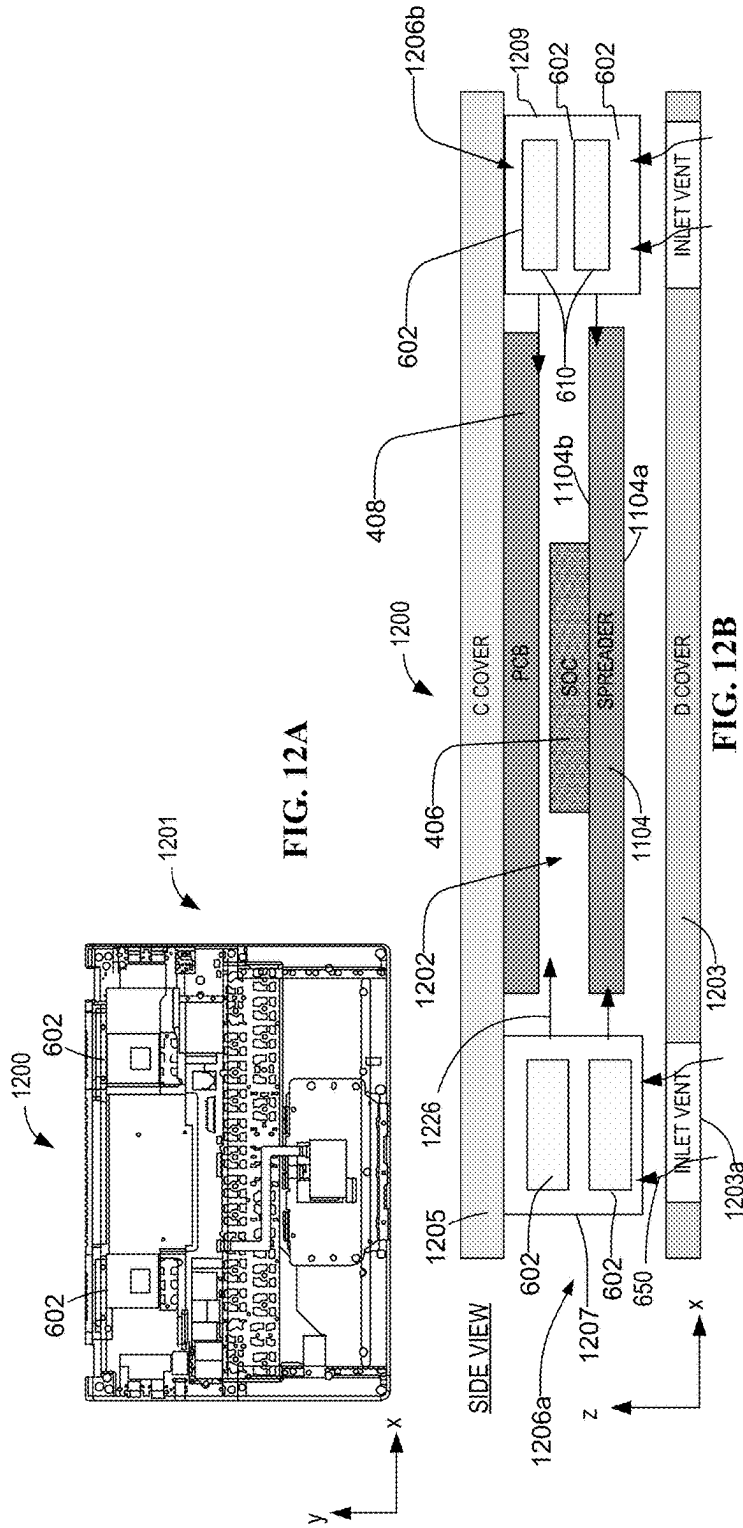


FIG. 12A

FIG. 12B

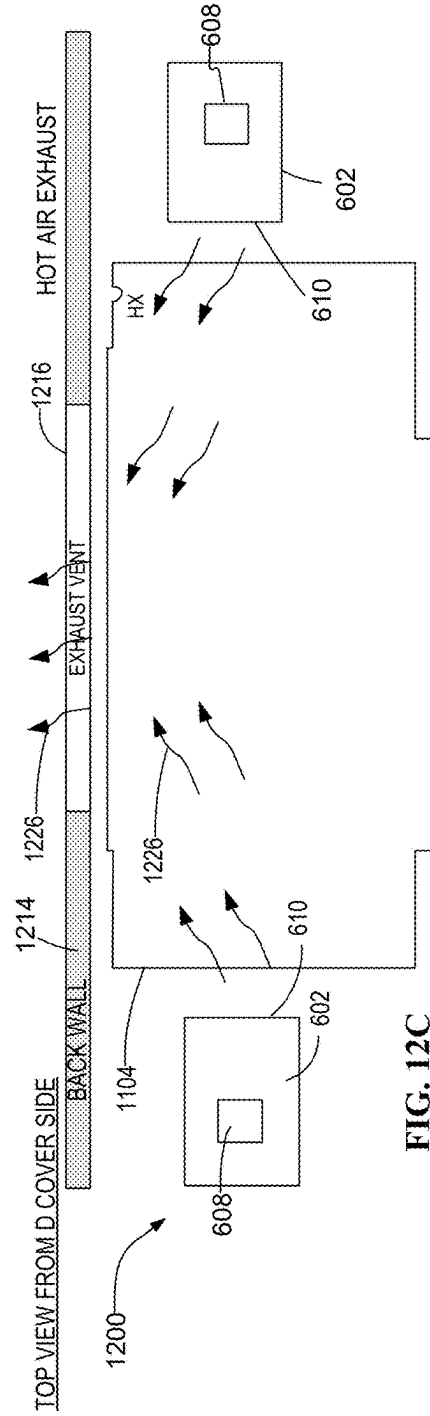


FIG. 12C

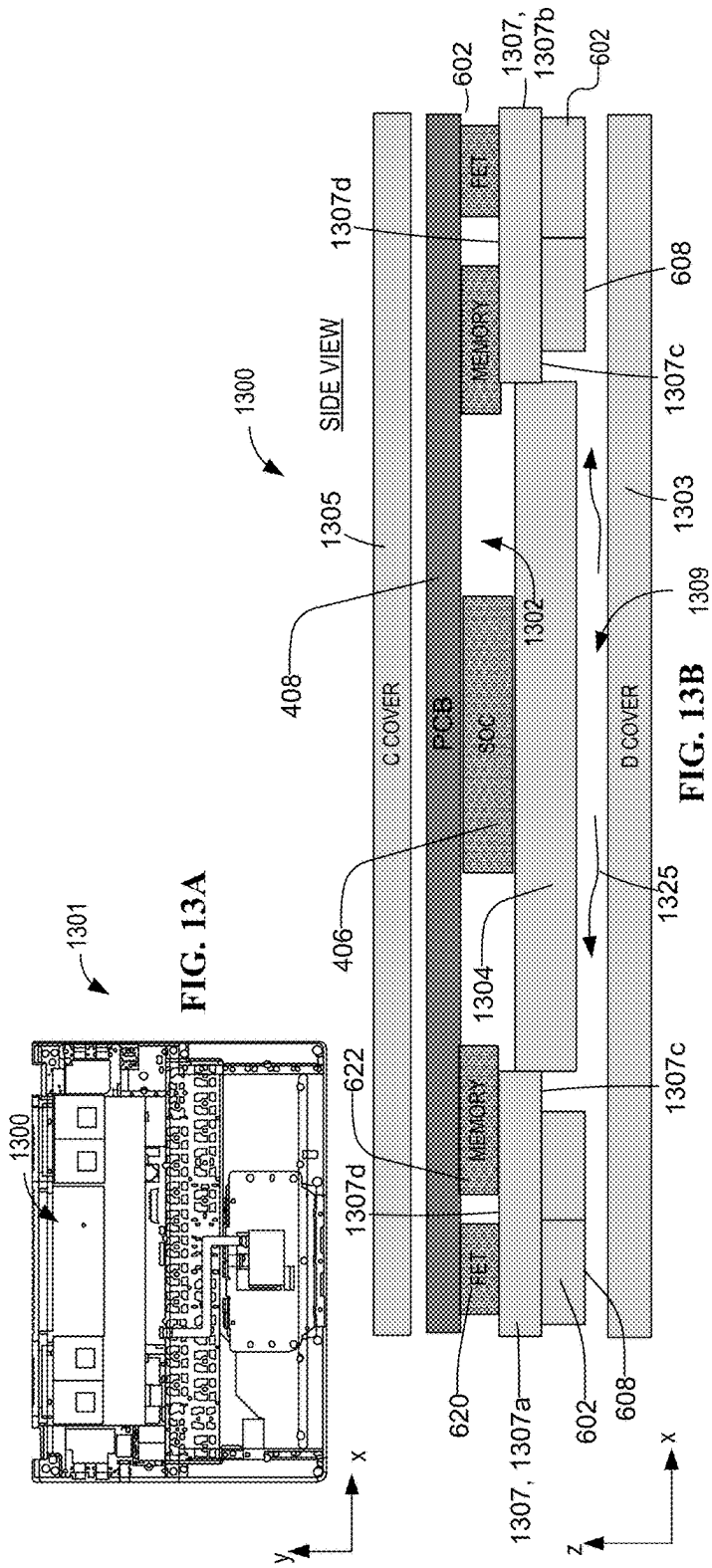
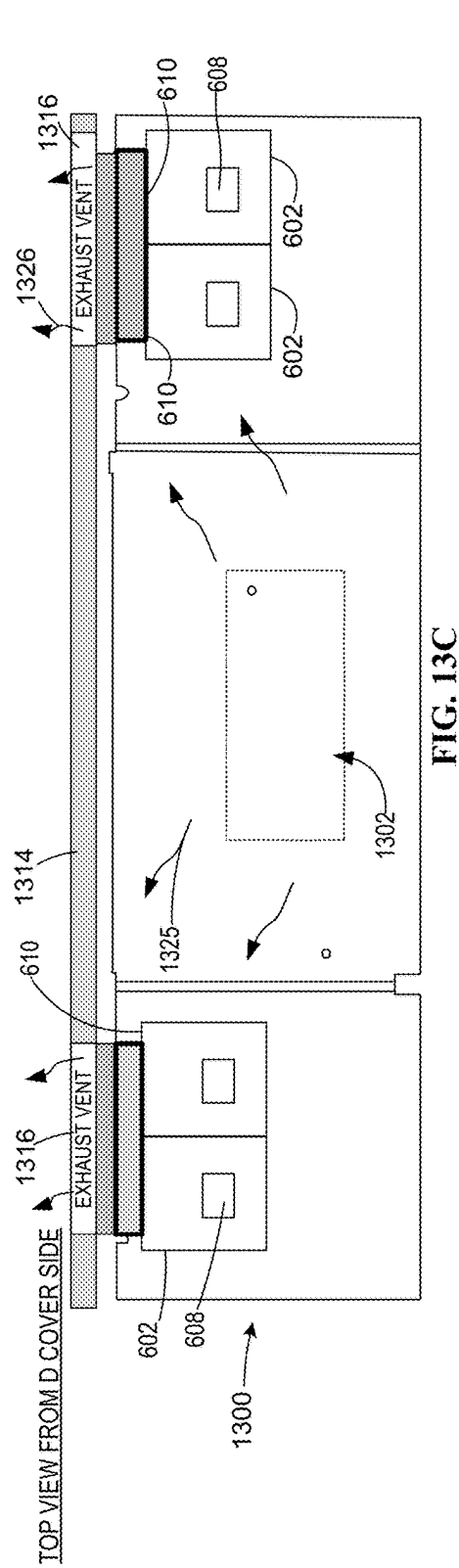
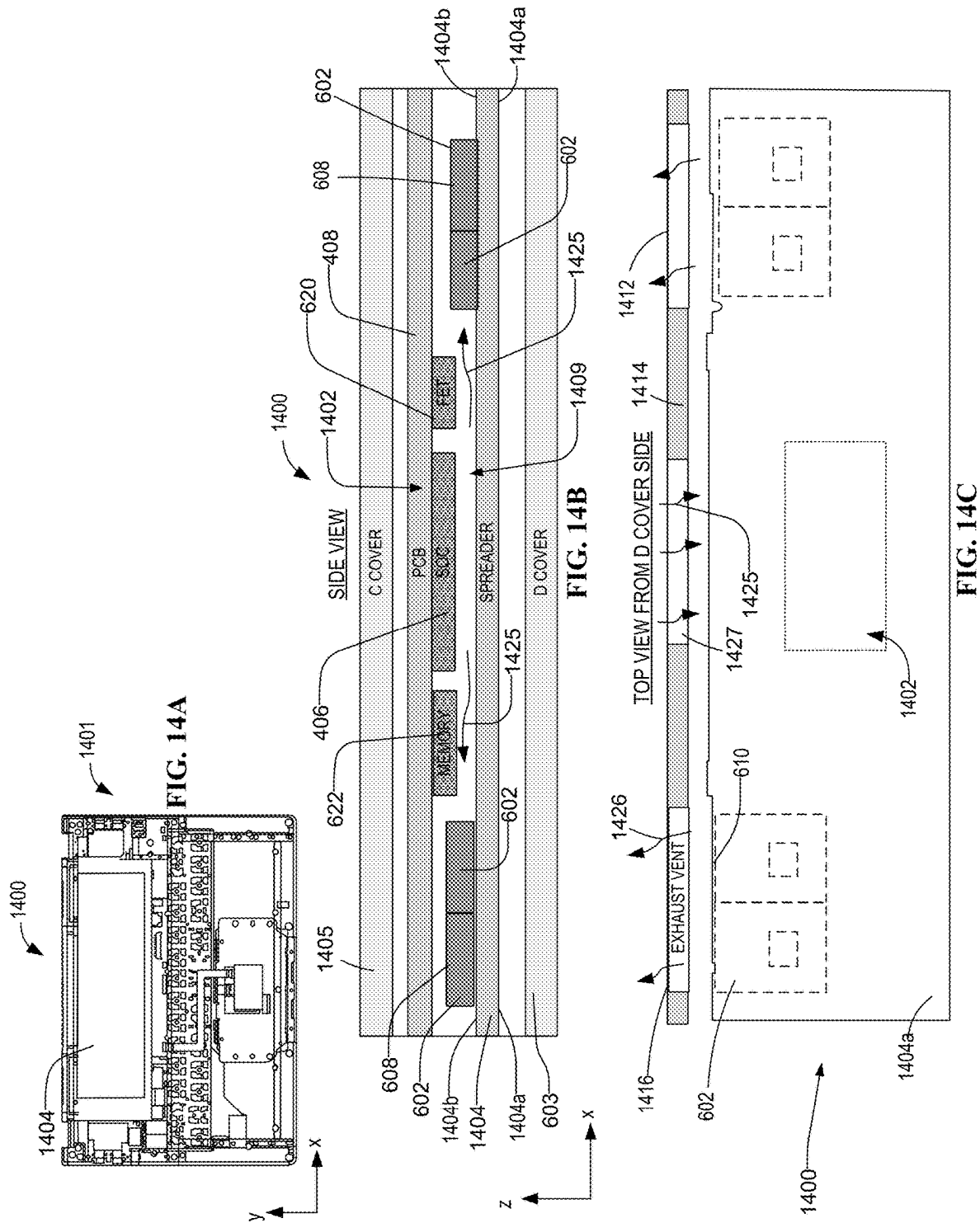


FIG. 13B





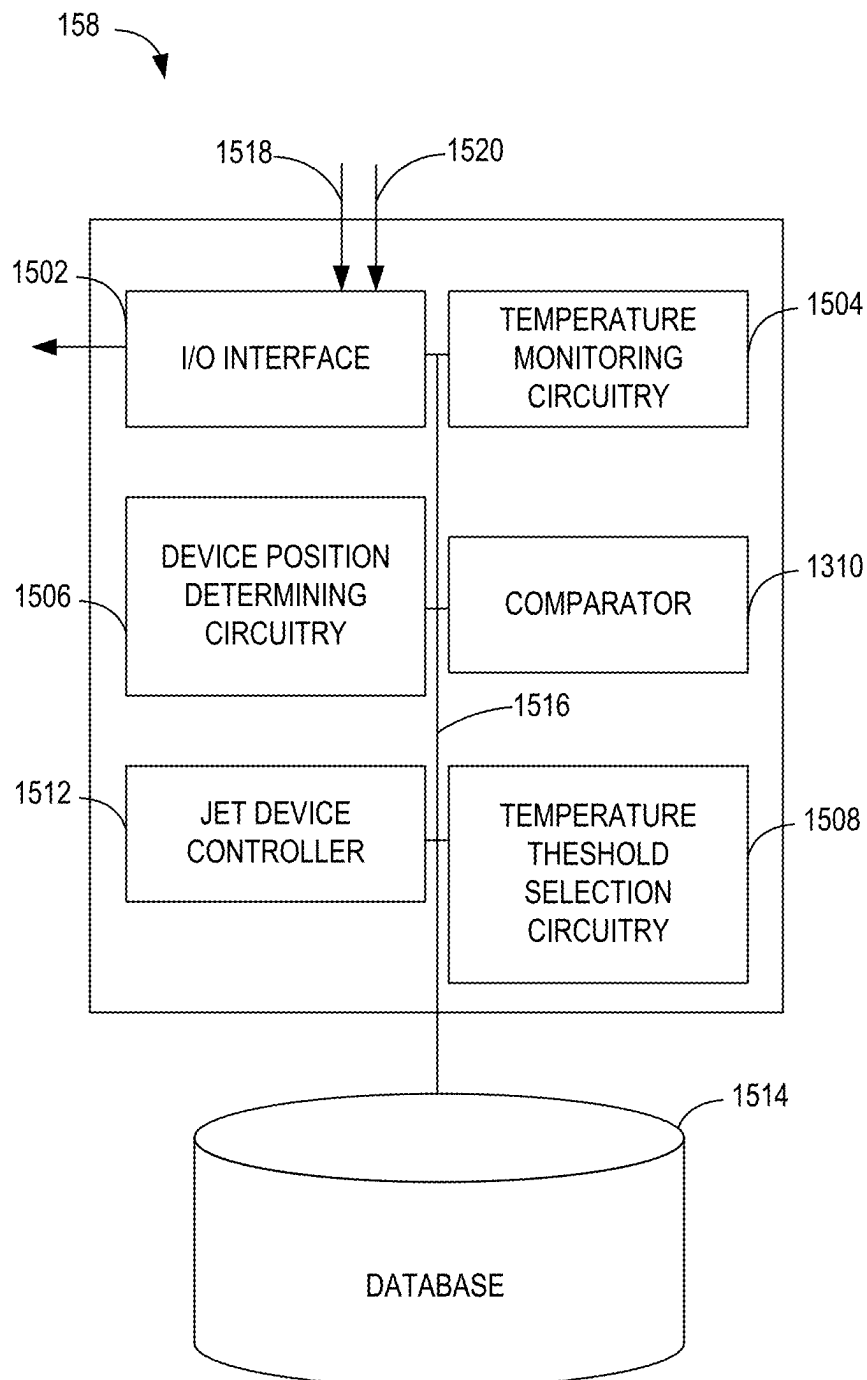


FIG. 15

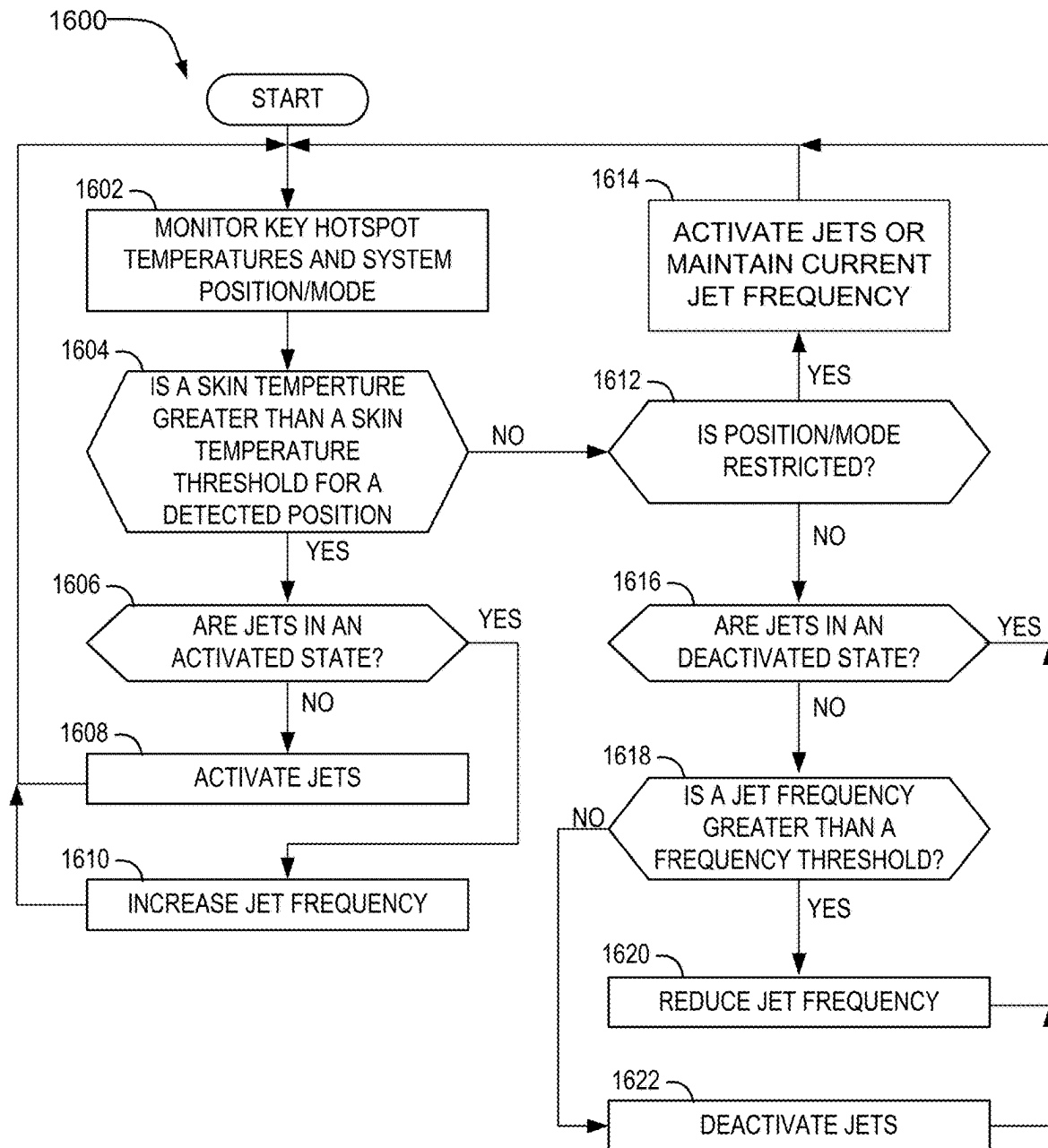


FIG. 16

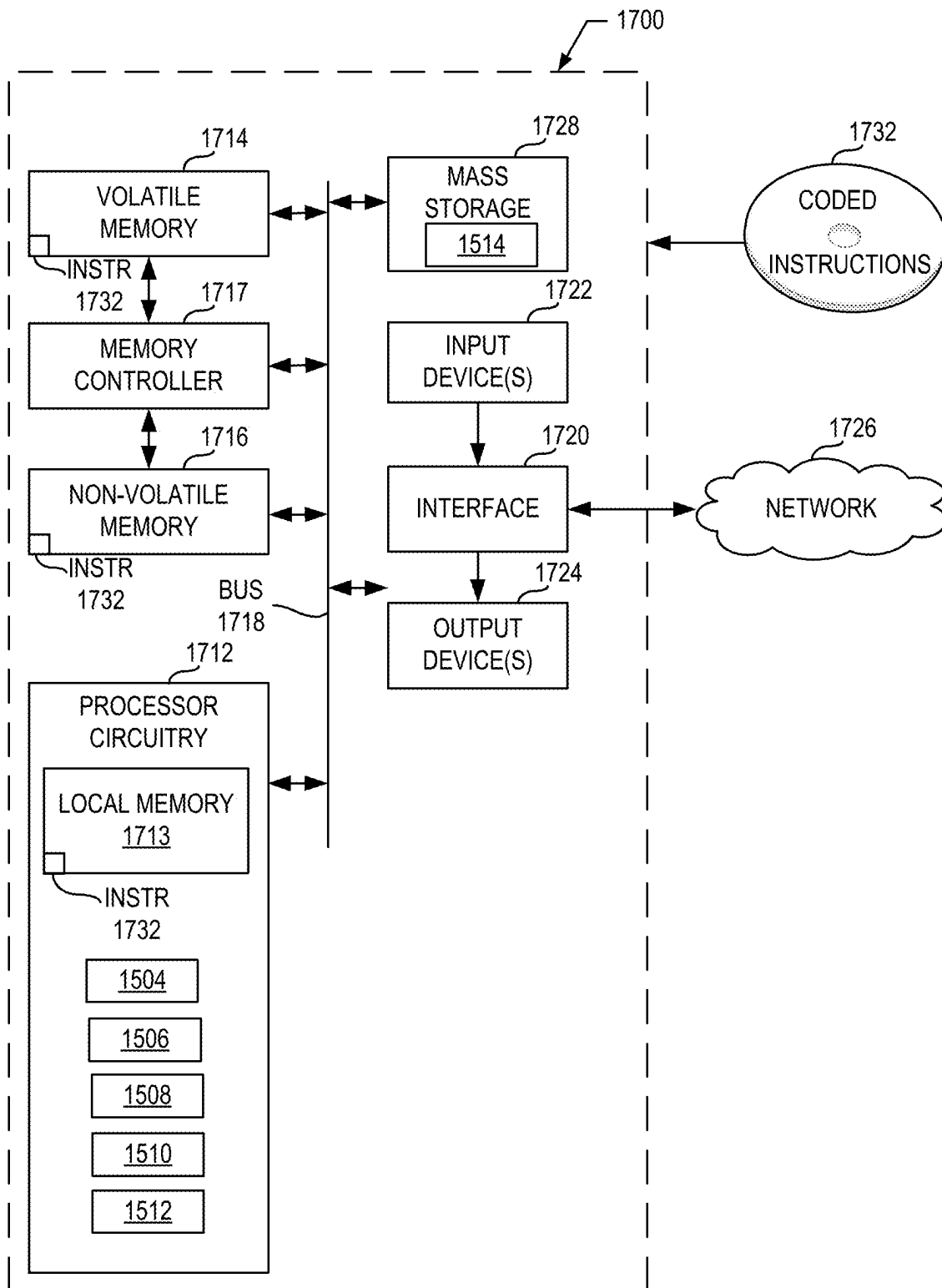


FIG. 17

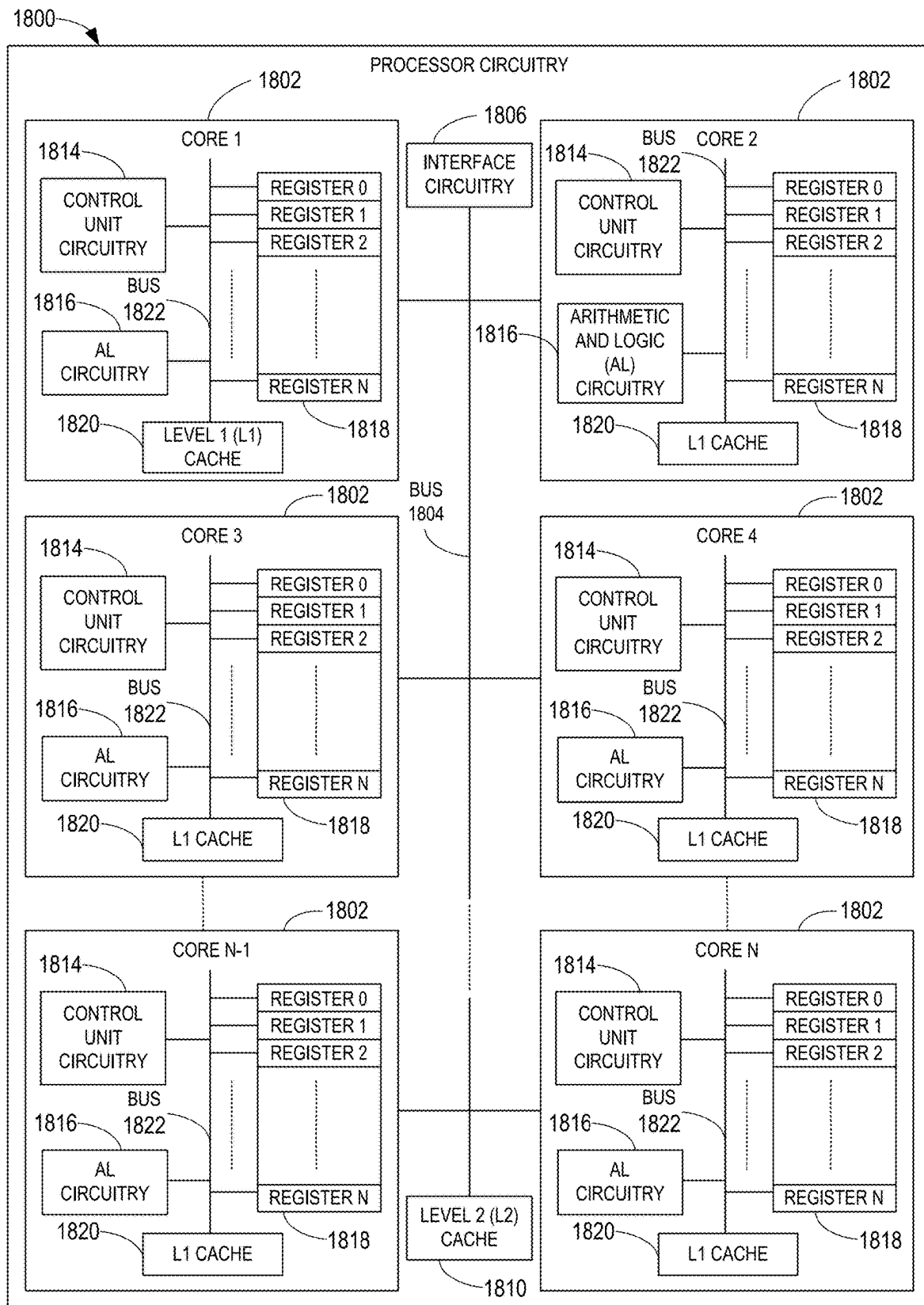
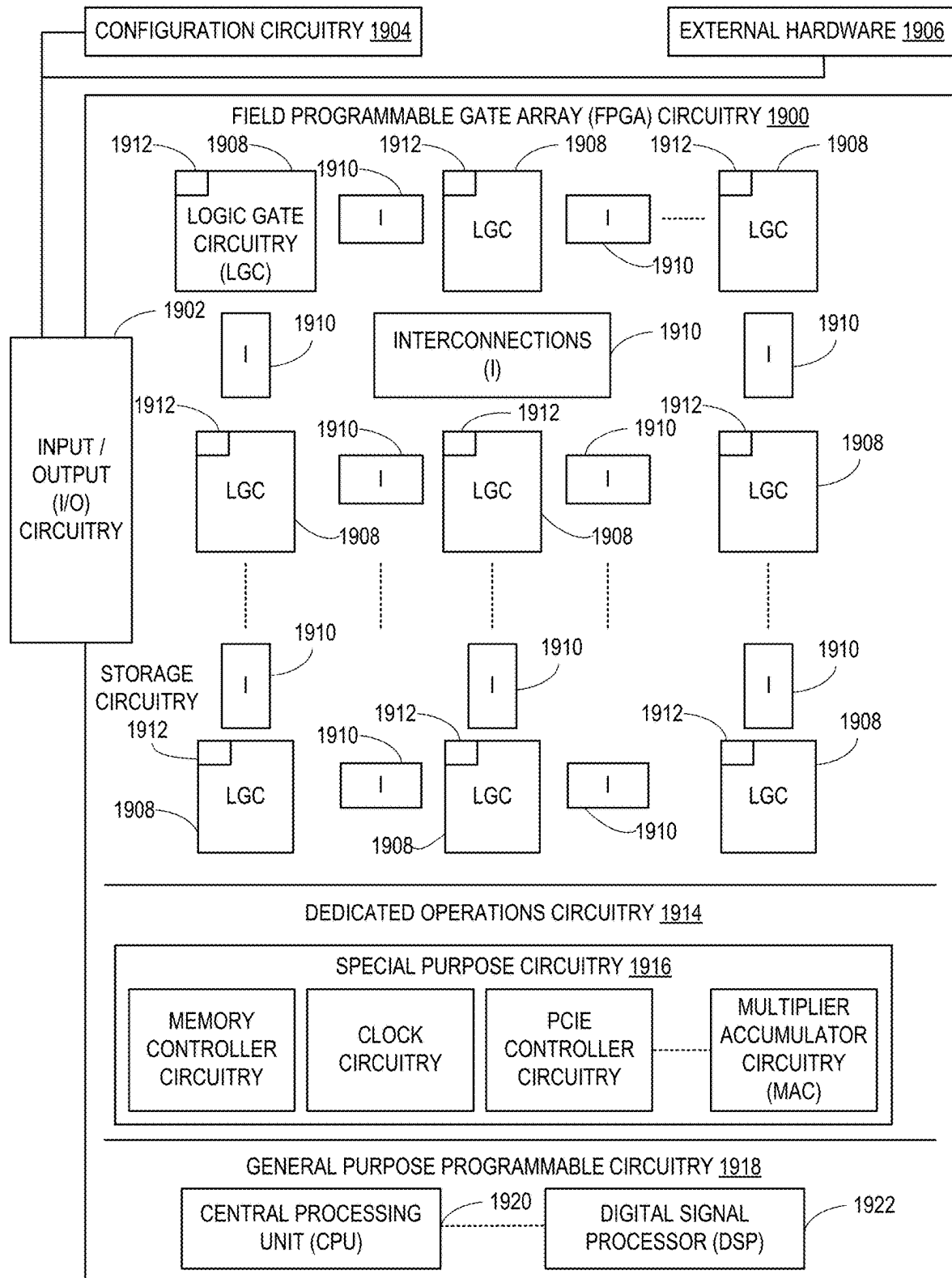


FIG. 18

**FIG. 19**

THERMAL MANAGEMENT SYSTEMS FOR ELECTRONIC DEVICES AND RELATED METHODS

CROSS-REFERENCE TO RELATED APPLICATION

This patent claims the benefit to Indian Provisional Patent Application No. 2021/41028573, which was filed on Jun. 25, 2021, and which is hereby incorporated herein by reference in its entirety. Priority to Indian Provisional Patent Application No. 2021/41028573 was filed with the Intellectual Property of India on Jun. 25, 2021, and is hereby claimed.

FIELD OF THE DISCLOSURE

This disclosure relates generally to electronic devices, and, more particularly, to thermal management systems for electronic devices and related methods.

BACKGROUND

Electronic devices employ thermal designs to manage thermal conditions. To manage thermal conditions, electronic devices employ thermal cooling systems that cool electronic components of the electronic devices during use.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A is an example electronic device constructed in accordance with teachings of this disclosure.

FIGS. 1B-1D illustrate different use case system mode of the example electronic device of FIG. 1A.

FIG. 2 is a partial, enlarged view of the example electronic device of FIG. 1A shown in an example open position.

FIG. 3 illustrates measured synthetic jet data for skin hot spot cooling of an example hot spot zone of the example electronic device of FIG. 1A.

FIGS. 4A-4F illustrate an example synthetic jet cooling system disclosed herein.

FIGS. 5A and 5B are illustrate example velocity maps of an example first exhaust flow flowing through an example first outlet of an example jet, and an example second exhaust flow flowing through an example second outlet of the example jet, respectively.

FIGS. 6A-6D illustrate other example synthetic jet system disclosed herein.

FIGS. 7A-7C illustrate other example synthetic jet system disclosed herein.

FIGS. 8A-8C illustrate other example synthetic jet system disclosed herein.

FIGS. 9A-9C illustrate other example synthetic jet system disclosed herein.

FIGS. 10A-10C illustrate other example synthetic jet system disclosed herein.

FIGS. 11A-11C illustrate other example synthetic jet system disclosed herein.

FIGS. 12A-12C illustrate other example synthetic jet system disclosed herein.

FIGS. 13A-13C illustrate other example synthetic jet system disclosed herein.

FIGS. 14A-14C illustrate other example synthetic jet systems disclosed herein.

FIG. 15 is a block diagram of an example implementation of a thermal management device of an example thermal

management system of the example electronic device of FIG. 1A constructed in accordance with teachings of this disclosure.

FIG. 16 is a flowchart representative of example machine readable instructions and/or example operations that may be executed by example processor circuitry to implement the example thermal management device of FIG. 15.

FIG. 17 is a block diagram of an example processing platform including processor circuitry structured to execute the example machine readable instructions and/or the example operations of FIG. 16 to implement the thermal management device of FIG. 15.

FIG. 18 is a block diagram of an example implementation of the processor circuitry of FIG. 17.

FIG. 19 is a block diagram of another example implementation of the processor circuitry of FIG. 17.

In general, the same reference numbers will be used throughout the drawing(s) and accompanying written description to refer to the same or like parts. The figures are not to scale. Instead, the thickness of the layers or regions may be enlarged in the drawings. Although the figures show layers and regions with clean lines and boundaries, some, or all of these lines and/or boundaries may be idealized. The boundaries and/or lines may be unobservable, blended, and/or irregular.

As used herein, unless otherwise stated, the term “above” describes the relationship of two parts relative to Earth. A first part is above a second part, if the second part has at least one part between Earth and the first part. Likewise, as used herein, a first part is “below” a second part when the first part is closer to the Earth than the second part. As noted above, a first part can be above or below a second part with one or more of: other parts therebetween, without other parts therebetween, with the first and second parts touching, or without the first and second parts being in direct contact with one another.

As used in this patent, stating that any part (e.g., a layer, film, area, region, or plate) is in any way on (e.g., positioned on, located on, disposed on, or formed on, etc.) another part, indicates that the referenced part is either in contact with the other part, or that the referenced part is above the other part with one or more intermediate part(s) located therebetween.

As used herein, connection references (e.g., attached, coupled, connected, and joined) may include intermediate members between the elements referenced by the connection reference and/or relative movement between those elements unless otherwise indicated. As such, connection references do not necessarily infer those two elements are directly connected and/or in fixed relation to each other. As used herein, stating that any part is in “contact” with another part is defined to mean that there is no intermediate part between the two parts.

Unless specifically stated otherwise, descriptors such as “first,” “second,” “third,” etc., are used herein without imputing or otherwise indicating any meaning of priority, physical order, arrangement in a list, and/or ordering in any way, but are merely used as labels and/or arbitrary names to distinguish elements for ease of understanding the disclosed examples. In some examples, the descriptor “first” may be used to refer to an element in the detailed description, while the same element may be referred to in a claim with a different descriptor such as “second” or “third.” In such instances, it should be understood that such descriptors are used merely for identifying those elements distinctly that might, for example, otherwise share a same name.

As used herein, “approximately” and “about” refer to dimensions that may not be exact due to manufacturing

tolerances and/or other real—world imperfections. For example, the dimensions may be within a tolerance of plus or minus ten percent (10%).

As used herein, “processor circuitry” is defined to include (i) one or more special purpose electrical circuits structured to perform specific operation(s) and including one or more semiconductor-based logic devices (e.g., electrical hardware implemented by one or more transistors), and/or (ii) one or more general purpose semiconductor-based electrical circuits programmed with instructions to perform specific operations and including one or more semiconductor-based logic devices (e.g., electrical hardware implemented by one or more transistors). Examples of processor circuitry include programmed microprocessors, Field Programmable Gate Arrays (FPGAs) that may instantiate instructions, Central Processor Units (CPUs), Graphics Processor Units (GPUs), Digital Signal Processors (DSPs), XPU s, or microcontrollers and integrated circuits such as Application Specific Integrated Circuits (ASICs). For example, an XPU may be implemented by a heterogeneous computing system including multiple types of processor circuitry (e.g., one or more FPGAs, one or more CPUs, one or more GPUs, one or more DSPs, etc., and/or a combination thereof) and application programming interface(s) (API(s)) that may assign computing task(s) to whichever one(s) of the multiple types of the processing circuitry is/are best suited to execute the computing task(s).

DETAILED DESCRIPTION

During operation of an electronic device (e.g., a laptop, a tablet, etc.), hardware components such as a processor, graphics card, semiconductor devices, logic circuitry, analog and/or digital circuitry and/or one or more batteries, disposed in a body or housing of the device generate heat. Heat generated by the hardware components of the electronic device can cause a temperature of one or more electronic components to exceed maximum operating temperature limits of the one or more electronic components and/or cause a temperature of a skin enclosure to approach and/or exceed a desired maximum allowable temperature. To prevent overheating of the hardware components and/or skin enclosure temperatures from exceeding a desired skin temperature threshold, electronic devices include thermal management systems to dissipate heat from the electronic devices. Example thermal management systems can include passive cooling systems or active cooling systems.

Passive cooling systems employ natural convection and heat dissipation by utilizing heat spreaders or heat sinks to increase (e.g., maximize) radiation and convection heat transfer. For instance, passive cooling systems do not employ external devices such as fans or blowers that would otherwise force airflow to exhaust heat from the housing of the electronic device. Instead, passive cooling systems rely on material characteristic(s) to provide heat transfer pathways between electronic components and outer surfaces or skins of the electronic devices. Additionally, or alternatively, passive cooling can be implemented by a throttling policy of an electronic device. For instance, to cool one or more electronic components using passive cooling throttling, power to one or more electronic components can be reduced and/or throttled to cool the hardware components and/or skin enclosure. Passive cooling systems are significantly less expensive than active cooling systems and/or do not generate noise during operation. However, passive cooling sys-

tems may not often manage heat dissipation requirements during certain operations performed by electronic devices (e.g., high speed computing).

Active cooling systems employ forced convection methods to increase a rate of fluid flow, which increases a rate of heat removal. For example, to exhaust heat generated within the body of the electronic device and cool the electronic device, active cooling systems often employ external devices such as fans or blowers, forced liquid, thermoelectric coolers, etc. For certain electronic devices (e.g., laptops, tablets, mobile devices, etc.), a fan can provide a thermal solution to safeguard system performance to a target thermal design power. However, active cooling systems employing fans often require additional space. Placement of a fan can be challenging in electronic devices due to space constraints within housings of the electronic devices. Additionally, active cooling systems such as fans can sometimes increase noise to undesired levels during use, which can lead to poor user experience.

Other thermal solutions employ ultra-thin fans (e.g., fans having a thickness of less than 3 millimeters). However, ultra-thin fans may not be effective when used in space constrained systems and, thus, can perform less than desired. For example, traditional ultra-thin blower fans response can be slow in a small zone of influence due to low airflow exit velocities, thereby reducing cooling performance. Additionally, because ultra-thin fan cooling underperforms, SoC components and non-SoC components are often throttled to avoid overheating. Thus, thermal management systems employing ultra-thin fans can lead to poor user experience.

Other example thermal management solution includes synthetic jet cooling. Unlike traditional fans, synthetic jets generate high velocity pulsating jets to create local turbulent flows that can instantly and/or rapidly cool hot spots on a skin, a glass cover and/or other component of the electronic device generated during use.

Furthermore, synthetic jet systems enable products to achieve a small form factor and provide a power density associated with active cooling, without a low system Mean Time Between Failure (MTBF) (e.g., an average time that equipment is operating between breakdowns or stoppages) of fans and/or space requirements of fans. Additionally, synthetic jet devices have relatively low acoustic output.

In operation, synthetic jets provide an unsteady, pulsating airflow or jet that sweeps over or across a heated surface. Vortices inherent to flow create fluid mixing. Synthetic jets inherently have high stream velocities that create local turbulent flow (e.g., even for very-low profile and/or smaller thickness geometries) compared to traditional fans. Specifically synthetic jets employ flexible diaphragms to generate a turbulent, pulsating airflow that can be directed at precise locations for hot-spot cooling. The oscillating diaphragms create high-velocity pulses of air. The high-velocity pulses of air remove heat conducted by the heat sink and pull entrained air from an area in its wake. This fluid mass flow carries heat away from a heat sink.

Some example synthetic jet systems include jets positioned or located directly above (e.g., on top of) a system on chip (SoC) assembly. Such an assembly typically includes a heat spreader. However, the heated area immediately below the heat spreader is typically the hottest region. As a result, air pulled into a flow path of the jets is relatively hot air, which cannot absorb additional heat and, thus, does not provide much cooling effect.

Examples disclosed herein include synthetic jet systems and related methods. Specifically, example synthetic jet cooling systems disclosed herein advantageously modify

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and/or improve thermal performance based on system ergonomic specifications and/or performance expectations that vary on usage modes and system positions. For example, a notebook computer or laptop when used in an on-table mode has a higher on-table performance expectation compared to an on-lap performance when the laptop is used in an on-lap mode. The example synthetic jet cooling systems and related methods disclosed herein account for different ergonomic touch temperature specifications and/or performance expectations.

Example synthetic jet systems disclosed herein employ one or more synthetic jets. Specifically, unlike known synthetic jet systems, example synthetic jet systems disclosed herein employ one or more jets positioned or located away from a core processor and/or system on chip (SoC). Additionally, example synthetic jet systems disclosed herein include fluid flow redistribution systems. For example, fluid flow redistribution systems disclosed herein can include one or more features (e.g., openings or slots) in a heat spreader that allows relatively high temperature generated by a processor and/or a processor region, thereby increasing cool airflow in a core zone (e.g., at the SoC region containing the processor circuitry) to improve heat transfer efficiency. In particular, redistribution features disclosed herein increase airflow through a heat exchanger region, thereby improving cooling compared to systems that employ synthetic jets without example redistribution features disclosed herein. For instance, by redistributing the flow, example redistribution features disclosed herein can increase synthetic jet cooling capability by at least 40% compared to one or more prior art systems.

Additionally, some examples disclosed herein employ synthetic jet configurations to improve heat transfer and/or heat removal from processor circuitry, SoC components, and/or non-system on chip components such as, for example, off chip memory. Specifically, synthetic jets disclosed herein are mounted strategically relative to a core processor to improve cooling (e.g., hot spot cooling). In some examples, synthetic jet systems and related methods disclosed herein employ one or more synthetic jet(s) for impingement cooling to almost instantaneously mitigate a skin hot spot based on a detected physical position of the electronic device. Example synthetic jet cooling systems and related methods disclosed herein operate in an active cooling mode based on a detected position (e.g., on-desk vs on-lap, clamshell position, tent, etc.) of the electronic device and skin hotspot temperature(s) detected by (e.g., one or more sensors of) the synthetic jet cooling system. Based on a detected position of the electronic device, example cooling systems and related methods disclosed herein can be operate under an active cooling scheme (e.g., synthetic jet "OFF"), allowing a higher power capability for higher on-table skin temperature specifications or, alternatively, operate under an active cooling scheme or capability (synthetic jet "ON") to instantly cool the system to meet lower skin temperature (e.g., on-lap) specification when a control system (e.g., via a sensor) detects a change in usage position or mode (e.g., a clamshell mode, a laptop mode, etc.).

In some instances, example synthetic jet systems and related methods disclosed herein can provide up to 5° C. (Celsius) difference for on-table configurations versus on-lap configurations, which can provide up to 5-6 Watts (W) additional power (e.g., which is approximately between 40-60% greater system thermal capability compared to a typical passive design or known synthetic jet systems). In some examples, on-lap mode performance can further be enhanced by gradually updating a first power limit and/or a

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second power limit (PL1/PL2) settings (from higher to required lower) while adjusting a jet frequency of one or more synthetic jets. As a result, example synthetic jet cooling systems and related methods disclosed herein can provide full power wireless wide area network (WWAN) (e.g., up to 5 W) and solid-state storage device (SSD) (e.g., 5-8 W) performance without compromising processor performance by using targeted synthetic jet skin cooling. In some examples, dynamic control using frameworks like Intel® Dynamic Tuning Technology (DDT) can be employed to automatically and/or dynamically allocate power between an Intel® processor and an Intel® Discrete Graphics Card to improve (e.g., optimize) performance and/or improve battery life.

Additionally, example cooling systems and related methods disclosed herein provide a thermal control scheme. Specifically, the thermal control scheme can implement a quasi-active cooling system that can meet high performance requirements without user experience trade-off during position and/or mode change. For example, one or more synthetic jets (e.g., small size jets having between about 20-30 millimeters (mm) in length and less than about 1 mm in thickness) can be mounted in an electronic device to target skin hotspots in a processor (e.g., a core) region and/or other key component like WWAN and SSD. The example one or more synthetic jets can be triggered on or off (trigger ON/OFF) at a desired frequency based on: (1) a hotspot temperature feedback control; and (2) a device position/mode feedback control. In some examples, Intel's Dynamic Tuning Technology framework can be used for a feedback-based control. Virtual sensors (e.g., correlated with thermistors of a motherboard) can function as proxy for the skin hotspots. Example cooling systems disclosed herein can provide high performance processing when the electronic device is configured for on-table use by utilizing higher temperature threshold specifications. Example synthetic jets disclosed herein can be turned ON intermittently or selectively to give a performance boost for select activities. Based on sensor feedback, as a position/mode change indicative of a change from on-table to on-lap use (which have lower thermal threshold specification than the on-table thermal threshold specification), example synthetic jets disclosed herein can be turned ON for instant temperature reduction to meet lower thermal specifications for on-lap use. Processor frequencies can also be gradually lowered to required/lower performance levels. Such gradual lowering of the power limits can enhance on-lap performance and/or user experience.

FIG. 1A is an example electronic device **100** constructed in accordance with teachings of this disclosure. The electronic device **100** of the illustrated example is a mobile device (e.g., a rugged laptop, a laptop, etc.). The electronic device **100** of the illustrated example includes a first housing **102** coupled to a second housing **104** via a hinge **106** (e.g., a 360 degree hinge). To enable user inputs, the first housing **102** of the illustrated example includes a keyboard **108** and a track pad **110**. For example, the keyboard **108** and the track pad **110** are exposed at an upper surface **112** of the second housing **104** (e.g., opposite a bottom surface **120** of FIGS. 1B-1D). The second housing **104** carries a display **114**, a camera **116** and a microphone **118**. The first housing **102** houses electronic components of the electronic device **100**. The hinge **106** enables the second housing **104** to rotate or fold relative to first housing **102** between a stored position (e.g., where the second housing **104** is aligned or parallel with the first housing **102**) and one or more open positions. In the example of FIG. 1A, the electronic device **100** is

shown in an example laptop mode **122**. In the laptop mode **122**, the second housing **104** is rotated relative to the first housing **102** about the hinge **106** to a desired viewing angle (e.g., a viewing angle of between 90 degrees and 115 degrees relative to the upper surface **112** of the first housing **102**) with the keyboard **108** exposed in front of the display **114**. For example, during use in the laptop mode **122**, the electronic device **100** in a laptop mode can be positioned on a user's body (e.g., a user's lap).

FIGS. 1B-1D illustrate different use positions (e.g., open positions) of the electronic device **100**. In some examples, electronic device **100** enables, via the hinge **106**, the second housing **104** to move to other open positions including, but not limited to, a tablet mode **124** as shown in FIG. 1B (e.g., with the keyboard **108** and the display **114** on opposite sides of the device and the display **114** exposed to the user), a kiosk mode **126** as shown in FIG. 1C, and a tent mode **128** as shown in FIG. 1D. In the tablet mode **124**, the second housing **104** is rotated approximately 360 degrees from the closed position such that the second housing **104** is positioned on the first housing **102** with the display **114** exposed for user use (e.g., an outer surface **130** of the second housing **104** engages the bottom surface **120** of the first housing **102**). In the kiosk mode **126**, the second housing **104** is rotated to a viewing angle of greater than 180 degrees relative to the upper surface **112** of the first housing **102**, and the keyboard **108** can engage and/or be adjacent a support surface (e.g., a table or desk) or a user's lap. In the tent mode **128**, the second housing **104** is rotated relative to the first housing **102** to an angle that is greater than 270 degrees from the closed position such that respective edges of the first housing **102** and the second housing **104** can engage a support surface (e.g., a table or desk).

Referring to FIG. 1A, to remove heat from electronic components of the electronic device **100**, the electronic device **100** of the illustrated example includes a thermal management system **150**. The thermal management system **150** of the illustrated example includes one or more temperature sensor(s) **152** (e.g., a plurality of temperature sensors), one or more position sensor(s) **154**, a synthetic jet system **156**, and a thermal management device **158**. The thermal management system **150** of the illustrated example includes temperature sensor(s) **152** (e.g., thermocouples) to measure temperature(s) associated with the hardware components of the electronic device **100**, skin enclosure, and/or other component(s). The position sensor(s) **154** are employed to detect an open position and/or use mode configuration (e.g., the laptop mode **122**, the tablet mode **124**, the kiosk mode **126**, and the tent mode **128** of the electronic device **100**). The thermal management system **150** controls one or more of the synthetic jet system **156** based on temperatures provided by the temperature sensor(s) **152** and an open position or use mode orientation of the electronic device **100** as detected by the position sensor(s) **154**. In some examples, the thermal management system **150** (e.g., synthetic jet system **156**) of the illustrated example can be located in the first housing **102**, the second housing **104** and/or any other location of the electronic device **100**.

Although the electronic device **100** of the illustrated example is a laptop, in some examples, the electronic device **100** can be a tablet, a desktop, a mobile device, a cell phone, a smart phone, a hybrid or convertible PC, a personal computing (PC) device, a server, a modular computing device, a digital picture frame, a graphic calculator, a smart watch, and/or any other electronic device that employs active cooling. For example, employing a mobile device (e.g., a tablet, etc.) can include a device orientation sensor to

detect an orientation of the mobile device including, for example, upright orientation or stand position, a lap orientation, a hand-held orientation, etc.

FIG. 2 is a partial, enlarged view of the electronic device **100** shown in an open position (e.g., the laptop mode **122**). Specifically, FIG. 2 illustrates an example heat distribution map **200** for heat generated in the first housing **102** during operation. The example heat distribution map **200** is overlaid on the first housing **102** of the electronic device **100** of FIG. 2. The heat distribution map **200** shows a hot spot zone **202** (e.g., an elevated heated zone), a medium heat zone **204** (e.g., a non-hot spot zone), and a cool zone **206** (e.g., a non-hot spot zone). In the illustrated example, the thermal management system **150** reduces a temperature of the hot spot zone **202** to a desired or specification temperature threshold. For example, the synthetic jet system **156** can be positioned adjacent an upper right corner **208** of the first housing **102** that can provide instant skin hot spot cooling to enhance system performance and/or user experience. In particular, to reduce the temperature of the hot spot zone **202**, the synthetic jet system **156** of the illustrated example employs turbulent jets **210** (e.g., synthetic jets) directed towards the hot spot zone **202**. The hot spot zone **202** of the illustrated example is 80 millimeters by 100 millimeters. However, in other examples, the hot spot zone **202** can be any sized area.

FIG. 3 illustrates infrared images of measured synthetic jet data **300** for skin hot-spot cooling of the hot spot zone **202** of FIG. 2. A first infrared image **302** illustrates a heat map of the hot spot zone **202** (e.g., an elevated temperature hot zone) when the synthetic jet system **156** is in a passive condition (e.g., a "turned off" condition or state) and a second infrared image **304** juxtaposed relative to the first infrared image **302** illustrates a heat map (e.g., a cooled zone) of the hot spot zone **202** when the synthetic jet system **156** is in an active condition (e.g., a "turned on" condition or state). Thus, in the illustrated example, the synthetic jet system **156** cooled the hot spot zone **202** from a temperature of approximately 52.2 degrees Celsius to a temperature of approximately 39.5 degrees Celsius. For example, in the illustrated example, the synthetic jet **156** has a dimension of 40 millimeters by 1 millimeter. The synthetic jet system **156** of the illustrated example can cool the hot spot zone **202** (e.g., a skin zone) by approximately 10° C. within 3-5 seconds (e.g., for an area of approximately 100 millimeters by 80 millimeters). The example of FIG. 3 can provide significantly more cooling by increasing a size of the synthetic jet system **156**.

FIGS. 4A-4F illustrate an example synthetic jet system **400** constructed in accordance with teachings disclosed herein that can implement the example synthetic jet system **156** and/or the thermal management system **150** of FIGS. 1-3. FIG. 4A is a bottom view of an example electronic device **401** that includes the example synthetic jet system **400**. FIG. 4B is a top view of the example synthetic jet system **400** constructed in accordance with teachings disclosed herein. FIG. 4C is a front view of the example synthetic jet system **400** of FIG. 4B. FIG. 4D is a top view of the example synthetic jet system **400** of FIG. 4B but shown without a heat spreader **404**, a processor **406** and a printed circuit board **408**. FIG. 4E is a cross-sectional view taken along line 4D-4D of FIG. 4B. FIG. 4F is a partial, cross-sectional view of the example synthetic jet system **400** of FIG. 4E.

Referring to FIG. 4A, the electronic device **401** is a housing **403** (e.g., a top view of the first housing **102** of FIG. 1) shown without a cover to illustrate the positioning of the

synthetic jet system **400** relative to a cavity **405** of the housing **403**. The synthetic jet system **400** of the illustrated example is positioned or located adjacent an edge **407** (e.g., a back wall) of the housing **403** that includes a hinge (e.g., the hinge **106** of FIG. 1) and/or a detachable mechanism (e.g., a tablet). In some examples, the housing **403** can be a housing (e.g., the second housing **104** of FIG. 1A) to house a display (e.g., a tablet, a touchscreen display or any other electronic device).

Referring to FIGS. 4B and 4C, the synthetic jet system **400** of the illustrated example includes a plurality of synthetic jets **402** (e.g., four jets), the heat spreader **404**, a processor **406** (system on chip (SoC) or other semiconductor package), a printed circuit board **408**, and heat exchangers **410**. The heat spreader **404** is positioned adjacent (e.g., above) the processor **406**. When the synthetic jet system **400** is assembled, the heat spreader **404** is positioned between the heat exchangers **410** and the synthetic jets **402**, and the heat exchangers **410** are positioned between the heat spreader **404** and the circuit board **408**. Specifically, the synthetic jets **402** are positioned adjacent (e.g., directly on) a first side or surface **404a** (e.g., an upper surface) of the heat spreader **404** and the heat exchangers **410** are positioned adjacent (e.g., directly on) a second side or surface **404b** (e.g., a lower surface) of the heat spreader **404** opposite the first surface **404a**. In the illustrated example, a first heat exchanger **410a** is associated with a first set **402a** (e.g., pair) of the synthetic jets **402** and a second heat exchanger **410b** is associated with a second set **402b** (e.g., a pair) of the synthetic jets **402**. In some examples, each jet includes a dedicated heat exchanger. In some examples, any number of heat exchangers and/or jets can be employed. In some examples, the heat exchangers **410** are not included.

Additionally, the synthetic jets **402** of the illustrated example are positioned on or above (e.g., directly in contact with an upper surface of) the heat spreader **404**. The heat spreader **404** of the illustrated example is a plate capable of spreading heat generated by the processor **406** and/or other electronic components of the circuit board **408**. In some examples, the heat spreader **404** can be a vapor chamber, a heat exchanger, a plate, a heat plate and/or any other heat spreader. In some examples, the heat spreader **404** is a planar heat pipe, which can spread heat in two dimensions (e.g., across a surface area of the vapor chamber). The heat spreader **404** of the illustrated example can be composed of brass, copper and/or any other suitable material(s) for transferring and/or spreading heat. The synthetic jets **402** of the illustrated example are positioned adjacent or laterally relative to (e.g., away from) the processor **406**. In other words, the synthetic jets **402** are not positioned directly above the processor **406** in a stack-up direction or z-direction (e.g., FIG. 4C). In the illustrated example, the synthetic jets **402** are positioned adjacent lateral sides or edges **406a** of the processor **406**. For instance, the synthetic jets **402** are positioned at a location away from the processor **406** in the y-direction and/or the x-direction (e.g., not directly above the processor **406** in the z-direction).

Referring to FIGS. 4D-4E, the synthetic jets **402** of the illustrated example each have a body **412** defining a fluid passageway **414** between an inlet **416** and a first outlet **418** and a second outlet **420**. A diaphragm or actuator **422** is positioned in the fluid passageway **414** of the body **412**. The actuator **422** is actuated (e.g., via an electrical signal or command from the thermal management system **150**) to induce fluid flow from the inlet **416** to the first outlet **418** and the second outlet **420**.

Referring to FIGS. 4B-4E, the body **412** of the illustrated example has a rectangular shape that includes a first wall **423** (e.g., an upper or top wall), side walls **412a** (e.g., four side walls), and a third wall **427** (e.g., a lower or bottom wall **412c**) that define a cavity **412b** to provide a portion of the fluid passageway **414**. The first wall **423** of the body **412** includes a first aperture **423a** to define the inlet **416**, a second wall **425** (e.g., a front wall or one of the side walls **412a**) includes a second aperture **425a** to define the first outlet **418**, and a third wall **427** defining the first opening **433** to provide the bypass **430** and/or the second outlet **420**. The second wall **425** of the illustrated example is perpendicular relative to the first wall **423** and the third wall **427**, and the first wall **423** is substantially parallel relative to the third wall **427**. The third wall **427** is oriented toward the first surface **404a** of the heat spreader **404** and the second wall **425** is oriented in a direction away from the first surface **404a** of the heat spreader **404**.

The fluid passageway **414** of respective ones of the synthetic jets **402** of the illustrated example bifurcates or splits the airflow **424** in the fluid passageway **414**. Specifically, the synthetic jets **402** and the heat spreader **404** define the fluid passageway **414** between the inlet **416** and the second outlet **420**. For instance, an airflow **424** in the fluid passageway **414** that is received by the inlet **416** is separated or split into a first exhaust **426a** and a second exhaust **426b**. Specifically, the first exhaust **426a** flows across the first surface **404a** of the heat spreader **404** and the second exhaust **426b** flows across the second surface **404b** of the heat spreader **404**. In particular, the second exhaust **426b** flows through the heat exchanger **410** associated with or positioned adjacent the respective pair of the synthetic jets **402**. In other words, the airflow **424** flows both above the heat spreader **404** (e.g., the first surface **404a**) and below the heat spreader **404** (e.g., the second surface **404b**).

Referring to FIG. 4D-4F, to split, divide or bifurcate the airflow **424** through the fluid passageway **414**, the synthetic jet system **400** of the illustrated example employs a flow redistributor or bypass **430**. The bypass **430** of the illustrated example communicatively and/or fluidly couples the first surface **404a** of the heat spreader and the second surface **404b** of the heat spreader **404**. Thus, the bypass **430** bifurcates airflow from the inlet **416**. To provide the bypass **430**, the heat spreader **404** and the body **412** of a respective one of the synthetic jets **402** bifurcate airflow from the inlet **416** to the first outlet **418** and the second outlet **420**. In particular, to bifurcate the airflow **424** into the first exhaust **426a** and the second exhaust **426b**, the body **412** includes a first opening **433** and the heat spreader **404** includes a second opening **436** (e.g., a second pass through aperture, hole, channel, slot, etc.) in communication with the fluid passageway **414** via the first opening **433** (e.g., a slot) formed in the body **412** of a respective one of the synthetic jets **402**. The second opening **436** of the illustrated example is the opening of a hole **436a** extending between the first surface **404a** and the second surface **404b**. Specifically, the first opening **433**, the hole **436a**, and the second opening **436** communicatively couple the first surface **404a** of the heat spreader **404** and the second surface **404b** of the heat spreader **404**.

The bypass **430** of the illustrated example includes a gasket **432** (e.g., a heat exchanger bypass gasket duct) positioned (e.g., sandwiched) between the body **412** and the heat spreader **404**. The gasket **432** of the illustrated example includes a slot **434** (e.g., a first pass through aperture, hole, channel, opening, etc.) to fluidly couple the first opening **433** and the second opening **436**. The gasket **432** has a body or solid layer **432a** that blocks part of first exhaust **426a** flow

from channeling through the bypass **430** and allows the first exhaust **426a** to exit the first outlet **418**. The gasket **432** and the heat spreader **404**, via the bypass **430** formed by the first opening **433** and the second opening **436** allow the second exhaust **426b** (e.g., a portion of the airflow **424**) to flow through to the heat spreader **404** and to the heat exchanger **410**. The second exhaust **426b** exits the synthetic jet system **400** via the heat exchangers **410**, enhancing overall cooling effectiveness of the synthetic jet system **400**. In other words, the flow distribution provided by the bypass **430** improves overall thermal solution effectiveness. The bypass **430** (e.g., the opening or features provided by the heat spreader **404** and the gasket **432** allows airflow (e.g., the second exhaust **426b**) for the heat exchanger **410** and allows airflow (e.g., the first exhaust **426a**) to be exhaust through the first opening **433** and/or a vent of the housing **403** (FIG. 4A). In the illustrated example, a thermal conductive material or layer **438** (e.g., graphite) is positioned between the gasket **432** and the heat spreader **404** and/or on the second surface **404b** of the heat spreader **404**. The thermal conductive layer **438** dissipates and/or spreads heat from processor **406** to the heat spreader **404**. Each of the layers **328** includes an openings **439** that aligns with the first opening **433** and the second opening **436** (e.g., and the slot **434**) to allow the second exhaust **426b** to flow between the first surface **404a** and the second surface **404b** of the heat spreader **404**. A thermal interface material (TIM) **435** (e.g., a thermal pad, silicon material, thermal gap filler, etc.) is positioned between the synthetic jets **402** and the heat spreader **404**. In some examples, the layers **438** are not included. In some examples, one of the layers **438** is provided only on the first surface **404a** or the second surface **404b** of the heat spreader **404**. In the illustrated example, the synthetic jet system **400** includes a duct **441** to channel the second exhaust **426b** to the heat exchanger **410**. However, in some examples, the duct **441** is not provided.

Referring to FIGS. 4A-4F, in operation, the synthetic jets **402**, via actuation of the actuator **422** in the body **412**, generate a suction at the inlet **416** to draw airflow **424** into the inlet **416** and the fluid passageway **414**. The actuator **422** creates a pressure differential (e.g., a suction or vacuum at the inlet **416**) in the fluid passageway **414** to cause the airflow **424** to flow toward the first outlet **418** and the second outlet **420** via the fluid passageway **414** and the bypass **430**. Additionally, the airflow **424** at the inlet **416** is air received from above the first surface **404a** of the heat spreader **404** (e.g., heat generated from the processor **406** that is spread via the heat spreader **404**). Thus, the position and/or location of the synthetic jets **402** adjacent the processor **406** enables heated air directly above the processor **406** to be pulled away from the processor **406** and toward the inlets **416** of the synthetic jets **402**. The airflow **424**, which includes heated air generated from the processor **406** and dissipated by the heat spreader **404**, flows through the synthetic jets **402** and bifurcates into the first exhaust **426a** and the second exhaust **426b**, thereby removing the heat from the processor **406** and/or a processor zone **411** (e.g., a system on chip (SoC) area) and/or cooling a hot spot region (e.g., the hot spot zone **202** of FIG. 2). As noted, the example synthetic jet system **400** of the illustrated example bifurcates or otherwise separates the airflow **424** from the inlet **416** into at least two separate airflow streams (e.g., the first exhaust **426a** and the second exhaust **426b**). In other words, the first exhaust **426a** exits the first outlet **418** of the synthetic jets **402**, and a second exhaust **426b** is directed underneath the heat spreader **404** and toward the heat exchanger **410**. By redistributing the flow, a greater amount of airflow or air mass can

be moved from an area adjacent the processor **406** processor, which increases significantly synthetic jet cooling capability or characteristics. In some examples, the synthetic jets **402** increase cooling capability by as much as 40% compared to jets that do not have bifurcated flow paths.

In some instances, the first exhaust **426a** can exit the first outlet **418** and vents from the housing **403** of the electronic device **401**. In this example, the second exhaust **426b** flows through the heat exchanger **410**, which reduces or cools (e.g., remove heat from) the second exhaust **426b**. The second exhaust **426b** exiting the heat exchanger **410** can be directed or redirected to flow is directed toward the hot spot region, thereby cooling a hot spot region (e.g., the hot spot zone **202** of FIG. 2), and/or the processor to further cool the processor.

Further, in some examples, a first portion of the heat spreader **404** on which the synthetic jets **402** are positioned can have an offset elevation (e.g., in the z-direction of FIG. 1A) relative to a second portion of the heat spreader **404** that is above the processor **406**. For instance, a first upper surface of the first portion of the heat spreader **404** can be at a lower elevation relative to a second upper surface of the second portion of the heat spreader. In this manner, a greater gap is provided between a cover (e.g., a cover or skin of the first housing **102** of FIG. 1A) and the inlets of the synthetic jets **402**, enabling greater airflow towards inlets. In other words, if the synthetic jets **402** are positioned on the second portion directly above the processor, a gap (e.g., in the vertical or z-direction in the orientation of FIG. 1A) formed between the inlet and a cover of a housing (e.g., the first housing **102** of FIG. 1A) of an electronic device may be small and/or otherwise impeded airflow to the inlets of the synthetic jets **402**. In some examples, the inlet **416** can be provided on a side surface of the body **412** (on a surface perpendicular to a surface on which the inlet **416** is formed in FIGS. 4B and 4C).

FIG. 5A is an infrared velocity map **502** of the first exhaust **426a** flowing through a synthetic jet **501** (i.e., a respective one) of the synthetic jets **402** and to the first outlet **418**. FIG. 5B is an infrared velocity map **504** of the second exhaust **426b** flowing through the synthetic jet **501** and to the second outlet **420**. Referring to FIG. 5A, the velocity of the second exhaust **426b** flowing through the bypass **430** at to the heat exchanger **410** is approximately 10 meters/second (m/s). Similarly, the first exhaust **426a** flows through the first outlet **418** is approximately 10 m/s. The example synthetic jet system **400** increased cooling capability by 40 percent by redistributing the airflow **424** into the first exhaust **426a** and the second exhaust **426b**. In some examples, the first exhaust **426a** can have a greater mass flow rate and/or velocity than a mass flow rate or velocity of the second exhaust **426b**. In some examples, the second exhaust **426b** can have a greater mass flow rate and/or velocity than a mass flow rate or velocity of the first exhaust **426a**. In some examples, the first exhaust **426a** can have a mass flow rate and/or velocity that is substantially similar or identical to a mass flow rate or velocity of the second exhaust **426b**.

In some examples, the synthetic jets **402** can have different orientations. For example, the first outlet **418** and the second outlet **420** can be positioned in directions perpendicular to each other. In some examples, the first outlet **418** can be oriented substantially perpendicular or non-parallel relative to the second outlet **420**. Thus, the first exhaust **426a** can be oriented in a first direction, and the second exhaust **426b** can be oriented in a second direction different than (e.g., perpendicular to) the first direction. In some examples, the synthetic jets **402** can have any other orientation such

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that the exhausts **426a**, **426b** blow outwards, blowing inwards toward the processor **406**, a combination of inwards and outwards, in a horizontal direction relative to an upper surface of the circuit board **408**, in vertical direction relative to the circuit board **408**, and/or any other direction(s). In some examples, the synthetic jets **402** can be oriented and/or configured for different types of cooling: impingement flow; as flow mover; combination of impingement and flow mover, etc. This can be useful under various inlet(s) location (s) (e.g., D-cover, C-cover, side walls) and heat exchanger location(s) (e.g., rear, side etc.). In some examples, the heat exchangers **410** can be omitted and/or replaced with different spreaders (e.g., a heat pipe, a vapor chamber, a copper plate, a copper and graphite plate, high thermal conductivity material(s), etc.). In some examples, the synthetic jets **402** can be stacked on top of each other or can be single stack as shown in FIG. **4A**. In some examples, the synthetic jet system **400** of the illustrated example can be located in the first housing **102** of FIG. **1**, the second housing **104** of FIG. **1** and/or any other location(s).

FIGS. **6A-14C** illustrate other example electronic devices **601**, **701**, **801**, **901**, **1001**, **1101**, **1201**, **1301**, **1401** that include other example synthetic jet systems **600**, **700**, **800**, **900**, **1000**, **1100**, **1200**, **1300**, **1400** disclosed herein. The example synthetic jet systems **600**, **700**, **800**, **900**, **1000**, **1100**, **1200**, **1300**, **1400** can implement the electronic device **100** of FIG. **1**, a tablet, a laptop, a desktop, a mobile device, and/or any other electronic device that employs thermal management systems. Many of the components of the example synthetic jet systems **600**, **700**, **800**, **900**, **1000**, **1100**, **1200**, **1300**, **1400** are substantially similar or identical to the components described above in connection with FIGS. **1A-1D**, **2**, **3**, **4A-4F**, **5A** and **5B**. As such, those components will not be described in detail again below. Instead, the interested reader is referred to the above corresponding descriptions for a complete written description of the structure and operation of such components. To facilitate this process, similar or identical reference numbers will be used for like structures in FIGS. **6A-12C** as used in FIGS. **1A-1D**, **2**, **3**, **4A-4F**, **5A** and **5B**. For example, the electronic devices **601**, **701**, **801**, **901**, **1001**, **1101**, **1201**, **1301**, **1401** include a processor **406** and a circuit board **408**. The examples of FIGS. **6A-14C** illustrate different configurations and/or orientations of the synthetic jet systems **600**, **700**, **800**, **900**, **1000**, **1100**, **1200**, **1300**, **1400** relative to the processor **406** and the circuit board **408**.

FIG. **6A** is a cross-sectional view of another example synthetic jet **602** disclosed herein. The synthetic jet **602** of the illustrated example includes a body **604** defining a fluid passageway **606** between an inlet **608** and an outlet **610**. Unlike the synthetic jets **402** of FIGS. **4A-4F**, the synthetic jet **602** of the illustrated example does not include a bypass **430** as shown in FIGS. **4A-4F**. In other words, the synthetic jet **602** of FIG. **6A** is a unitary flow path (e.g., a single outlet similar to the first outlet **418** of FIGS. **4A-4E**). A bottom surface **611** of the body **604** mounts or couples to a heat spreader and/or other structure of an electronic device.

Each of the example synthetic jet systems **600**, **700**, **800**, **900**, **1000**, **1100**, **1200**, **1300**, **1400** of FIGS. **6B-14C** include a plurality of synthetic jets **602**. However, in some examples, the synthetic jet systems **600**, **700**, **800**, **900**, **1000**, **1100**, **1200**, **1300**, **1400** of FIGS. **6B-14C** can be configured with a plurality of synthetic jets **402** of FIGS. **4A-4E**. For example, the plurality of synthetic jets **402** can replace the plurality of synthetic jets **602** as shown in FIGS. **6B-14C**. In some examples, the synthetic jet systems **400**, **600**, **700**, **800**, **900**, **1000**, **1100**, **1200**, **1300**, **1400** disclosed herein can be

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configured with both the synthetic jets **402** and the synthetic jets **602**. Additionally, some examples shown herein include a certain number of synthetic jets **602** (e.g., four synthetic jets, eight synthetic jets, etc.). However, any number of synthetic jets can be used (e.g., less than four, five, six, more than eight, etc.).

FIG. **6B** is rear side view of an example electronic device **601** including another example synthetic jet system **600** disclosed herein. A first cover **612** (e.g., a D-cover or bottom cover) (FIG. **6C**) is removed for clarity. FIG. **6C** is a bottom view from the first cover **612** (e.g., a D cover side, which is not shown in FIG. **6C** for clarity) of the example synthetic jet system **600**. FIG. **6D** is a bottom view of the example synthetic jet system **600**. The first cover **612** is not shown in FIGS. **6B** and **6C** for clarity.

Referring to FIG. **6C** and, the electronic device **601** includes the printed circuit board **408** and the processor **406** positioned between the first cover **612** (e.g., a D-cover or bottom cover or frame) and a second cover **614** (e.g., a C-cover or top cover or frame). The example synthetic jet system **600** includes a plurality of synthetic jets **602** (e.g., four jets) coupled to a heat spreader **616**. Specifically, the synthetic jets **602** are positioned on a first surface **616a** and the processor **406** is oriented toward (e.g., or, alternatively, coupled to) a second surface **616b** of the heat spreader **616**. Thus, the synthetic jets **602** are positioned between the first surface **616a** of the heat spreader **616** and the first cover **612** and the processor **406** is positioned between the circuit board **408** and the second surface **616b** of the heat spreader **616**. Additionally, the synthetic jets **602** are positioned away from a processor zone **618** (e.g., a core region of processor **406**). For example, the processor zone **618** can be defined by a perimeter of the processor **406**, where a boundary or zone defined by the perimeter of the processor **406** extends (e.g., vertically) between the first cover **612** and the second cover **614**. The heat spreader **616** of the illustrated example is extended for non-system on chip (non-SOC) component cooling (e.g., FET **620**, memory **622**, etc.). In operation, airflow **650** (e.g., cool air) flows in the inlet **608** of the respective ones of the synthetic jets **602** via inlet vents **626** formed in the first cover **612** and exits from the outlet **610** of the respective ones of the synthetic jets **602**. Airflow or exhaust exiting the outlet **610** impinges on the heat spreader **616** as it exits the synthetic jets **602** and exits through exhaust vents **628** formed in a back wall **630** (e.g., a side wall) of the electronic device **601**. In some examples, duct **632** (e.g., downstream from the outlet **610**) fluidly couples the outlets **610** and the exhaust vents **628**. Thus, in the illustrated example, the outlets **610** are directed away from the processor **406** and/or the processor zone **618** (e.g., a core region) and synthetic jets **602** provide cooling for non-system on chip components (e.g., FET **620**, memory **622**, etc.).

FIG. **7A** is a bottom view of an example electronic device **701** including another example synthetic jet system **700** disclosed herein. FIG. **7B** is a rear side view of the example synthetic jet system **700**. FIG. **7C** is a bottom view of the example synthetic jet system **700** of FIG. **7A**. Referring to FIGS. **7B** and **7C**, the electronic device **701** includes the printed circuit board **408** and the processor **406** positioned between a first cover **703** (e.g., a D-cover or bottom cover or frame) and a second cover **705** (e.g., a C-cover or top cover or frame). The first cover **703** is not shown in FIGS. **7A** and **7C** for clarity. The example synthetic jet system **700** includes a plurality of synthetic jets **602** (e.g., eight jets) coupled to a heat spreader **704**. In particular, the synthetic jets **602** (e.g., eight jets) of the illustrated example are

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positioned on both sides of a heat spreader **704** (e.g., a first side **704a** and a second side **704b** opposite the first side). In the illustrated example, a first plurality of synthetic jets **602** (e.g., four jets) are positioned on the first side **704a** and a second plurality of synthetic jets **602** (e.g., four jets) are positioned on the second side **704b** opposite the first side **704a**. The synthetic jets **602** are placed in an evacuative layout away from a processor zone **702** (e.g., a system on chip (SoC) zone) and/or the processor **406** such that the synthetic jets **602** are not positioned directly over the processor **406** (e.g., in the z-direction or stack-up direction) but are positioned on lateral sides of the processor zone **702** and/or the processor **406**. Thus, the heat spreader **704** is an extended spreader for non-SOC component cooling. In other words, the outlets **610** of the synthetic jets **602** are oriented in a direction away from the processor **406** (e.g., toward a back wall **707** of the electronic device **701**).

In operation, airflow **650** is drawn into the inlets **608** of the synthetic jets **602** via inlet vents **703a** formed in the first cover **703** and/or inlet vents **705a** formed in the second cover **705**. As air exits the outlets **610** of the synthetic jets **602**, air exiting the outlets **610** impinges the heat spreader **704**. Additionally, the synthetic jet system **700** of the illustrated example includes a heat exchanger **706** in communication with the outlets **610** of the respective ones of the synthetic jets **602**. The synthetic jet system **700** can include a duct **709** to channel the airflow from the heat exchanger **706** to an exhaust vent **712** formed in a back wall **714** (e.g., a side wall) of the electronic device **701**. The heat exchanger **706** can be added on either of the first side **704a** or the second side **704b** of the heat spreader **704** based on system design feasibility. The outlets **610** of the example synthetic jets **602** of FIGS. 7A-7C are directed away from the processor **406**. Thus, in the illustrated example, the outlets **610** are directed away from the processor **406** and/or the processor zone **702** and can effectively cool non-system on chip components (e.g., FET **620**, memory **622**, etc.).

FIG. 8A is a bottom view of an example electronic device **801** including another example synthetic jet system **800** disclosed herein. FIG. 8B is a rear side view of the example synthetic jet system **800**. FIG. 8C is a bottom view from a first cover **803** (e.g., a D cover side) of the example synthetic jet system **800**. Referring to FIGS. 8B and 8C, the electronic device **801** includes the printed circuit board **408** and the processor **406** positioned between the first cover **803** (e.g., a D-cover or bottom cover or frame) and a second cover **805** (e.g., a C-cover or top cover or frame). The first cover **803** is not shown in FIGS. 8A and 8C for clarity. The example synthetic jet system **800** includes a plurality of synthetic jets **602** (e.g., four jets) coupled to a first side **804a** of a heat spreader **804** opposite a second side **804b**, where the second side **804b** is oriented toward the processor **406**. The synthetic jet system **800** of the illustrated example includes a first set of jets **602a** and a second set of jets **602b** placed in different orientations (e.g., an ester island layout). For example, the first set of jets **602a** are oriented non-parallel (e.g., perpendicular) relative to the second set of jets **604a**. More specifically, respective first outlets **610a** of the first set of jets **602a** are positioned non-parallel (e.g., perpendicular) relative to respective second outlets **610b** of the second set of jets **602b**. In particular, the first outlets **610a** of the first set of jets **602a** are oriented or directed toward a back wall **814** of the electronic device **801** and the second outlets **610b** of the second set of jets **602b** are oriented toward the processor **406** and/or a processor zone **802**. In the illustrated example, exhaust **826a** exiting the first outlets **610a** of the first set of jets **602a** is non-parallel (e.g., perpendicular)

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relative to exhaust **826b** exiting the second outlets **610b** of the second set of jets **602b**. In operation, the first set of jets **602a** and the second set of jets **602b** draw airflow **650** into the inlets **608** via inlet vents **703a** of the first cover **703**. Airflow or exhaust **826a** exits the first outlets **610a**, flows across the heat exchangers **806**, and through exhaust vents **812** of the back wall **814** (e.g., for non-SOC component cooling). Additionally, airflow or exhaust **826b** exits the second outlets **610b** of the second set of jets **602b** and impinges on and/or flows over the heat spreader **804** across the processor zone **802** (e.g., the SoC components). The exhaust **826b** and exits through an exhaust vent **816** of a central system **818** (e.g., an exhaust system for the processor zone **802**) formed in the back wall **814**. The central system **818** can include a heat exchanger **820**.

FIG. 9A is a bottom view of an example electronic device **901** including another example synthetic jet system **900** disclosed herein. FIG. 9B is a side view of the example synthetic jet system **900**. FIG. 9C is a bottom side view from a first cover **903** (e.g., a D cover side) of the example synthetic jet system **900**. Referring to FIGS. 9B and 9C, the electronic device **901** includes the printed circuit board **408** and the processor **406** positioned between the first cover **903** (e.g., a D-cover or bottom cover or frame) and a second cover **905** (e.g., a C-cover or top cover or frame). The first cover **903** is not shown in FIGS. 9A and 9C for clarity. The example synthetic jet system **900** includes a plurality of synthetic jets **602** (e.g., four jets) coupled to a first side **904a** of a heat spreader **904** opposite a second side **904b**. The first side **904a** of the heat spreader **904** is oriented toward the first cover **903** and the second side **904b** is oriented toward the processor **406**. The synthetic jet system **900** includes a plurality of synthetic jets **602** (e.g., four jets) positioned on the first side **904a** of a heat spreader **904** opposite a second side **904b** having the processor **406** (SoC). Specifically, the outlets **610** of respective ones of the synthetic jets **602** are oriented toward the processor **406** and/or a processor zone **902** (e.g. an SoC zone) of the heat spreader **904**. The synthetic jets **602** of the illustrated example are placed in Hyperbaric Layout away from processor zone **902**, but the outlets **610** are oriented to direct the airflow **926** toward the processor zone **902**. In operation, airflow **650** (e.g., cool air) flows in the inlet **608** of the respective ones of the synthetic jets **602** via inlet vents **903a** formed in the first cover **903** and exits from the outlet **610** of the respective ones of the synthetic jets **602**. Airflow **926** exiting the outlets **610** impinges on the heat spreader **904** as the airflow flows across the processor zone **618**. The airflow **926** exits the electronic device **901** through an exhaust vent **916** formed in a back wall **914** (e.g., a side wall) of the electronic device **901**. A heat exchanger **906** is provided downstream from the outlets **610** and upstream from the exhaust vent **916**. Thus, in the illustrated example, the outlets **610** are directed toward from the processor **406** and/or the processor zone **902** to cool the processor **406**.

FIG. 10A is a bottom view of an example electronic device **1001** including the example synthetic jet system **1000** disclosed herein. FIG. 10B is a rear side view of the example synthetic jet system **1000**. FIG. 10C is a bottom side view of the example synthetic jet system **1000**. Referring to FIGS. 10B and 10C, the electronic device **1001** includes the printed circuit board **408** and the processor **406** positioned between a first cover **1003** (e.g., a D-cover or bottom cover or frame) and a second cover **1005** (e.g., a C-cover or top cover or frame). The first cover **1003** is not shown in FIGS. 10A and 10C for clarity. The example synthetic jet system **900** includes a plurality of synthetic jets **602** (e.g., four jets)

coupled to a first side **1004a** of a heat spreader **1004** opposite a second side **1004b**. The first side **1004a** of the heat spreader **1004** is oriented toward the first cover **1003** and the second side **1004b** is oriented toward the processor **406**. The synthetic jets **602** (e.g., four jets) of the illustrated example are positioned adjacent (e.g., above or below) the processor **406** and/or the processor zone **1002**. In other words, the synthetic jets **602** of the illustrated example are positioned on the first side **1004a** of the heat spreader **1004** at least partially overlapping a perimeter of the processor **406** on the second side **1004b** of the heat spreader **1004**. In operation, airflow **650** enters the inlets **608** of the respective synthetic jets **602** via an inlet vent **1003a** formed in the first cover **1003**. As the airflow **1026** exits the outlets **610**, the air impinges and/or flows across the heat spreader **1004** and exits through a vent **1016** of a central system **1008** formed in a back wall **1014** of the electronic device **1001**. Heat exchangers **1006** is provided at the outlets **610** of the synthetic jets **602**. A duct **1018** is provided between the heat exchangers **1006** and the vent **1016**. In this example, each outlet **610** of the synthetic jets **602** is directed toward the heat exchangers **1006**. Thus, in the illustrated example, although the outlets **610** are not directed or oriented toward the processor **406**, the synthetic jets **602** are positioned to overlap (e.g., are positioned over or below) the processor **406** to cool the processor **406**.

FIG. **11A** is a bottom view of an example electronic device including another example synthetic jet system **1100** disclosed herein. FIG. **11B** is a rear side view of the example synthetic jet system **1100**. FIG. **11C** is a bottom view of the example synthetic jet system **1100**. Referring to FIGS. **11B** and **11C**, the electronic device **1101** includes the printed circuit board **408** and the processor **406** positioned between a first cover **1103** (e.g., a D-cover or bottom cover or frame) and a second cover **1105** (e.g., a C-cover or top cover or frame). The first cover **1103** is not shown in FIGS. **11A** and **11C** for clarity. The example synthetic jet system **1100** includes a plurality of synthetic jets **602** (e.g., four jets) coupled to a first side **1104a** of a heat spreader **1104** opposite a second side **1104b**. The first side **1104a** of the heat spreader **1104** is oriented toward the first cover **1103** and the second side **1104b** is oriented toward the processor **406**. The synthetic jet system **1100** of the illustrated example includes a plurality of synthetic jets **602** (e.g., four synthetic jets) placed away from the processor **406** and on only one side of processor **406** or the processor zone **1102**. For example, the synthetic jets **602** are positioned adjacent a lateral edge **1104c** (e.g., a side edge, a right side edge in the orientation of FIG. **10C**). In some examples, the synthetic jets **602** are positioned on the second side **1104b** of the heat spreader **1104** (e.g., on the same side as the processor **406**). In operation, airflow **650** enters the inlets **608** of the respective synthetic jets **602** via an inlet vent **1103a** formed in the first cover **1103**. The airflow **1126** exits the outlets **610** exits through a vent **1116** formed in a back wall **1114** of the electronic device **1101**. Heat exchangers **1106** are provided at the outlets **610** of the synthetic jets **602**. A duct **1118** is provided between the heat exchangers **1106** and the vent **1016**.

FIG. **12A** is a bottom view of an example electronic device **1201** including another example synthetic jet system **1200** disclosed herein. FIG. **12B** is a rear side view of the example synthetic jet system **1200**. FIG. **12C** is a front view of the example synthetic jet system **1200**. Referring to FIGS. **12B** and **12C**, the electronic device **1201** includes the printed circuit board **408** and the processor **406** positioned between a first cover **1203** (e.g., a D-cover or bottom cover or frame)

and a second cover **1205** (e.g., a C-cover or top cover or frame). The first cover **1203** is not shown in FIGS. **12A** and **12B** for clarity. The example synthetic jet system **1200** includes a plurality of synthetic jets **602** (e.g., four jets) are placed in hyperbaric layout away from the processor **406** and/or a processor zone **1202** (e.g., a SOC zone). In contrast to the foregoing examples, the synthetic jets **602** of the illustrated example are not positioned on or coupled to a heat spreader **1204**. The heat spreader **1204** of the illustrated example is positioned between the processor **406** and the first cover **1203** (e.g., vertically or in the z-direction in the illustrated example) and also between a first set **1206a** of synthetic jets **602** and a second set **1206b** of synthetic jets **602** (e.g., laterally, horizontally, or in the x-direction in the illustrated example). Specifically, the synthetic jets **602** are positioned on respective sides of the heat spreader **1204**. The first set **1206a** of synthetic jets **602** of the illustrated example is supported by a first frame **1207** and the second set **1206b** of the synthetic jets **602** is supported by a second frame **1209**. The first frame **1207** and the second frame **1209** are coupled to (directly or indirectly to) the second cover **1205**. In some examples, the first frame **1207** and/or the second frame **1209** can be any structure of the electronic device **1201**. Additionally, the synthetic jets **602** are oriented or stacked in the z-direction (e.g., a vertical direction). In other words, the synthetic jets **602** of the first set **1206a** of synthetic jets **602** are in a stacked orientation such that a first one of the synthetic jets **602** of the first set **1206a** is positioned above (or below) a second one of the synthetic jets **602** of the first set **1206a**. Likewise, the synthetic jets **602** of the second set **1206b** of synthetic jets **602** are in a stacked orientation such that a first one of the synthetic jets **602** of the second set **1206b** is positioned above (or below) a second one of the synthetic jets **602** of the second set **1206b**. Furthermore, the outlets **610** of the synthetic jets **602** are directed toward the processor **406**. In operation, airflow **650** enters the inlets **608** of the respective synthetic jets **602** via inlet vents **1203a** formed in the first cover **1203**. Airflow **1226** exits the outlets **610** exits through an exhaust vent **1216** formed in a back wall **1214** of the electronic device **1201**. In contrast to some of the examples noted above, the airflow does not impinge the heat spreader **1204** because the synthetic jets **602** are not positioned on the heat spreader **1204**.

FIG. **13A** is a bottom view of the example electronic device **1301** having the example synthetic jet system **1300** disclosed herein. FIG. **13B** is a rear side view of the example synthetic jet system **1300**. FIG. **13C** is a bottom view of the example synthetic jet system **1300** of FIG. **13A**. Referring to FIGS. **13A-13C**, the electronic device **1301** includes the printed circuit board **408** and the processor **406** positioned between a first cover **1303** (e.g., a D-cover or bottom cover or frame) and a second cover **1305** (e.g., a C-cover or top cover or frame). A main heat spreader **1304** is coupled to the processor **406**. The first cover **1303** is not shown in FIGS. **13A** and **13C** for clarity.

The example synthetic jet system **1300** includes a heat spreader **1307** (e.g., a Z-bend heat spreader) to support the plurality of synthetic jets **602** (e.g., four jets). For example, a first side **1307a** (e.g., a first Z-bend) of the heat spreader **1307** supports a first plurality **1306a** of the synthetic jets **602** and a second side **1307b** of the heat spreader **1307** supports a second plurality **1306b** of the synthetic jets **602**. In particular, the synthetic jets **602** (e.g., eight jets) of the illustrated example are positioned on a first surface **1307c** (e.g., opposite a second surface **1307d**) of the heat spreader **1307**. The heat spreader **1307** of the illustrated example includes a main heat spreader portion **1304** positioned

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between the first side **1307a** and the second side **1307b** in a first direction (e.g., the x-direction) and positioned between the processor **406** and the first cover **1303** in a second direction (e.g., the z-direction). The synthetic jets **602** are placed in an evacuative layout away from a processor zone **1302** (e.g., a system on chip (SoC) zone) and/or the processor **406** such that the synthetic jets **602** are not positioned directly over or under the processor **406** but are positioned on lateral sides of the processor zone **1302** and/or the processor **406**. Thus, the heat spreader **1307** is extended for non-SOC component cooling. In other words, the outlets **610** of the synthetic jets **602** are oriented in a direction away from the processor **406** (e.g., toward a back wall **1314** of the electronic device **1201**). In other words, airflow **1325** (e.g., heated air) internally of the electronic device **701** (e.g., from the cavity **1309** and/or adjacent the processor **406**) is drawn into the synthetic jets **602** and the exhaust **1326** exiting the outlets **610** is exhausted via exhaust vents **1316** formed in a back wall **1314** of the electronic device **1301**. The outlets **610** of the example synthetic jets **602** of FIGS. **13A-13C** are directed away from the processor **406**. Thus, in the illustrated example, the outlets **610** are directed away from the processor **406** and/or the processor zone **1302** and can cool the non-system on chip components (e.g., FET **620**, memory **622**, etc.).

FIG. **14A** is a bottom view of an example electronic device including another example synthetic jet system **1400** disclosed herein. FIG. **14B** is a rear side view of the example synthetic jet system **1400**. FIG. **14C** is a Bottom view of the example synthetic jet system **1400**. Referring to FIGS. **14B** and **14C**, the electronic device **1401** includes the printed circuit board **408** and the processor **406** positioned between a first cover **1403** (e.g., a D-cover or bottom cover or frame) and a second cover **1405** (e.g., a C-cover or top cover or frame). The first cover **1403** is not shown in FIGS. **14A** and **14C** for clarity. The example synthetic jet system **1400** includes a plurality of synthetic jets **602** (e.g., four jets) coupled to a second side **1404b** of a heat spreader **1404** opposite a first side **1404a**. The first side **1404a** of the heat spreader **1404** is oriented toward the first cover **1403** and the second side **1404b** is oriented toward the processor **406**. Thus, the synthetic jets **602** and the processor **406** are positioned between the heat spreader **1404** and the circuit board **408**. In operation, airflow **1425** is drawn from an inlet **1427** formed in a back wall **1414** of the electronic device **1401**. The airflow **1425** flows within a cavity **1409** between the first cover **1403** and the second cover **1405** (e.g., from between the heat spreader **1404** and the circuit board **408**). The airflow **1425** is drawn into the inlets **608** of the synthetic jets **602**. In other words, cold or ambient temperature airflow **1425** (e.g., cold air) is drawn into the inlets **608** of the synthetic jets **602** via the inlet **1427** and the airflow **1426** exiting the outlets **610** is exhausted via exhaust vents **1416** formed in the back wall **1414** of the electronic device **1401**. The outlets **610** of the example synthetic jets **602** of FIGS. **14A-14C** are directed away from the processor **406** (e.g., toward the back wall **1414**). In the illustrated example, the synthetic jets **602** provide cooling to the processor **406** and/or the processor region **1402** and non-system on chip components (e.g., FET **620**, memory **622**, etc.).

The foregoing examples of the synthetic jet systems can be cooling systems. Although each example synthetic jet systems disclosed above have certain features, it should be understood that it is not necessary for a particular feature of one example to be used exclusively with that example. Instead, any of the features described above and/or depicted in the drawings can be combined with any of the examples,

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in addition to or in substitution for any of the other features of those examples. One example's features are not mutually exclusive to another example's features. Instead, the scope of this disclosure encompasses any combination of any of the features. Any one of the synthetic jet systems disclosed herein can implement the example thermal management system **150** and/or the electronic device **100** of FIG. **1A**. For example, any one of the examples shown in FIGS. **6A-14C** can be implemented with the bypass **430** shown in FIGS. **4A-4F**. In some of these examples, some of the synthetic jets **602** of examples of FIGS. **6A-14C** can be removed or eliminated. Additionally, although most of the examples disclosed herein includes four jets, any number of synthetic jets **402** and/or **602** can be used in a synthetic jet system. Additionally, the synthetic jets **402** and/or **602** can be positioned on a first side of a heat spreader, a second side of a heat spreader, a combination of a first side and a second side of a heat spreader, and/or on a support structure or frame of an electronic device located adjacent to the heat spreader. In some examples, a synthetic jet system can include a combination of one or more synthetic jets **402** of FIGS. **4A-4E** and one or more synthetic jets **602** of FIGS. **6A-14C**. In some examples, the synthetic jets **402** and/or **602** can include any number of jets. Anyone of the synthetic jet systems **400-1400** can implement the electronic device **100** of FIG. **1A** and/or can be controlled by the example thermal management system **150** of FIG. **1A**.

FIG. **17** is a block diagram of an example implementation of the example thermal management device **158** of the example thermal management system **150** of FIG. **1A** in accordance with teachings of this disclosure to control an operation of the synthetic jets **402** of the synthetic jet system **156**. In some examples, the thermal management device **158** can implement the example synthetic jet systems of FIGS. **6A-14C**. The thermal management device **158** of FIG. **15** may be instantiated (e.g., creating an instance of, bring into being for any length of time, materialize, implement, etc.) by processor circuitry such as a central processing unit executing instructions. Additionally or alternatively, the thermal management device **158** of FIG. **15** of FIG. **2** may be instantiated (e.g., creating an instance of, bring into being for any length of time, materialize, implement, etc.) by an ASIC or an FPGA structured to perform operations corresponding to the instructions. It should be understood that some or all of the circuitry of FIG. **15** may, thus, be instantiated at the same or separate times. Some or all of the circuitry may be instantiated, for example, in one or more threads executing concurrently on hardware and/or in series on hardware. Moreover, in some examples, some or all of the circuitry of FIG. **15** may be implemented by one or more virtual machines and/or containers executing on the micro-processor.

The thermal management device **158** of the illustrated example includes an example I/O interface **1502**, example temperature monitoring circuitry **1504**, example device position determining circuitry **1506**, example temperature threshold selection circuitry **1508**, example comparator **1510** and example synthetic jet controller **1512**, all of which are communicatively coupled (e.g., sharing data via memory) via a bus **1516**, etc. A database **1514** stores a plurality of temperature thresholds or specifications for different or distinct positions of the electronic device **100**.

The I/O interface **1502** provides or establishes communication between the temperature sensor(s) **152**, the position sensor(s) **154** and/or other components of the electronic device **100** with the thermal management device **158**. For example, the thermal management device **158** receives tem-

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perature signals **1518** from the temperature sensor(s) **152** and position signals **1520** from the position sensor(s) **154** via the I/O interface **1502**. The I/O interface **1502** may be implemented by an API, for example.

The temperature monitoring circuitry **1504** receives one or more temperature signals **1518** from the temperature sensor(s) **152** via the I/O output interface **1502**. The temperature monitoring circuitry **1504** converts the temperature signal(s) from the temperature sensor(s) **152** for comparison by the comparator **1510**. For example, the temperature monitoring circuitry **1504** obtains or determines a measured temperature (e.g., a first temperature) of a hot spot skin or other area, a second measured temperature of electronic components (e.g., the processor **406**), etc.

The device position determining circuitry **1506** receives one or more signals from the position sensor(s) **154** via the I/O output interface **1502**. The device position determining circuitry **1506** determines an open position of the electronic device **100**. For example, the position sensor(s) **154** can determine or measure an angle between the first housing **102** and the second housing **104**. In some examples, based on the signal(s) **1520** from the position sensor(s) **154**, the device position determining circuitry **1506** determines if the electronic device **100** is in the laptop mode **122**, the tablet mode **124**, the kiosk mode **126**, or the tent mode **128**. In some examples, the position sensor(s) **154** (e.g., via gyroscope, accelerometer, and/or other sensor) can detect whether the laptop is on a body of a person (e.g., a lap, an arm, a hand, etc.). The device position determining circuitry **1506** communicates the device position to the temperature threshold selection circuitry **1508**.

The temperature threshold selection circuitry **1508** obtains from the database **1514** a temperature threshold associated with the determined position of the electronic device provided by the device positioning determining circuitry **1506**. For example, the electronic device **100** and/or the thermal management system **150** can employ different temperature threshold specifications based on a position of the electronic device **100**. For example, a first temperature threshold can be associated with a hot spot region corresponding to the electronic device **100** being in the laptop mode **122** or the tablet mode **124**. For example, a second temperature threshold can be associated with the hot spot region corresponding to the electronic device **100** being in the kiosk mode **126** or the tent mode **128**. Based on the device position, the temperature threshold selection circuitry **1508** obtains a temperature threshold from the database **1514** (e.g., via a look-up table) that is associated with the device position and provides the selected temperature value to the comparator **1510**.

The comparator **1510** compares the measured temperature and the selected temperature threshold. The jet device controller **1512** activates the synthetic jet system **156** (e.g., one or more synthetic jets **402**) when the measured temperature exceeds the selected temperature threshold. The jet device controller **1512** deactivates the synthetic jet system **156** (e.g., one or more synthetic jets **402**) when the measured temperature does not exceed the selected temperature threshold.

In some examples, the thermal management device **158** and/or the jet device controller **1512** can activate select ones of the synthetic jets **402** of a synthetic jet system **156**. For example, the jet device controller **1512** can activate select ones of the synthetic jets **402** of a synthetic jet system **156** when monitoring multiple hot spot areas or zones. In some examples, the jet device controller **1512** activates two or more of the synthetic jets **402** of a synthetic jet system **156**

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in a pattern (e.g., activate the first set **402a** of synthetic jets **402** for a first duration, activate the second set **402b** of synthetic jets **402** for a second duration, offset activations of the first set **402a** and the second set **402b** of synthetic jets **402**, etc.

While an example manner of implementing the thermal management device **158** of FIG. **1A** is illustrated in FIG. **15**, one or more of the elements, processes, and/or devices illustrated in FIG. **15** may be combined, divided, re-arranged, omitted, eliminated, and/or implemented in any other way. Further, the example I/O interface **1502**, the example temperature monitor circuitry **1504**, the example device position determiner circuitry **1506**, the example temperature threshold selection circuitry **1508**, the example comparator **1510** and the example synthetic jet controller **1512** and/or, more generally, the example thermal management device **158** of FIG. **15**, may be implemented by hardware alone or by hardware in combination with software and/or firmware. Thus, for example, any of the example I/O interface **1502**, the example temperature monitor circuitry **1504**, the example device position determiner circuitry **1506**, the example temperature threshold selection circuitry **1508**, the example comparator **1510** and the example synthetic jet controller **1512** and/or, more generally, the example thermal management device **158**, could be implemented by processor circuitry, analog circuit(s), digital circuit(s), logic circuit(s), programmable processor(s), programmable microcontroller(s), graphics processing unit(s) (GPU(s)), digital signal processor(s) (DSP(s)), application specific integrated circuit(s) (ASIC(s)), programmable logic device(s) (PLD(s)), and/or field programmable logic device(s) (FPLD(s)) such as Field Programmable Gate Arrays (FPGAs). Further still, the example thermal management device **158** of FIG. **1A** may include one or more elements, processes, and/or devices in addition to, or instead of, those illustrated in FIG. **15**, and/or may include more than one of any or all of the illustrated elements, processes and devices.

In some examples, the thermal management device **158** includes means for receiving, retrieving and/or otherwise obtaining temperature signals (e.g., output signals) from the temperature sensor(s) **152**. For example, the means for receiving, retrieving and/or otherwise obtaining temperature signals (e.g., output signals) from the temperature sensor(s) **152** may be implemented by the temperature monitoring circuitry **1504**. In some examples, the temperature monitoring circuitry **1504** may be instantiated by processor circuitry such as the example processor circuitry **1712** of FIG. **17**. For instance, the temperature monitoring circuitry **1504** may be instantiated by the example general purpose processor circuitry **1800** of FIG. **18** executing machine executable instructions such as that implemented by at least blocks **1602**, **1604** of FIG. **16**.

In some examples, the thermal management device **158** includes means for receiving, retrieving and/or otherwise obtaining position signals (e.g., output signals) from the position sensor(s) **154**. For example, the means for receiving, retrieving and/or otherwise obtaining position signals (e.g., output signals) from the temperature sensor(s) **152** may be implemented by the device position determining circuitry **1506**. In some examples, the device position determining circuitry **1506** may be instantiated by processor circuitry such as the example processor circuitry **1712** of FIG. **17**. For instance, the device position determining circuitry **1506** may be instantiated by the example general purpose processor circuitry **1800** of FIG. **18** executing machine executable instructions such as that implemented by at least block **1612** of FIG. **16**.

In some examples, the thermal management device **158** includes means for controlling one or more synthetic jets **402** of the synthetic jet system **156**. For example, means for controlling one or more synthetic jets **402** of the synthetic jet system **156** may be implemented by the jet device controller **1512**. In some examples, the jet device controller **1512** may be instantiated by processor circuitry such as the example processor circuitry **1712** of FIG. **17**. For instance, the jet device controller **1512** may be instantiated by the example general purpose processor circuitry **1800** of FIG. **18** executing machine executable instructions such as that implemented by at least blocks **1606**, **1608**, **1610**, **1614**, **1616**, **1618**, **1620**, **1622**, **1604** of FIG. **16**.

In some examples, the thermal management device **158** includes means for selecting and/or otherwise obtaining a threshold temperature associated with a detected position of the electronic device. For example, the means for selecting and/or otherwise obtaining a threshold temperature associated with a detected position of the electronic device may be implemented by the temperature threshold selection circuitry **1508**. In some examples, the temperature threshold selection circuitry **1508** may be instantiated by processor circuitry such as the example processor circuitry **1712** of FIG. **17**. For instance, the temperature threshold selection circuitry **1508** may be instantiated by the example general purpose processor circuitry **1800** of FIG. **18** executing machine executable instructions such as that implemented by at least block **1602** of FIG. **16**.

In some examples, the thermal management device **158** includes means for comparing a threshold temperature and a measured temperature. For example, the means for comparing a threshold temperature and a measured temperature may be implemented by the comparator **1510**. In some examples, the comparator **1510** may be instantiated by processor circuitry such as the example processor circuitry **1712** of FIG. **17**. For instance, the comparator **1510** may be instantiated by the example general purpose processor circuitry **1800** of FIG. **18** executing machine executable instructions such as that implemented by at least block **1602** of FIG. **16**.

In some examples, the temperature monitoring circuitry **1504**, the device position determining circuitry **1506**, the comparator **1510**, the jet device controller **1512** and/or the temperature threshold selection circuitry **1508** may be instantiated by hardware logic circuitry, which may be implemented by an ASIC or the FPGA circuitry **1900** of FIG. **19** structured to perform operations corresponding to the machine readable instructions. Additionally or alternatively, the temperature monitoring circuitry **1504**, the device position determining circuitry **1506**, the comparator **1510**, the jet device controller **1512** and/or the temperature threshold selection circuitry **1508** may be instantiated by any other combination of hardware, software, and/or firmware. For example, the temperature monitoring circuitry **1504**, the device position determining circuitry **1506**, the comparator **1510**, the jet device controller **1512** and/or the temperature threshold selection circuitry **1508** may be implemented by at least one or more hardware circuits (e.g., processor circuitry, discrete and/or integrated analog and/or digital circuitry, an FPGA, an Application Specific Integrated Circuit (ASIC), a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) structured to execute some or all of the machine readable instructions and/or to perform some or all of the operations corresponding to the machine readable instructions without executing software or firmware, but other structures are likewise appropriate.

A flowchart representative of example hardware logic circuitry, machine readable instructions, hardware implemented state machines, and/or any combination thereof for implementing the thermal management device **158** of FIG. **15** is shown in FIG. **16**. The machine readable instructions may be one or more executable programs or portion(s) of an executable program for execution by processor circuitry, such as the processor circuitry **1712** shown in the example processor platform **1700** discussed below in connection with FIG. **17** and/or the example processor circuitry discussed below in connection with FIGS. **18** and/or **19**. The program may be embodied in software stored on one or more non-transitory computer readable storage media such as a compact disk (CD), a floppy disk, a hard disk drive (HDD), a solid-state drive (SSD), a digital versatile disk (DVD), a Blu-ray disk, a volatile memory (e.g., Random Access Memory (RAM) of any type, etc.), or a non-volatile memory (e.g., electrically erasable programmable read-only memory (EEPROM), FLASH memory, an HDD, an SSD, etc.) associated with processor circuitry located in one or more hardware devices, but the entire program and/or parts thereof could alternatively be executed by one or more hardware devices other than the processor circuitry and/or embodied in firmware or dedicated hardware. The machine readable instructions may be distributed across multiple hardware devices and/or executed by two or more hardware devices (e.g., a server and a client hardware device). For example, the client hardware device may be implemented by an endpoint client hardware device (e.g., a hardware device associated with a user) or an intermediate client hardware device (e.g., a radio access network (RAN)) gateway that may facilitate communication between a server and an endpoint client hardware device). Similarly, the non-transitory computer readable storage media may include one or more mediums located in one or more hardware devices. Further, although the example program is described with reference to the flowchart illustrated in FIG. **17**, many other methods of implementing the example thermal management device **158** may alternatively be used. For example, the order of execution of the blocks may be changed, and/or some of the blocks described may be changed, eliminated, or combined. Additionally, or alternatively, any or all of the blocks may be implemented by one or more hardware circuits (e.g., processor circuitry, discrete and/or integrated analog and/or digital circuitry, an FPGA, an ASIC, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) structured to perform the corresponding operation without executing software or firmware. The processor circuitry may be distributed in different network locations and/or local to one or more hardware devices (e.g., a single-core processor (e.g., a single core central processor unit (CPU)), a multi-core processor (e.g., a multi-core CPU), etc.) in a single machine, multiple processors distributed across multiple servers of a server rack, multiple processors distributed across one or more server racks, a CPU and/or a FPGA located in the same package (e.g., the same integrated circuit (IC) package or in two or more separate housings, etc.).

The machine readable instructions described herein may be stored in one or more of a compressed format, an encrypted format, a fragmented format, a compiled format, an executable format, a packaged format, etc. Machine readable instructions as described herein may be stored as data or a data structure (e.g., as portions of instructions, code, representations of code, etc.) that may be utilized to create, manufacture, and/or produce machine executable instructions. For example, the machine readable instructions

may be fragmented and stored on one or more storage devices and/or computing devices (e.g., servers) located at the same or different locations of a network or collection of networks (e.g., in the cloud, in edge devices, etc.). The machine readable instructions may require one or more of installation, modification, adaptation, updating, combining, supplementing, configuring, decryption, decompression, unpacking, distribution, reassignment, compilation, etc., in order to make them directly readable, interpretable, and/or executable by a computing device and/or other machine. For example, the machine readable instructions may be stored in multiple parts, which are individually compressed, encrypted, and/or stored on separate computing devices, wherein the parts when decrypted, decompressed, and/or combined form a set of machine executable instructions that implement one or more operations that may together form a program such as that described herein.

In another example, the machine readable instructions may be stored in a state in which they may be read by processor circuitry, but require addition of a library (e.g., a dynamic link library (DLL)), a software development kit (SDK), an application programming interface (API), etc., in order to execute the machine readable instructions on a particular computing device or other device. In another example, the machine readable instructions may need to be configured (e.g., settings stored, data input, network addresses recorded, etc.) before the machine readable instructions and/or the corresponding program(s) can be executed in whole or in part. Thus, machine readable media, as used herein, may include machine readable instructions and/or program(s) regardless of the particular format or state of the machine readable instructions and/or program(s) when stored or otherwise at rest or in transit.

The machine readable instructions described herein can be represented by any past, present, or future instruction language, scripting language, programming language, etc. For example, the machine readable instructions may be represented using any of the following languages: C, C++, Java, C#, Perl, Python, JavaScript, HyperText Markup Language (HTML), Structured Query Language (SQL), Swift, etc.

As mentioned above, the example operations of FIG. 16 may be implemented using executable instructions (e.g., computer and/or machine readable instructions) stored on one or more non-transitory computer and/or machine readable media such as optical storage devices, magnetic storage devices, an HDD, a flash memory, a read-only memory (ROM), a CD, a DVD, a cache, a RAM of any type, a register, and/or any other storage device or storage disk in which information is stored for any duration (e.g., for extended time periods, permanently, for brief instances, for temporarily buffering, and/or for caching of the information). As used herein, the terms non-transitory computer readable medium and non-transitory computer readable storage medium is expressly defined to include any type of computer readable storage device and/or storage disk and to exclude propagating signals and to exclude transmission media.

"Including" and "comprising" (and all forms and tenses thereof) are used herein to be open ended terms. Thus, whenever a claim employs any form of "include" or "comprise" (e.g., comprises, includes, comprising, including, having, etc.) as a preamble or within a claim recitation of any kind, it is to be understood that additional elements, terms, etc., may be present without falling outside the scope of the corresponding claim or recitation. As used herein, when the phrase "at least" is used as the transition term in,

for example, a preamble of a claim, it is open-ended in the same manner as the term "comprising" and "including" are open ended. The term "and/or" when used, for example, in a form such as A, B, and/or C refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, or (7) A with B and with C. As used herein in the context of describing structures, components, items, objects and/or things, the phrase "at least one of A and B" is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing structures, components, items, objects and/or things, the phrase "at least one of A or B" is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. As used herein in the context of describing the performance or execution of processes, instructions, actions, activities and/or steps, the phrase "at least one of A and B" is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing the performance or execution of processes, instructions, actions, activities and/or steps, the phrase "at least one of A or B" is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B.

As used herein, singular references (e.g., "a", "an", "first", "second", etc.) do not exclude a plurality. The term "a" or "an" object, as used herein, refers to one or more of that object. The terms "a" (or "an"), "one or more", and "at least one" are used interchangeably herein. Furthermore, although individually listed, a plurality of means, elements or method actions may be implemented by, e.g., the same entity or object. Additionally, although individual features may be included in different examples or claims, these may possibly be combined, and the inclusion in different examples or claims does not imply that a combination of features is not feasible and/or advantageous.

FIG. 16 is a flowchart representative of example machine readable instructions and/or example operations 1600 that may be executed and/or instantiated by processor circuitry to enable thermal control scheme and/or to implement the thermal management device described above. The machine readable instructions and/or operations 1600 of FIG. 16 begin at block 1602, at which the temperature monitoring circuitry 1504 monitors key hotspot temperatures (e.g., measured temperatures) and the device position determining circuitry 1506 determines a position of the electronic device 100.

At block 1604, the comparator 1510 compares the measured temperature and a selected temperature threshold associated with the detected position. To determine the temperature threshold, for example, the temperature threshold selection circuitry 1508 obtains, receives and/or otherwise retrieves the measured temperature provided by the temperature monitoring circuitry 1504 and the determined position of the electronic device 100 provided by the device position determining circuitry 1506. The temperature threshold selection circuitry 1508 selects a temperature threshold from the database 1514 associated or corresponding with the determined device position and/or the measured temperature.

If at block 1604 the measured temperature (e.g., skin temperature) exceeds the selected threshold temperature (e.g., the current spec), the jet device controller 1512 determines if the one or more synthetic jets 402 are in an activated

state (block 1606). If the jet device controller 1512 determines that the jets are not in an activate state at block 1606, the jet device controller 1512 activates the one or more synthetic jets 402 of the synthetic jet system 156 to increase an airflow through the electronic device 100 (block 1608). If the jet device controller 1512 determines that the synthetic jets 402 are in an activated state at block 1604, the jet device controller 1512 increases a frequency of the one or more synthetic jets 402 to increase an airflow through the electronic device 100 (block 1610).

If at block 1604 the measured temperature (e.g., skin temperature) does not exceed the selected threshold temperature (e.g., the current spec), the device position determining circuitry 1506 determines whether the position of the electronic device 100 is in a restricted position (e.g., a user touch position, laptop mode 122, tablet mode 124, a stricter position/mode)(block 1612).

If at block 1612 the device position determining circuitry 1506 determines that the position of the electronic device 100 is in a restricted position (e.g., a user touch position, laptop mode 122, tablet mode 124, a stricter position/mode), the jet device controller 1512 activates the synthetic jet system 156 and/or maintains a frequency of one or more synthetic jets 402 of the synthetic jet system 156 to increase airflow through the electronic device 100 (block 1614).

If at block 1612 the device position determining circuitry 1506 determines that the position of the electronic device 100 is not in a restricted position (e.g., a non-user touch position, a tent mode 128, a kiosk mode 126, a relaxed position), the jet device controller 1512 determines if the synthetic jets 402 are in a deactivated state (block 1616).

If at block 1616 the jet device controller 1512 determines that the jets are in a deactivated state, control returns to block 1602.

If at block 1616 the jet device controller 1512 determines that the jets are not a deactivated state, the jet device controller 1512 determines if a jet frequency is greater than frequency threshold (block 1618).

If at block 1618 the jet device controller 1512 determines that the jet frequency is greater than frequency threshold, the jet device controller 1512 reduces the jet frequency and control returns to block 1602.

If at block 1618 the jet device controller 1512 determines that the jet frequency is not greater than frequency threshold, the jet device controller 1512 deactivates the one or more synthetic jets 402 of the synthetic jet system 156 and control returns to block 1602.

FIG. 17 is a block diagram of an example processor platform 1700 structured to execute and/or instantiate the machine readable instructions and/or the operations of FIG. 16 to implement the thermal management device 158 of FIG. 15. The processor platform 1700 can be, for example, a server, a personal computer, a workstation, a self-learning machine (e.g., a neural network), a mobile device (e.g., a cell phone, a smart phone, a tablet such as an iPad™), a personal digital assistant (PDA), an Internet appliance, a headset (e.g., an augmented reality (AR) headset, a virtual reality (VR) headset, etc.) or other wearable device, or any other type of computing device.

The processor platform 1700 of the illustrated example includes processor circuitry 1712. The processor circuitry 1712 of the illustrated example is hardware. For example, the processor circuitry 1712 can be implemented by one or more integrated circuits, logic circuits, FPGAs microprocessors, CPUs, GPUs, DSPs, and/or microcontrollers from any desired family or manufacturer. The processor circuitry 1712 may be implemented by one or more semiconductor

based (e.g., silicon based) devices. In this example, the processor circuitry 1712 implements the example temperature monitoring circuitry 1504, the example device position determining circuitry 1506, the example temperature threshold selection circuitry 1508, the example comparator 1510 and the example synthetic jet controller 1512.

The processor circuitry 1712 of the illustrated example includes a local memory 1713 (e.g., a cache, registers, etc.). The processor circuitry 1712 of the illustrated example is in communication with a main memory including a volatile memory 1714 and a non-volatile memory 1716 by a bus 1718. The volatile memory 1714 may be implemented by Synchronous Dynamic Random Access Memory (SDRAM), Dynamic Random Access Memory (DRAM), RAMBUS® Dynamic Random Access Memory (RDRAM®), and/or any other type of RAM device. The non-volatile memory 1716 may be implemented by flash memory and/or any other desired type of memory device. Access to the main memory 1714, 1716 of the illustrated example is controlled by a memory controller 1717.

The processor platform 1700 of the illustrated example also includes interface circuitry 1720. The interface circuitry 1720 may be implemented by hardware in accordance with any type of interface standard, such as an Ethernet interface, a universal serial bus (USB) interface, a Bluetooth® interface, a near field communication (NFC) interface, a Peripheral Component Interconnect (PCI) interface, and/or a Peripheral Component Interconnect Express (PCIe) interface.

In the illustrated example, one or more input devices 1722 are connected to the interface circuitry 1720. The input device(s) 1722 permit(s) a user to enter data and/or commands into the processor circuitry 1712. The input device(s) 1722 can be implemented by, for example, an audio sensor, a microphone, a camera (still or video), a keyboard, a button, a mouse, a touchscreen, a track-pad, a trackball, an isopoint device, and/or a voice recognition system.

One or more output devices 1724 are also connected to the interface circuitry 1720 of the illustrated example. The output device(s) 1724 can be implemented, for example, by display devices (e.g., a light emitting diode (LED), an organic light emitting diode (OLED), a liquid crystal display (LCD), a cathode ray tube (CRT) display, an in-place switching (IPS) display, a touchscreen, etc.), and/or a tactile output device. The interface circuitry 1720 of the illustrated example, thus, typically includes a graphics driver card, a graphics driver chip, and/or graphics processor circuitry such as a GPU.

The interface circuitry 1720 of the illustrated example also includes a communication device such as a transmitter, a receiver, a transceiver, a modem, a residential gateway, a wireless access point, and/or a network interface to facilitate exchange of data with external machines (e.g., computing devices of any kind) by a network 1726. The communication can be by, for example, an Ethernet connection, a digital subscriber line (DSL) connection, a telephone line connection, a coaxial cable system, a satellite system, a line-of-site wireless system, a cellular telephone system, an optical connection, etc.

The processor platform 1700 of the illustrated example also includes one or more mass storage devices 1728 to store software and/or data. Examples of such mass storage devices 1728 include magnetic storage devices, optical storage devices, floppy disk drives, HDDs, CDs, Blu-ray disk drives, redundant array of independent disks (RAID) systems, solid state storage devices such as flash memory devices, and DVD drives.

The machine executable instructions **1732**, which may be implemented by the machine readable instructions of FIG. **15**, may be stored in the mass storage device **1728**, in the volatile memory **1714**, in the non-volatile memory **1716**, and/or on a removable non-transitory computer readable storage medium such as a CD or DVD.

FIG. **18** is a block diagram of an example implementation of the processor circuitry **1712** of FIG. **17**. In this example, the processor circuitry **1712** of FIG. **17** is implemented by a general purpose microprocessor **1800**. The general purpose microprocessor circuitry **1800** executes some or all of the machine readable instructions of the flowchart of FIG. **16** to effectively instantiate the circuitry of FIG. **15** as logic circuits to perform the operations corresponding to those machine readable instructions. In some such examples, the circuitry of FIG. **15** is instantiated by the hardware circuits of the microprocessor **1800** in combination with the instructions. For example, the microprocessor **1800** may implement multi-core hardware circuitry such as a CPU, a DSP, a GPU, an XPU, etc. Although it may include any number of example cores **1802** (e.g., 1 core), the microprocessor **1800** of this example is a multi-core semiconductor device including N cores. The cores **1802** of the microprocessor **1800** may operate independently or may cooperate to execute machine readable instructions. For example, machine code corresponding to a firmware program, an embedded software program, or a software program may be executed by one of the cores **1802** or may be executed by multiple ones of the cores **1802** at the same or separate times. In some examples, the machine code corresponding to the firmware program, the embedded software program, or the software program is split into threads and executed in parallel by two or more of the cores **1802**. The software program may correspond to a portion or all of the machine readable instructions and/or operations represented by the flowchart of FIG. **15**.

The cores **1802** may communicate by a first example bus **1804**. In some examples, the first bus **1804** may implement a communication bus to effectuate communication associated with one(s) of the cores **1802**. For example, the first bus **1804** may implement at least one of an Inter-Integrated Circuit (I2C) bus, a Serial Peripheral Interface (SPI) bus, a PCI bus, or a PCIe bus. Additionally, or alternatively, the first bus **1804** may implement any other type of computing or electrical bus. The cores **1802** may obtain data, instructions, and/or signals from one or more external devices by example interface circuitry **1806**. The cores **1802** may output data, instructions, and/or signals to the one or more external devices by the interface circuitry **1806**. Although the cores **1802** of this example include example local memory **1820** (e.g., Level 1 (L1) cache that may be split into an L1 data cache and an L1 instruction cache), the microprocessor **1800** also includes example shared memory **1810** that may be shared by the cores (e.g., Level 2 (L2) cache) for high-speed access to data and/or instructions. Data and/or instructions may be transferred (e.g., shared) by writing to and/or reading from the shared memory **1810**. The local memory **1820** of each of the cores **1802** and the shared memory **1810** may be part of a hierarchy of storage devices including multiple levels of cache memory and the main memory (e.g., the main memory **1714**, **1716** of FIG. **17**). Typically, higher levels of memory in the hierarchy exhibit lower access time and have smaller storage capacity than lower levels of memory. Changes in the various levels of the cache hierarchy are managed (e.g., coordinated) by a cache coherency policy.

Each core **1802** may be referred to as a CPU, DSP, GPU, etc., or any other type of hardware circuitry. Each core **1802**

includes control unit circuitry **1814**, arithmetic and logic (AL) circuitry (sometimes referred to as an ALU) **1816**, a plurality of registers **1818**, the L1 cache **1820**, and an example second bus **1822**. Other structures may be present. For example, each core **1802** may include vector unit circuitry, single instruction multiple data (SIMD) unit circuitry, load/store unit (LSU) circuitry, branch/jump unit circuitry, floating-point unit (FPU) circuitry, etc. The control unit circuitry **1814** includes semiconductor-based circuits structured to control (e.g., coordinate) data movement within the corresponding core **1802**. The AL circuitry **1816** includes semiconductor-based circuits structured to perform one or more mathematic and/or logic operations on the data within the corresponding core **1802**. The AL circuitry **1816** of some examples performs integer based operations. In other examples, the AL circuitry **1816** also performs floating point operations. In yet other examples, the AL circuitry **1816** may include first AL circuitry that performs integer based operations and second AL circuitry that performs floating point operations. In some examples, the AL circuitry **1816** may be referred to as an Arithmetic Logic Unit (ALU). The registers **1818** are semiconductor-based structures to store data and/or instructions such as results of one or more of the operations performed by the AL circuitry **1816** of the corresponding core **1802**. For example, the registers **1818** may include vector register(s), SIMD register(s), general purpose register(s), flag register(s), segment register(s), machine specific register(s), instruction pointer register(s), control register(s), debug register(s), memory management register(s), machine check register(s), etc. The registers **1818** may be arranged in a bank as shown in FIG. **18**. Alternatively, the registers **1818** may be organized in any other arrangement, format, or structure including distributed throughout the core **1802** to shorten access time. The second bus **1822** may implement at least one of an I2C bus, a SPI bus, a PCI bus, or a PCIe bus.

Each core **1802** and/or, more generally, the microprocessor **1800** may include additional and/or alternate structures to those shown and described above. For example, one or more clock circuits, one or more power supplies, one or more power gates, one or more cache home agents (CHAs), one or more converged/common mesh stops (CMSs), one or more shifters (e.g., barrel shifter(s)) and/or other circuitry may be present. The microprocessor **1800** is a semiconductor device fabricated to include many transistors interconnected to implement the structures described above in one or more integrated circuits (ICs) contained in one or more packages. The processor circuitry may include and/or cooperate with one or more accelerators. In some examples, accelerators are implemented by logic circuitry to perform certain tasks more quickly and/or efficiently than can be done by a general purpose processor. Examples of accelerators include ASICs and FPGAs such as those discussed herein. A GPU or other programmable device can also be an accelerator. Accelerators may be on-board the processor circuitry, in the same chip package as the processor circuitry and/or in one or more separate packages from the processor circuitry.

FIG. **19** is a block diagram of another example implementation of the processor circuitry **1712** of FIG. **17**. In this example, the processor circuitry **1712** is implemented by FPGA circuitry **1900**. The FPGA circuitry **1900** can be used, for example, to perform operations that could otherwise be performed by the example microprocessor **1800** of FIG. **18** executing corresponding machine readable instructions. However, once configured, the FPGA circuitry **1900** instantiates the machine readable instructions in hardware and,

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thus, can often execute the operations faster than they could be performed by a general purpose microprocessor executing the corresponding software.

More specifically, in contrast to the microprocessor **1800** of FIG. **18** described above (which is a general purpose device that may be programmed to execute some or all of the machine readable instructions represented by the flowchart of FIG. **15** but whose interconnections and logic circuitry are fixed once fabricated), the FPGA circuitry **1900** of the example of FIG. **6** includes interconnections and logic circuitry that may be configured and/or interconnected in different ways after fabrication to instantiate, for example, some or all of the machine readable instructions represented by the flowchart of FIG. **3**. In particular, the FPGA **1900** may be thought of as an array of logic gates, interconnections, and switches. The switches can be programmed to change how the logic gates are interconnected by the interconnections, effectively forming one or more dedicated logic circuits (unless and until the FPGA circuitry **1900** is reprogrammed). The configured logic circuits enable the logic gates to cooperate in different ways to perform different operations on data received by input circuitry. Those operations may correspond to some or all of the software represented by the flowchart of FIG. **15**. As such, the FPGA circuitry **1900** may be structured to effectively instantiate some or all of the machine readable instructions of the flowchart of FIG. **15** as dedicated logic circuits to perform the operations corresponding to those software instructions in a dedicated manner analogous to an ASIC. Therefore, the FPGA circuitry **1900** may perform the operations corresponding to the some or all of the machine readable instructions of FIG. **19** faster than the general purpose microprocessor can execute the same.

In the example of FIG. **19**, the FPGA circuitry **1900** is structured to be programmed (and/or reprogrammed one or more times) by an end user by a hardware description language (HDL) such as Verilog. The FPGA circuitry **1900** of FIG. **19**, includes example input/output (I/O) circuitry **1902** to obtain and/or output data to/from example configuration circuitry **1904** and/or external hardware (e.g., external hardware circuitry) **1906**. For example, the configuration circuitry **1904** may implement interface circuitry that may obtain machine readable instructions to configure the FPGA circuitry **1900**, or portion(s) thereof. In some such examples, the configuration circuitry **1904** may obtain the machine readable instructions from a user, a machine (e.g., hardware circuitry (e.g., programmed or dedicated circuitry) that may implement an Artificial Intelligence/Machine Learning (AI/ML) model to generate the instructions), etc. In some examples, the external hardware **1906** may implement the microprocessor **1800** of FIG. **18**. The FPGA circuitry **1900** also includes an array of example logic gate circuitry **1908**, a plurality of example configurable interconnections **1910**, and example storage circuitry **1912**. The logic gate circuitry **1908** and interconnections **1910** are configurable to instantiate one or more operations that may correspond to at least some of the machine readable instructions of FIG. **15** and/or other desired operations. The logic gate circuitry **1908** shown in FIG. **19** is fabricated in groups or blocks. Each block includes semiconductor-based electrical structures that may be configured into logic circuits. In some examples, the electrical structures include logic gates (e.g., And gates, Or gates, Nor gates, etc.) that provide basic building blocks for logic circuits. Electrically controllable switches (e.g., transistors) are present within each of the logic gate circuitry **1908** to enable configuration of the electrical structures and/or the logic gates to form circuits to perform desired

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operations. The logic gate circuitry **1908** may include other electrical structures such as look-up tables (LUTs), registers (e.g., flip-flops or latches), multiplexers, etc.

The interconnections **1910** of the illustrated example are conductive pathways, traces, vias, or the like that may include electrically controllable switches (e.g., transistors) whose state can be changed by programming (e.g., using an HDL instruction language) to activate or deactivate one or more connections between one or more of the logic gate circuitry **1908** to program desired logic circuits.

The storage circuitry **1912** of the illustrated example is structured to store result(s) of the one or more of the operations performed by corresponding logic gates. The storage circuitry **1912** may be implemented by registers or the like. In the illustrated example, the storage circuitry **1912** is distributed amongst the logic gate circuitry **1908** to facilitate access and increase execution speed.

The example FPGA circuitry **1900** of FIG. **19** also includes example Dedicated Operations Circuitry **1914**. In this example, the Dedicated Operations Circuitry **1914** includes special purpose circuitry **1916** that may be invoked to implement commonly used functions to avoid the need to program those functions in the field. Examples of such special purpose circuitry **1916** include memory (e.g., DRAM) controller circuitry, PCIe controller circuitry, clock circuitry, transceiver circuitry, memory, and multiplier-accumulator circuitry. Other types of special purpose circuitry may be present. In some examples, the FPGA circuitry **1900** may also include example general purpose programmable circuitry **1918** such as an example CPU **1920** and/or an example DSP **1922**. Other general purpose programmable circuitry **1918** may additionally or alternatively be present such as a GPU, an XPU, etc., that can be programmed to perform other operations.

Although FIGS. **18** and **19** illustrate two example implementations of the processor circuitry **1712** of FIG. **17**, many other approaches are contemplated. For example, as mentioned above, modern FPGA circuitry may include an on-board CPU, such as one or more of the example CPU **1920** of FIG. **19**. Therefore, the processor circuitry **1712** of FIG. **17** may additionally be implemented by combining the example microprocessor **1800** of FIG. **18** and the example FPGA circuitry **1900** of FIG. **19**. In some such hybrid examples, a first portion of the machine readable instructions represented by the flowchart of FIG. **15** may be executed by one or more of the cores **1802** of FIG. **18** and a second portion of the machine readable instructions represented by the flowchart of FIG. **15** may be executed by the FPGA circuitry **1900** of FIG. **19**, and/or a third portion of the machine readable instructions represented by the flowchart of FIG. **16** may be executed by an ASIC. It should be understood that some or all of the circuitry of FIG. **15** may, thus, be instantiated at the same or different times. Some or all of the circuitry may be instantiated, for example, in one or more threads executing concurrently and/or in series. Moreover, in some examples, some or all of the circuitry of FIG. **15** may be implemented within one or more virtual machines and/or containers executing on the microprocessor.

In some examples, the processor circuitry **1712** of FIG. **17** may be in one or more packages. For example, the processor circuitry **1800** of FIG. **18** and/or the FPGA circuitry **1900** of FIG. **19** may be in one or more packages. In some examples, an XPU may be implemented by the processor circuitry **1712** of FIG. **17**, which may be in one or more packages. For example, the XPU may include a CPU in one package, a

DSP in another package, a GPU in yet another package, and an FPGA in still yet another package.

Example methods, apparatus, systems, and articles of manufacture to analyze computer system attack mechanisms are disclosed herein. Further examples and combinations thereof include the following:

Example 1 includes an electronic device having a heat spreader including a first surface and a second surface opposite the first surface. A synthetic jet is coupled to the first surface of the heat spreader. The synthetic jet and the heat spreader define a fluid flow passageway having an inlet, a first outlet and a second outlet. The synthetic jet and the heat spreader to bifurcate airflow from the inlet such that the first outlet is to exhaust airflow from the passageway adjacent the first surface of the heat spreader and the second outlet is to exhaust airflow from the passageway adjacent the second surface of the heat spreader.

Example 2 includes the electronic device of example 1, where the heat spreader and the synthetic jet define a bypass to bifurcate the airflow.

Example 3 includes the electronic component of any one of the examples 1 and 2, where the bypass is formed via a first opening in the heat spreader between the first surface and the second surface and a second opening in a body of the synthetic jet, the first opening aligned with the second opening.

Example 4 includes the electronic component of any one of the examples 1-3, where the first outlet is defined by third opening in the body of the synthetic jet, and the second outlet is formed by the first opening and the second opening.

Example 5 includes the electronic component of any one of the examples 1-4, where the third opening is in a first wall of the body and the second opening is in a second wall of the body, the first wall is non-parallel relative to the second wall.

Example 6 includes the electronic component of any one of the examples 1-5, further including a gasket between the synthetic jet and the first surface of the heat spreader, the gasket includes a slot to fluidly couple the first opening and the second opening.

Example 7 includes the electronic component of any one of the examples 1-6, further including a heat exchanger downstream from the second outlet.

Example 8 includes the electronic component of any one of the examples 1-7, further including a duct to channel fluid flow from the first opening and the second opening toward the heat exchanger.

Example 9 includes the electronic component of any one of the examples 1-8, where the second surface of the heat spreader is oriented toward processor circuitry.

Example 10 includes the electronic component of any one of the examples 1-9, where at least one of the first outlet or the second outlet is oriented in a direction away from the processor circuitry.

Example 11 includes the electronic component of any one of the examples 1-10, where at least one of the first outlet or the second outlet is oriented in a direction towards the processor circuitry.

Example 12 includes the electronic component of any one of the examples 1-11, where the synthetic jet is spaced laterally relative to the processor circuitry such that the synthetic jet is positioned outside a perimeter of the processor circuitry.

Example 13 includes the electronic component of any one of the examples 1-12, where the synthetic jet is not positioned over the processor circuitry.

Example 14 includes a synthetic jet including a body having a plurality of walls to define to define a fluid flow

passageway, a first wall of the plurality of walls having a first opening to define an inlet of the fluid flow passageway, a second wall of the plurality of walls having a second opening to define a first outlet of the fluid flow passageway, a third wall of the plurality of walls having a third opening to define a second outlet of the fluid flow passageway, and an actuator positioned in the fluid flow passageway to generate a fluid flow from the inlet and to the first outlet and the second outlet.

Example 15 includes the electronic device of example 14, where the body is coupled to a heat spreader, the heat spreader including a fourth opening aligned with the third opening.

Example 16 includes the electronic device of any one of examples 14-15, where the second wall is perpendicular relative to the first wall and the third wall.

Example 17 includes the electronic device of any one of examples 14-16, where the first outlet is formed via a first aperture in a first wall of the body, and wherein the second outlet is formed via a second aperture in a second wall of the body.

Example 18 includes the electronic device of any one of examples 14-17, further including a gasket coupled to an outer surface of the third wall, the gasket having a slot aligned with the third opening.

Example 19 includes an electronic device having a processor circuitry, a heat spreader to dissipate heat from the processor circuitry, and a plurality of synthetic jets coupled to the heat spreader, the plurality of synthetic jets defining outlets.

Example 20 includes the electronic device of example 19, where the outlets of the synthetic jets are oriented in a direction away from the processor circuitry.

Example 21 includes the electronic device of any one of examples 19 or 20, where the outlets of the synthetic jets are oriented in a direction toward the processor circuitry.

Example 22 includes the electronic device of any one of examples 19-21, where the plurality of jets includes a first set of jets having first outlets oriented in a direction toward the processor circuitry and a second set of jets having second outlets oriented in a direction away from the processor circuitry.

Example 23 includes a method including positioning a heat spreader over processor circuitry of an electronic device to dissipate heat generated from the processor circuitry of an electronic device and positioning synthetic jets laterally relative to the processor circuitry such that the synthetic jets are not positioned over the processor circuitry.

Example 24 includes the method of example 23, further including orienting outlets of the synthetic jets in a direction away from the processor circuitry.

Example 25 includes the method of any one of examples 23 or 24, further including orientating the outlets of the synthetic jets in a direction toward the processor circuitry.

Example 26 includes the method of any one of examples 23-25, further including orientating first outlets of a first set of the synthetic jets in a direction toward the processor circuitry and orientating second outlets of a second set of the synthetic jets in a direction away from the processor circuitry.

The following claims are hereby incorporated into this Detailed Description by this reference, with each claim standing on its own as a separate embodiment of the present disclosure.

Although certain example systems, methods, apparatus, and articles of manufacture have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the

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contrary, this patent covers all systems, methods, apparatus, and articles of manufacture fairly falling within the scope of the claims of this patent.

What is claimed is:

1. An electronic device comprising:
 - a heat spreader including a first surface and a second surface opposite the first surface, the heat spreader includes a hole extending between the first surface and the second surface, the hole of the heat spreader having a first longitudinal axis that is non-parallel relative to the first surface and the second surface of the heat spreader;
 - a synthetic jet coupled to the first surface of the heat spreader, the synthetic jet defining a fluid passageway having an inlet, a first outlet and a second outlet, the first outlet having a second longitudinal axis, the second outlet having a third longitudinal axis, the second longitudinal axis of the first outlet being non-parallel relative to the third longitudinal axis of the second outlet, the hole of the heat spreader to axially align with the second outlet of the synthetic jet to define a bypass passageway through the heat spreader, the synthetic jet and the heat spreader to bifurcate airflow from the inlet such that the first outlet is to provide a first exhaust airflow from the fluid passageway adjacent the first surface of the heat spreader and the second outlet is to provide a second exhaust airflow from the fluid passageway adjacent the second surface of the heat spreader via the bypass passageway; and
 - a heat exchanger adjacent the second surface of the heat spreader, the heat exchanger to receive the second exhaust airflow from the bypass passageway.
2. The electronic device of claim 1, wherein the first outlet of the synthetic jet is in a first wall of a body of the synthetic jet and the second outlet of the synthetic jet is in a second wall of the body of the synthetic jet, the first wall is non-parallel relative to the second wall.
3. The electronic device of claim 1, including a gasket between the synthetic jet and the first surface of the heat spreader, the gasket includes a slot to fluidly couple the hole of the heat spreader and the second outlet of the synthetic jet.
4. The electronic device of claim 1, wherein the heat exchanger is downstream from the second outlet.
5. The electronic device of claim 4, including a duct to channel fluid flow from the bypass passageway toward the heat exchanger.
6. The electronic device of claim 1, wherein the second surface of the heat spreader is oriented toward processor circuitry.
7. The electronic device of claim 6, wherein at least one of the first outlet or the second outlet is oriented in a direction away from the processor circuitry.
8. The electronic device of claim 6, wherein at least one of the first outlet or the second outlet is oriented in a direction towards the processor circuitry.
9. The electronic device of claim 6, wherein the synthetic jet is spaced laterally relative to a board supporting the processor circuitry such that the synthetic jet is outside a perimeter of the board supporting the processor circuitry.
10. The electronic device of claim 9, wherein the synthetic jet is not positioned over the processor circuitry.
11. The electronic device of claim 1, wherein the exhaust airflow from the first outlet does not flow through the heat exchanger.

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12. The electronic device of claim 1, wherein the heat spreader is to distribute heat across a surface area of at least one of the first surface or the second surface to prevent localized hotspots.

13. The electronic device of claim 1, wherein the heat exchanger only receives the second exhaust airflow from the second outlet.

14. The electronic device of claim 1, wherein the heat spreader is between the heat exchanger and the synthetic jet.

15. The electronic device of claim 1, wherein the synthetic jet includes:

- a body having a plurality of walls defining the fluid passageway;
- a first wall of the plurality of walls having a first opening to define the inlet of the fluid passageway;
- a second wall of the plurality of walls having a second opening to define the first outlet of the fluid passageway;
- a third wall of the plurality of walls having a third opening to define the second outlet of the fluid passageway; and
- an actuator in the fluid passageway to generate a fluid flow from the inlet and to the first outlet and the second outlet.

16. The electronic device of claim 15, wherein the body is coupled to the heat spreader, the heat spreader including a fourth opening aligned with the third opening.

17. The electronic device of claim 15, wherein the second wall is perpendicular relative to the first wall and the third wall.

18. The electronic device of claim 15, including a gasket coupled to an outer surface of the third wall, the gasket having a slot aligned with the third opening.

19. The electronic device of claim 1, wherein the first longitudinal axis of the hole is normal to a parallel surface of the heat spreader.

20. The electronic device of claim 1, wherein the second longitudinal axis of the first outlet is perpendicular relative to the third longitudinal axis of the second outlet.

21. The electronic device of claim 1, wherein the first longitudinal axis of the hole of the heat spreader coaxially aligns with the third longitudinal axis of the second outlet of the synthetic jet.

22. An electronic device including:

- processor circuitry;
- a heat spreading plate to distribute heat from the processor circuitry across a larger surface area to prevent localized hotspots, the heat spreading plate including a plurality of holes extending through an upper heat spreading surface of the heat spreading plate and a lower heat spreading surface of the heat spreading plate, the holes being perpendicular to the upper heat spreading surface of the heat spreading plate;
- a plurality of synthetic jets coupled to the heat spreading plate, the plurality of synthetic jets defining a first set of outlets and a second set of outlets, the first set of outlets to exhaust a first airflow from the synthetic jets to at least a portion of the upper heat spreading surface of the heat spreading plate, the second outlets to align with respective ones of the holes of the heat spreading plate to provide respective ones of bypass passageways through the heat spreading plate to enable the second set of outlets to exhaust a second airflow from the synthetic jets to at least a portion of the lower heat spreading surface of the heat spreading plate, the first exhaust airflow separated from the second exhaust airflow by the heat spreading plate.

23. The electronic device of claim 22, wherein the first and second sets of outlets of the synthetic jets are oriented in a direction away from the processor circuitry.

24. The electronic device of claim 22, wherein the first and second sets of outlets of the synthetic jets are oriented 5 in a direction toward the processor circuitry.

25. The electronic device of claim 22, wherein a first plurality of the first and second outlets are oriented in a direction toward the processor circuitry and a second plurality of the first and second outlets are oriented in a 10 direction away from the processor circuitry.

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